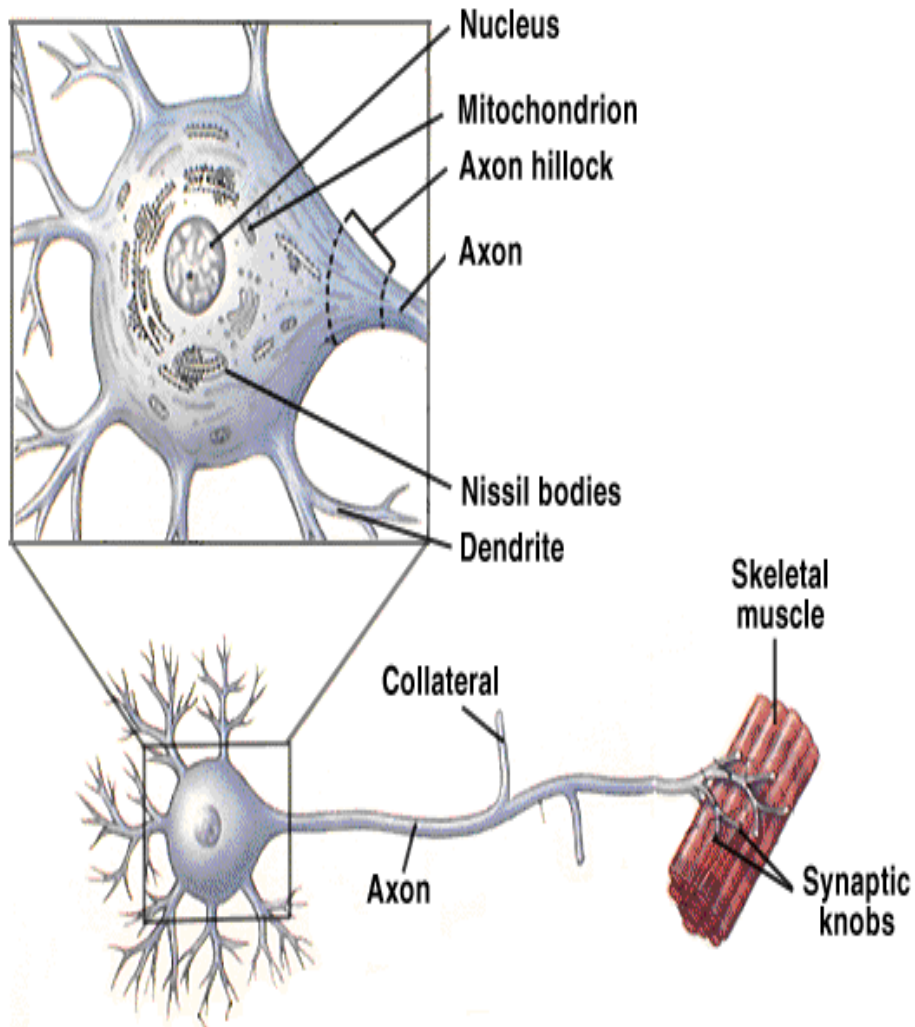


Anatomy of a Neuron



- Cell body: functional portion
- Dendrites: short extensions that receive signals
- Axon: long extension that transmits impulses away

Nervous System

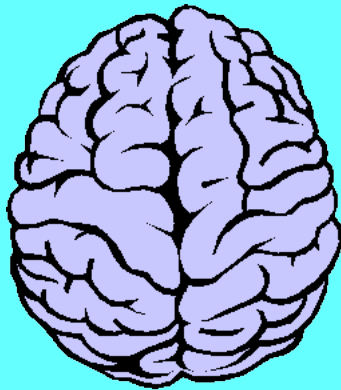
- Millions of interconnected neurons form the nervous system
- Neuron have three parts:
 - Cell body: functional portion
 - Dendrites: short extensions that receive signals
 - Axon: long extension that transmits impulses away

...

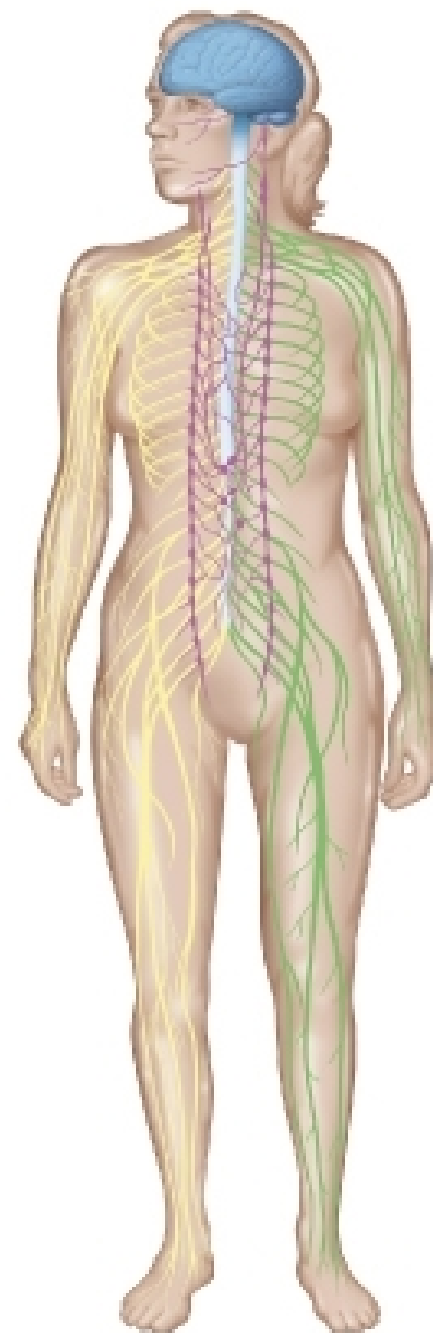
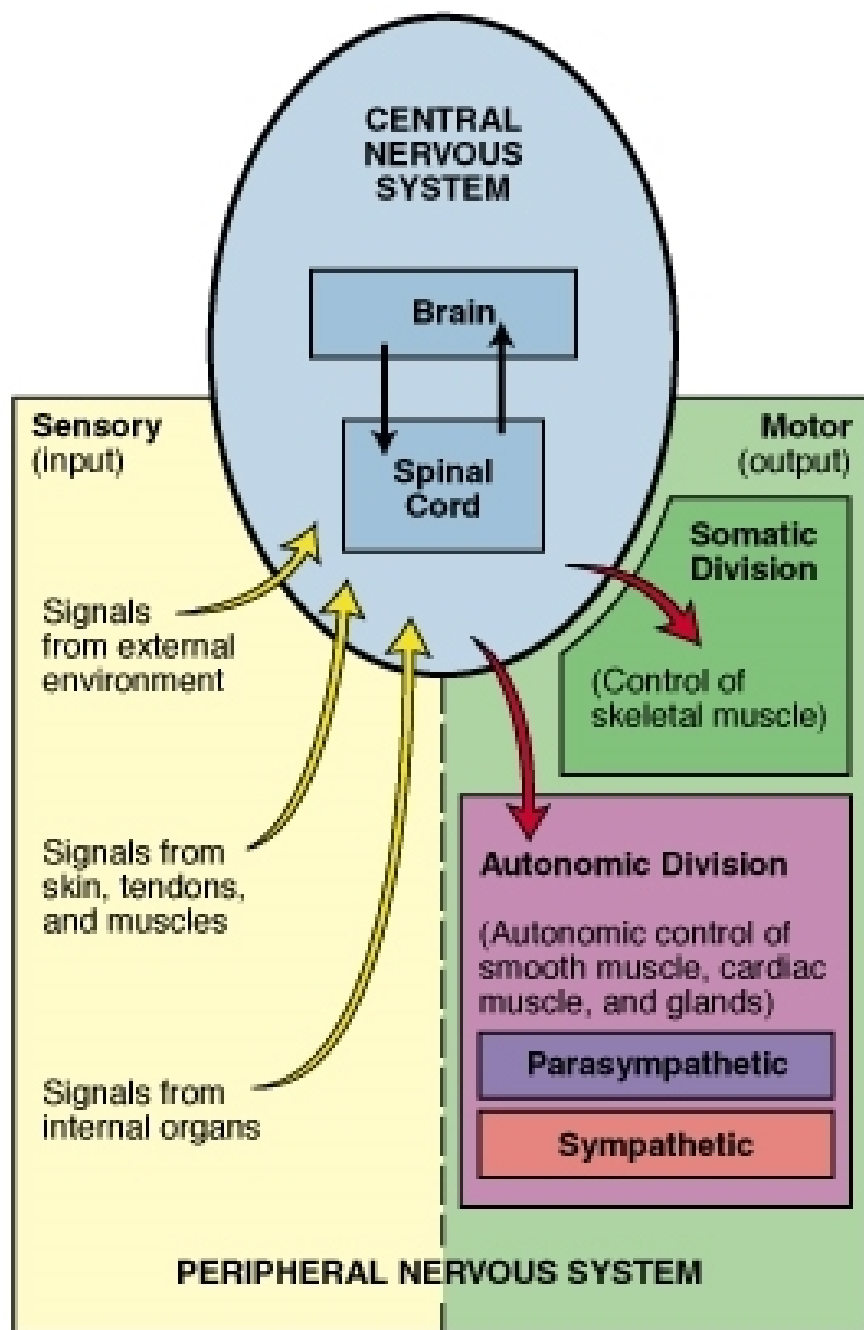
- Human nervous system have two major parts:
 - ü central nervous system
 - ü peripheral nervous system

Central Nervous System

The Central Nervous System is made of the brain and the spinal cord.



The Central Nervous System controls everything in the body.



Brain

An organ that controls your emotions, your thoughts, and every movement you make.



Peripheral Nervous System

The Peripheral Nervous System is made of the nerves and the sense organs.

Nerves



Sense organs



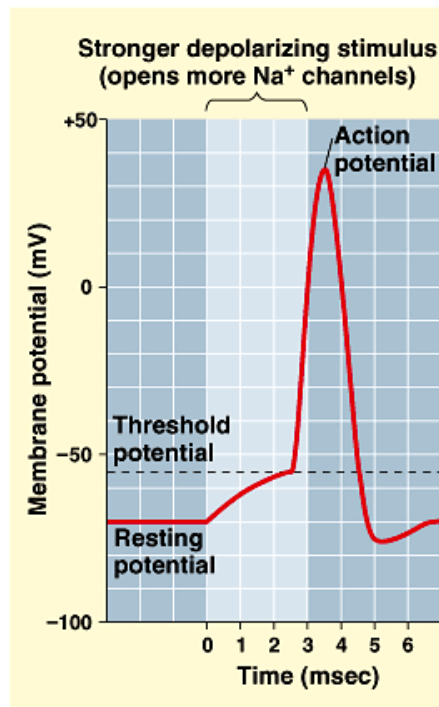
Nerve Impulse

Messages carried throughout the body by nerves.

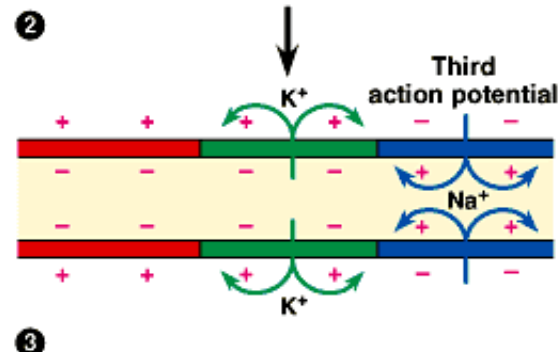
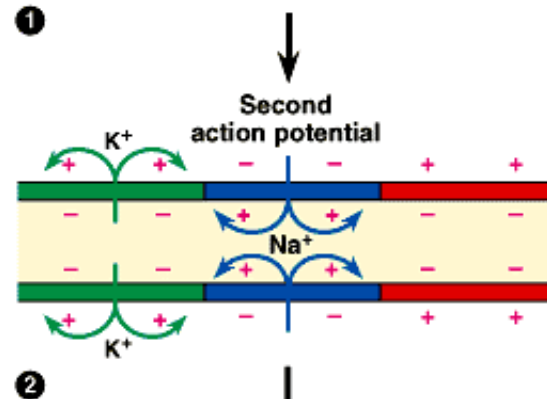
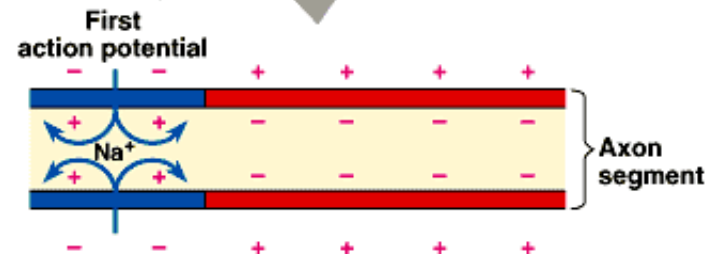
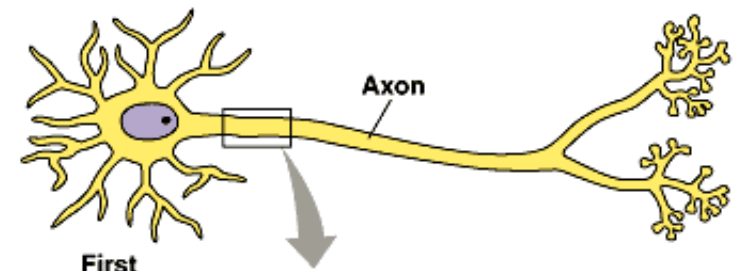
Nerve Impulse - The Action Potential

Threshold potential will trigger an action potential or nerve impulse

The action potential is an **all-or-none** response



(c) Action potential



How does a signal move from one neuron to another?

- A **synapse** divides 2 neurons
- The **action potential** will not move across the synapse
- Neuro transmitters
 - Released by the signal cell to the receiver cell
 - Move by diffusion

...

- **synapse is** junction between the dendrites of one neuron and the axon of a second neuron
- Nerves communicate by releasing chemical messenger at synapse

Types of chemical synapse

- Acetylcholine: neuromuscular junctions, glands, brain and spinal cord
- Norepinephrine: affects brain regions concerned with emotions, dreaming

The Central Nervous System

- * The Central Nervous System controls all of the body's activities.
- * The Central Nervous System is made of two main organs.
 1. The brain
 2. The spinal cord



The Spinal Cord



- * The spinal cord sends messages to the brain.
- * The spinal cord is the part of the nervous system that connects the brain to the rest of the nervous system.

The Brain

- * The brain controls everything in the body.
- * The brain is made of more than 10 billion nerves!
- * The brain is divided into three parts and is protected by the skull.

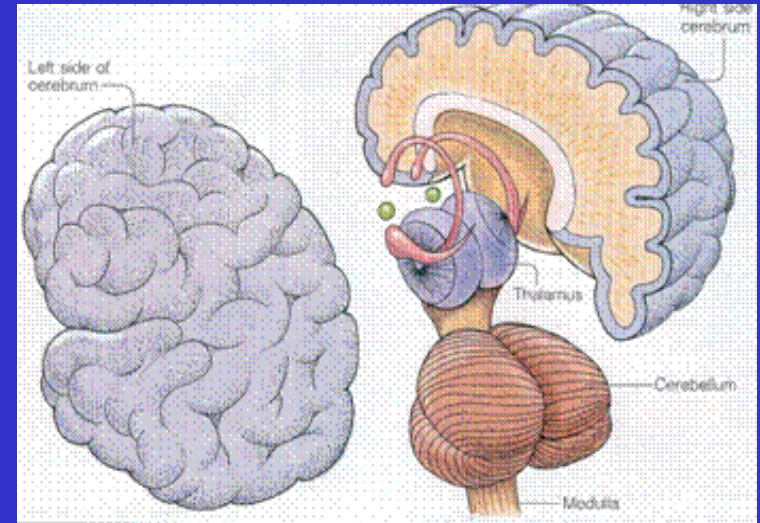


The Three Parts of the Brain



* The Brain has three main parts...

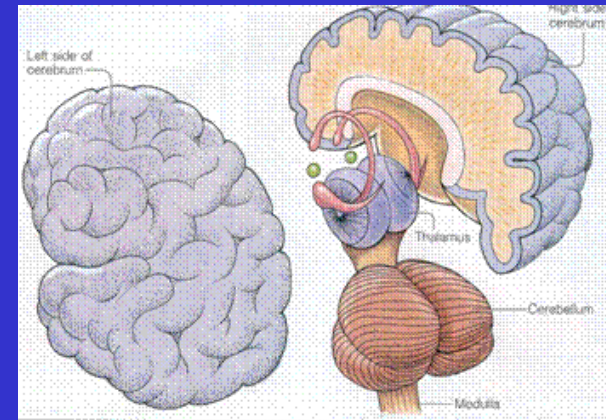
1. The Cerebrum
2. The Cerebellum
3. The Brain Stem

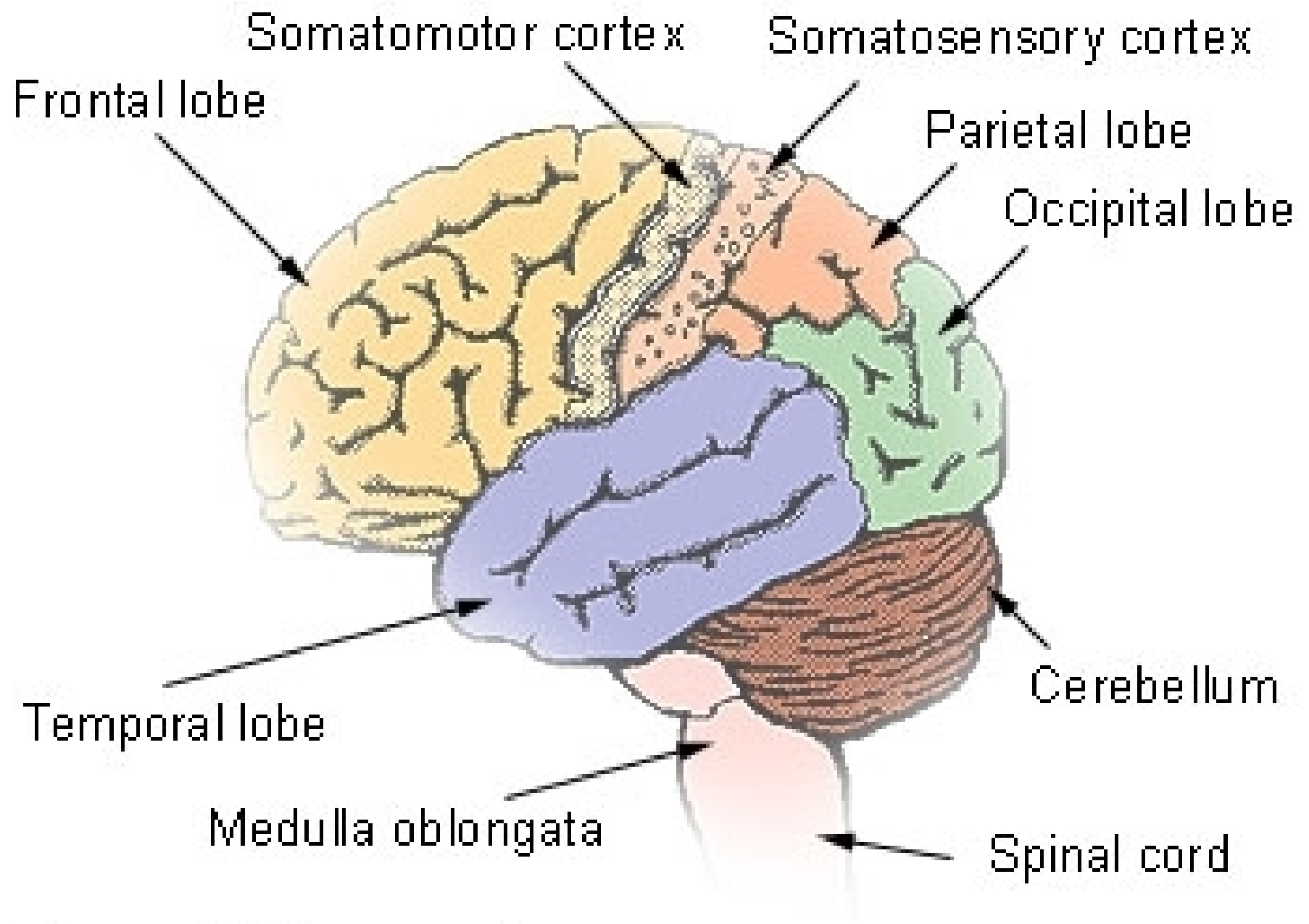


The Cerebrum

* The Cerebrum is the largest part of the brain.

1. The cerebrum controls your thinking.
2. The cerebrum controls your memory.
3. The cerebrum controls your speaking.
4. The cerebrum controls your movement and identifies the information gathered by your sense organs.



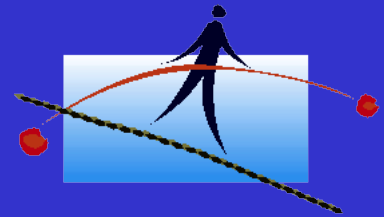


Lobes of the cerebrum

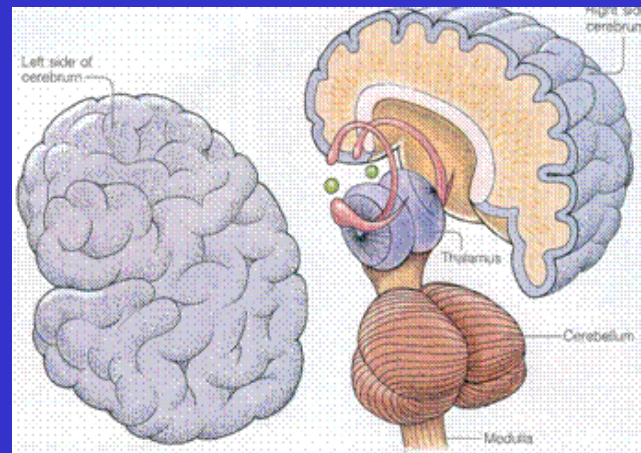
The Cerebellum

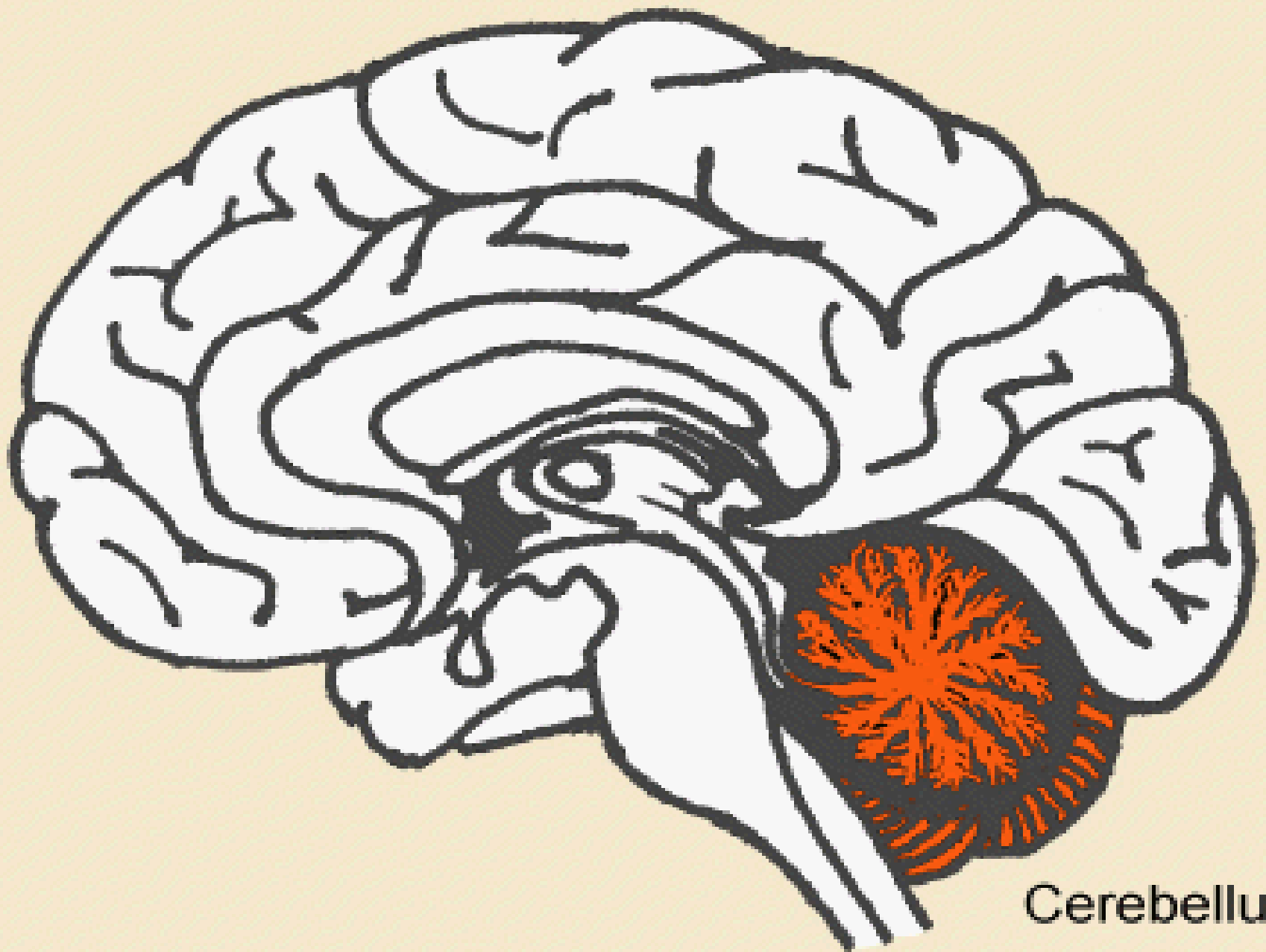
* The cerebellum is below and to the back of the cerebrum.

1. The cerebellum controls you balance.



2. The cerebellum controls your posture.

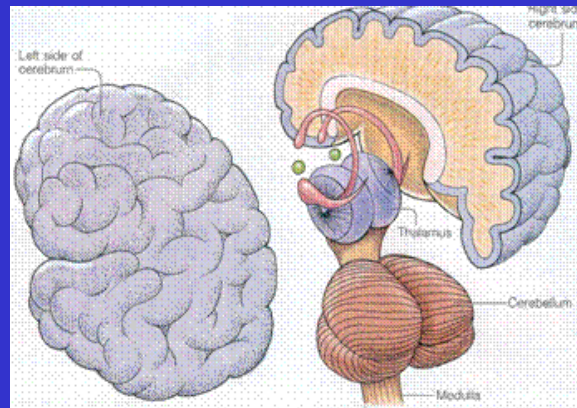


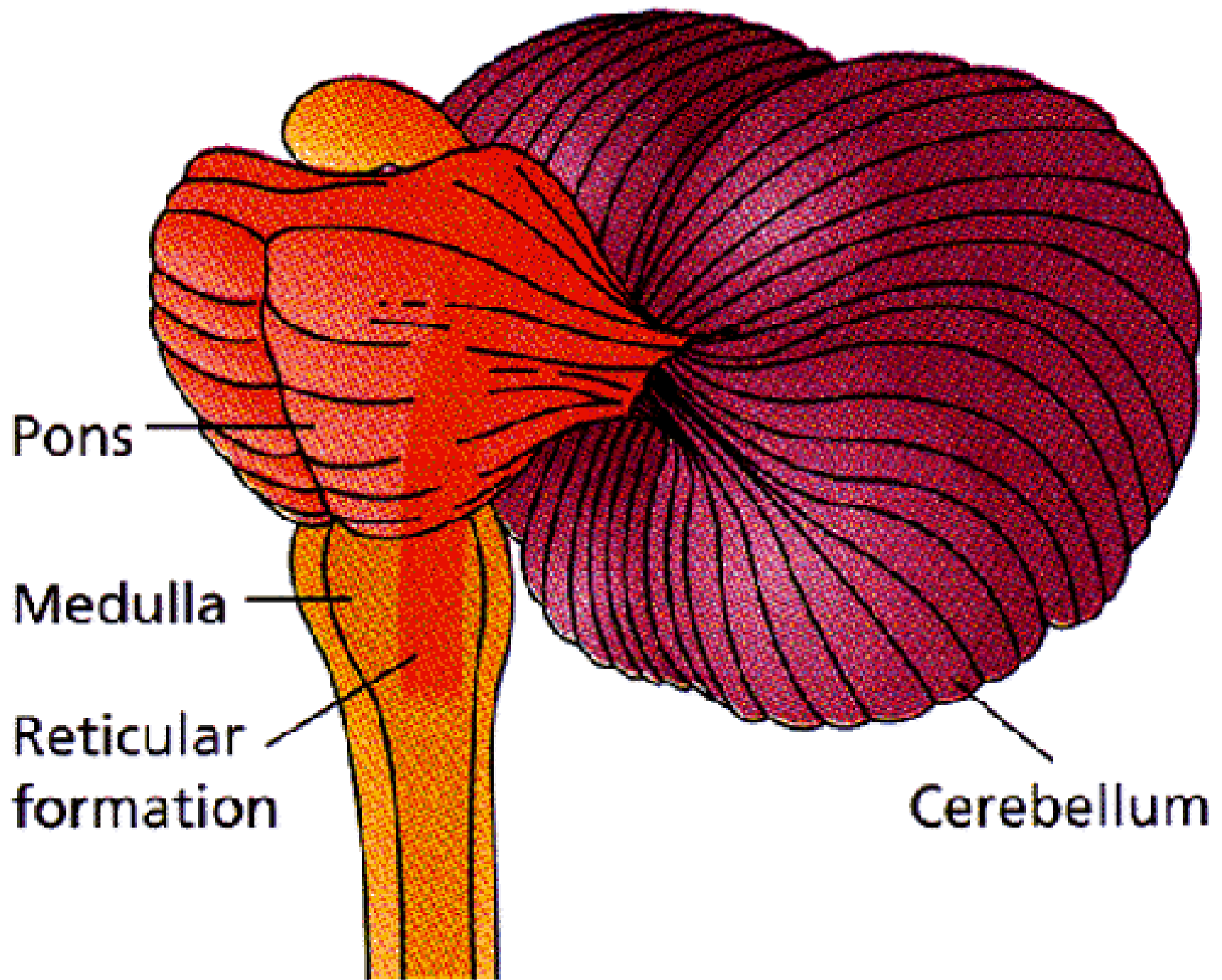


Cerebellum

The Brain Stem

- * The Brain Stem connects the brain to the spinal cord.
- * The nerves in the brain stem control your heartbeat, breathing, and blood pressure.





The Vertebrae

- * The vertebrae are the many bones that protect the nerves in the spinal cord.



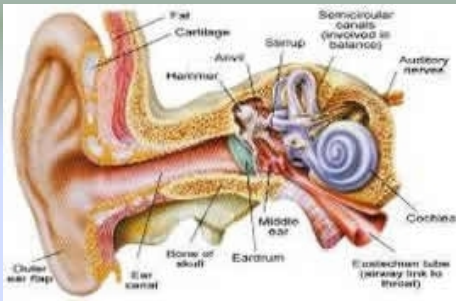
The Peripheral Nervous System

Peripheral Nervous System

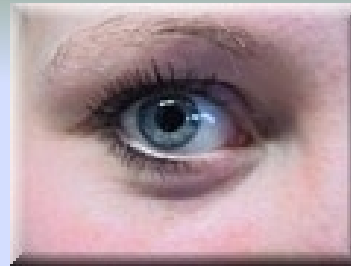
- * The outer nervous system carries messages between the central nervous system and the rest of the body.
- * The Outer Nervous System's job is to connect the Central Nervous System to the rest of the body.

Peripheral Nervous System

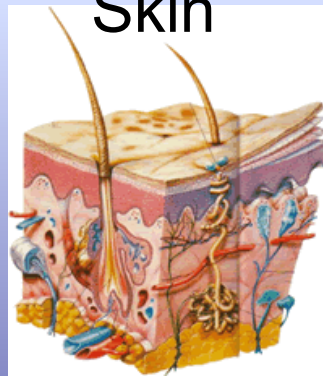
- * The peripheral nervous system is made of the nerves and the sense organs.



Ear



Eye



Skin

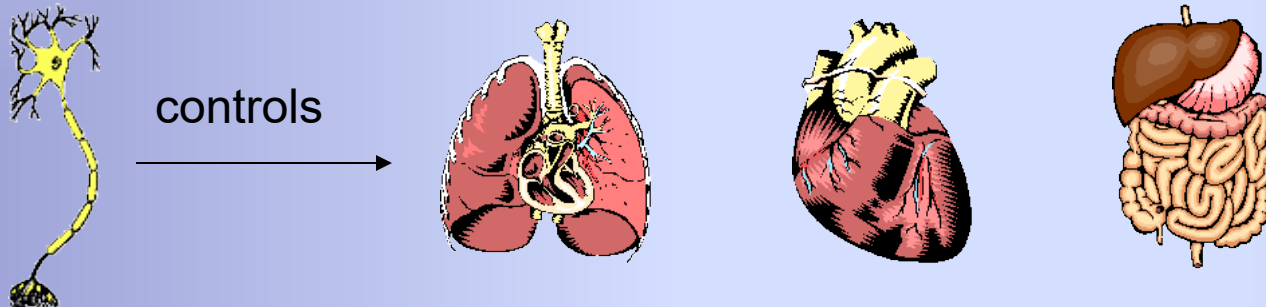
Nerves



Tongue

Automatic Nervous System

- * The outer nervous system controls the body's activities that you don't think about.
- * The outer nervous system controls activities in your small intestine, your breathing, and your heartbeat.



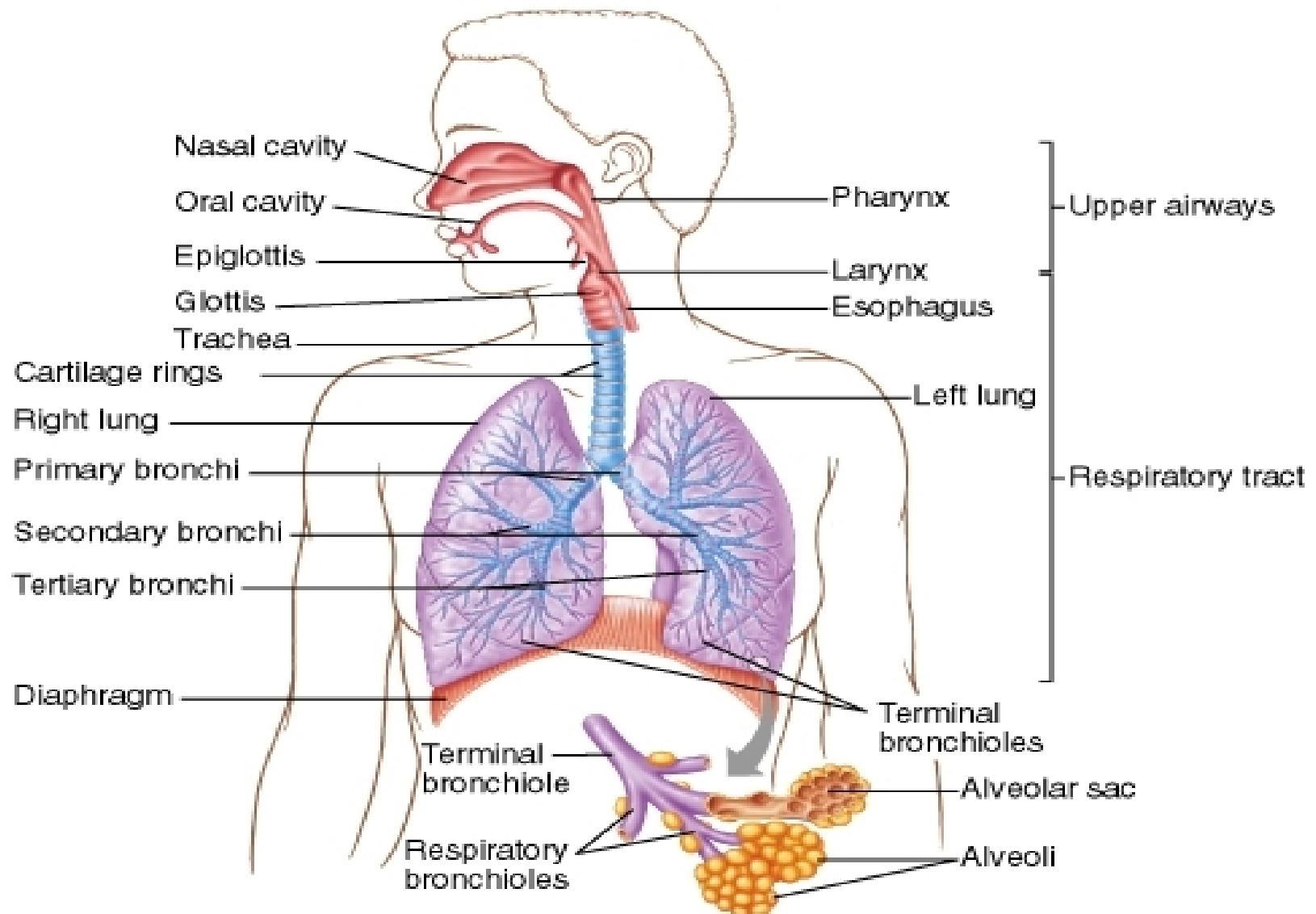
Sense Organs

The eyes, ears, nose, tongue, and skin are examples of sense organs.

The sense organs gather information (light, sound, heat, and pressure) from the environment.

Respiratory System

For Human Nutrition students



Functions of the Respiratory System

- Gas Exchange
 - O_2 , CO_2
- Acid-base balance
 - $CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow H^+ + HCO_3^-$
- Pulmonary defense
- Pulmonary metabolism and handling of bioactive materials

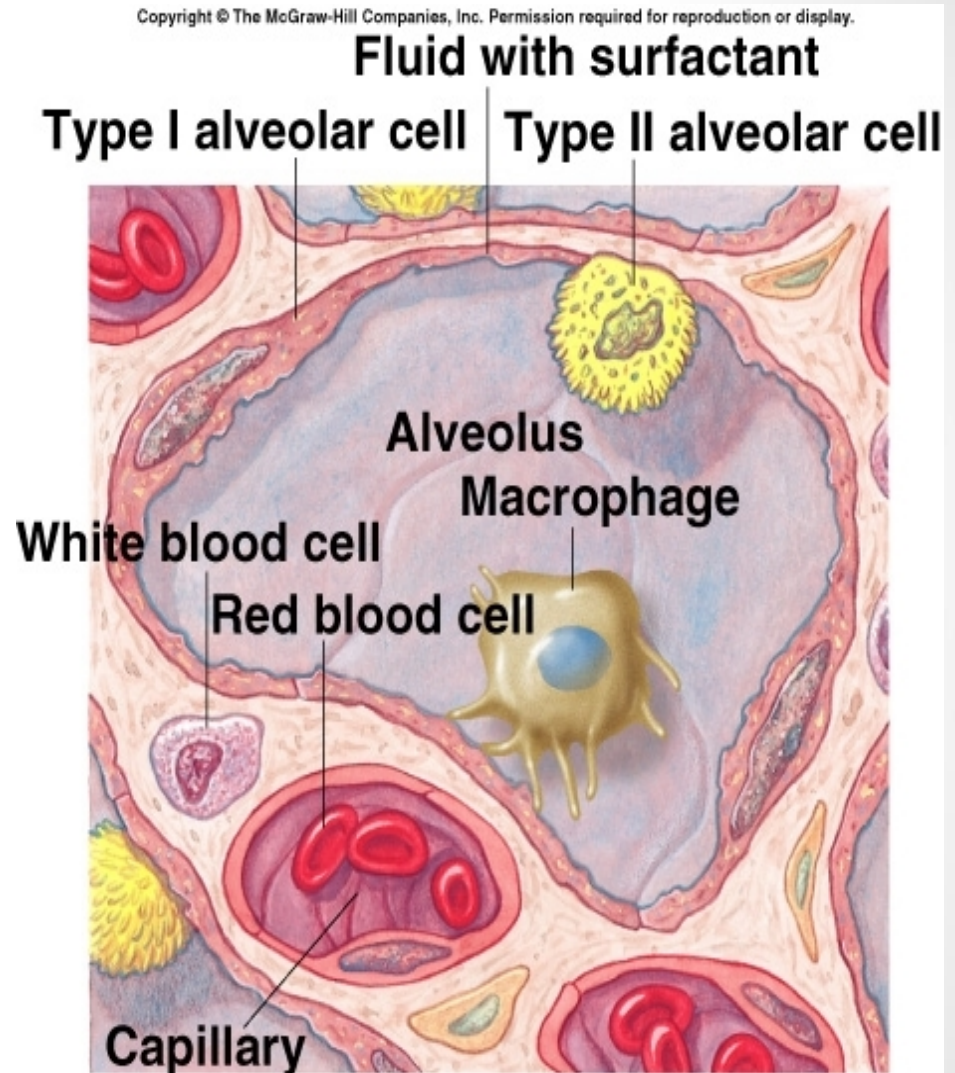
Respiration

The term respiration includes 3 separate functions:

- Ventilation:
 - Breathing.
- Gas exchange:
 - Between air and capillaries in the lungs.
 - Between systemic capillaries and tissues of the body.
- O_2 utilization:
 - Cellular respiration.

Ventilation

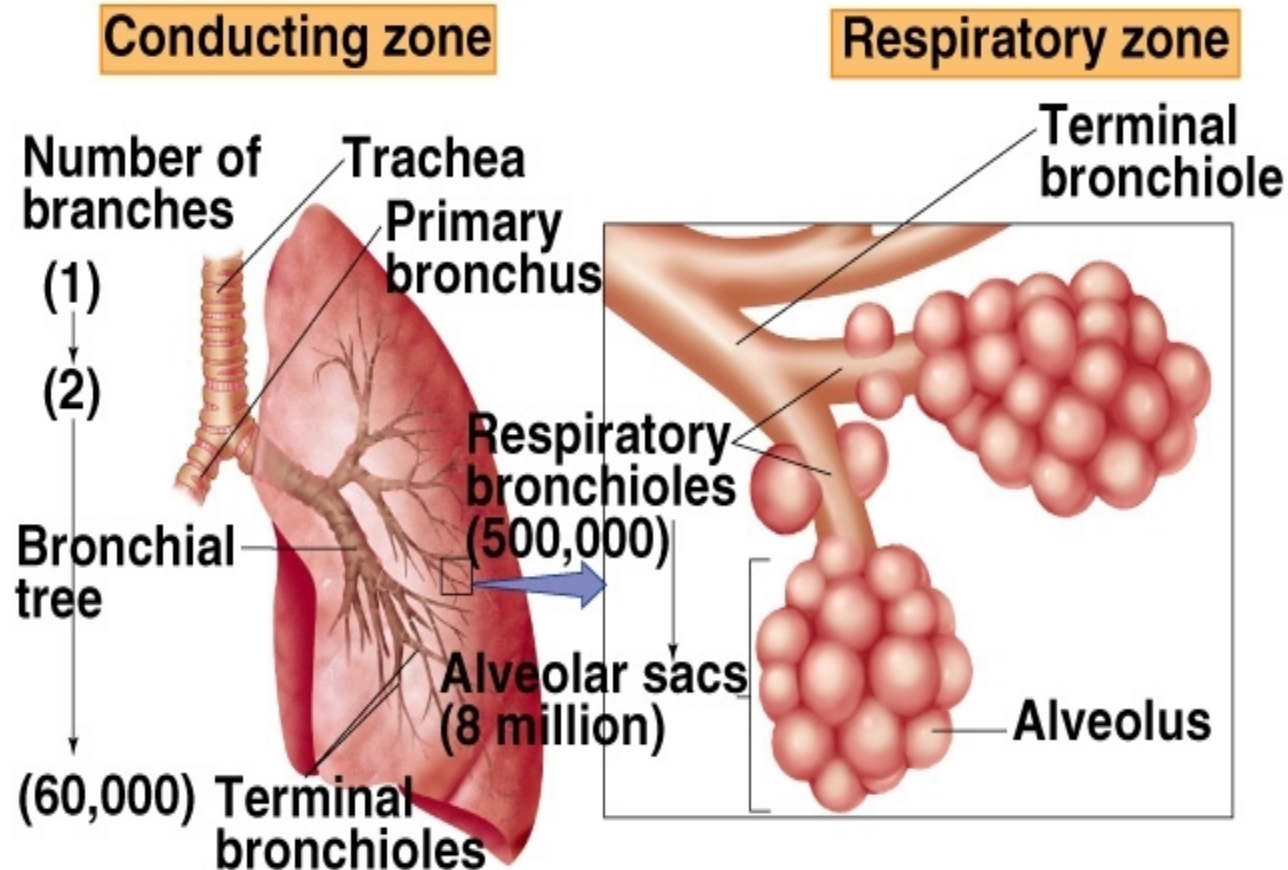
- Mechanical process that moves air in and out of the lungs.
- $[O_2]$ of air is higher in the lungs than in the blood, O_2 diffuses from air to the blood.
- CO_2 moves from the blood to the air by diffusing down its concentration gradient.
- Gas exchange occurs entirely by diffusion:
 - Diffusion is rapid because of the large surface area and the small diffusion distance.



Respiratory Zone

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- Region of gas exchange between air and blood.
- Includes respiratory bronchioles and alveolar sacs.
- Must contain alveoli.



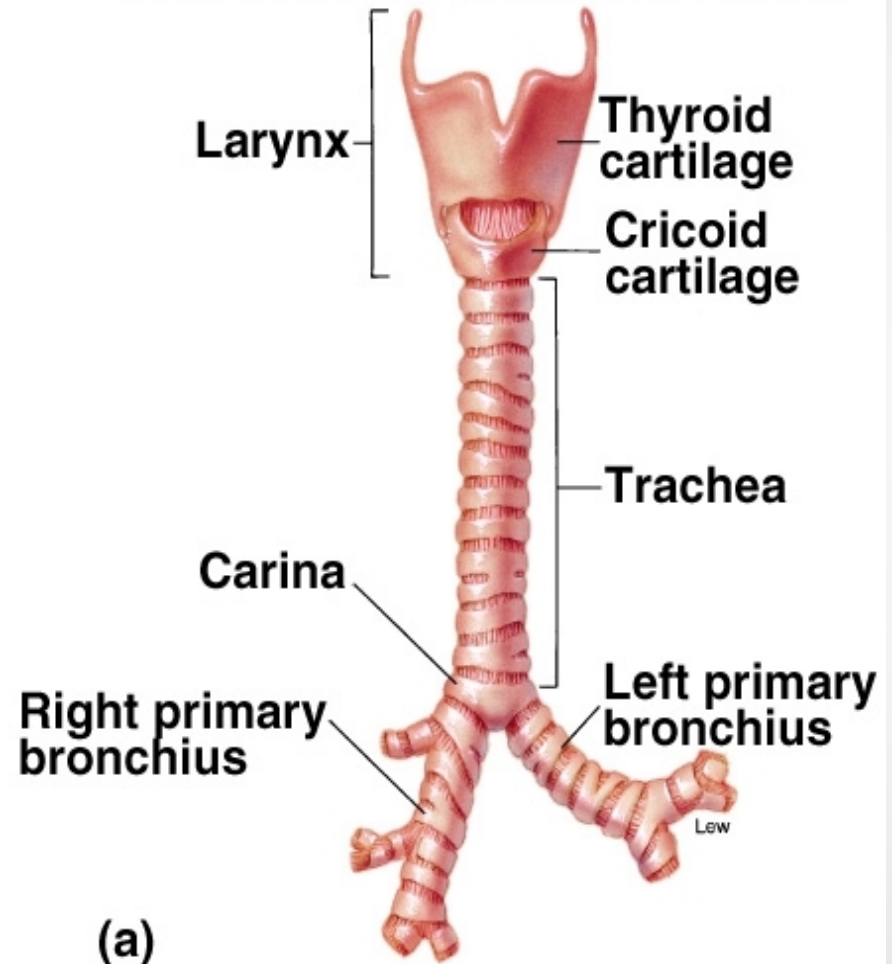
Alveoli

- Polyhedral in shape and clustered like units of honeycomb.
- ~ 300 million air sacs (alveoli).
 - Large surface area (60–80 m²).
 - Each alveolus is 1 cell layer thick.
 - Total air barrier is 2 cells across (2 mm).
- 2 types of cells:
 - Alveolar type I:
 - Structural cells.
 - Alveolar type II:
 - Secrete surfactant.

Conducting Zone

- All the structures air passes through before reaching the respiratory zone.
- Warms and humidifies inspired air.
- Filters and cleans:
 - Mucus secreted to trap particles in the inspired air.
 - Mucus moved by cilia to be expectorated.

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Tracheobronchial Tree:

a) **Trachea**- conduit for ventilation

- Clearance of tracheal & bronchial secretions
- Begins at the lower border of the cricoid cartilage and extends to the level of the carina
- Average length of 10–13 cm
- The external diameters of the trachea is approximately 2.3 cm coronally and 1.8 cm sagittally in men and 2.0 cm & 1.4 cm, respectively, in women

■ ■ ■

- The trachea bifurcates at the carina into the right and left main stem bronchi
- Dichotomous division, starting with the trachea and ending in alveolar sacs, is estimated to involve 23 divisions.
- An estimated 300 million alveoli provide an enormous membrane (50–100 m²) for gas exchange in the average adult
- Gas exchange can occur only across the flat epithelium, which begins to appear on respiratory bronchioles

Pulmonary Circulation & Lymphatics

- The lungs are supplied by two circulations, pulmonary and bronchial
- The bronchial circulation arises from the heart
- Along their courses, the bronchial vessels anastomose with the pulmonary arterial circulation and continue as far as the alveolar duct.
- The pulmonary circulation normally receives the total output of the right heart via the pulmonary artery, which divides into right and left branches to supply each lung

■

- Deoxygenated blood passes through the pulmonary capillaries, where O₂ is taken up and CO₂ is eliminated
- The oxygenated blood is then returned to the It heart by 4 main pulmonary veins (two from each lung)

Innervation

- The diaphragm is innervated by the phrenic nerves, which arise from the C3–C5 nerve roots.
- U/L phrenic nerve palsy only modestly reduces most indices of pulmonary function (~ 25%).
- B/L phrenic nerve palsies produce more severe impairment
- The vagus nerves provide sensory innervation to the tracheobronchial tree

Thoracic Cavity

- Diaphragm:
 - Sheets of striated muscle divides anterior body cavity into 2 parts.
- Above diaphragm: thoracic cavity:
 - Contains heart, large blood vessels, trachea, esophagus, thymus, and lungs.
- Below diaphragm: abdominopelvic cavity:
 - Contains liver, pancreas, GI tract, spleen, and genitourinary tract.
- Intrapleural space:
 - Space between visceral and parietal pleurae.

Intrapulmonary and Intra-pleural Pressures

- Visceral and parietal pleurae are flush against each other.
 - The intrapleural space contains only a film of fluid secreted by the membranes.
- Lungs normally remain in contact with the chest walls.
- Lungs expand and contract along with the thoracic cavity.
- Intrapulmonary pressure:
 - Intra-alveolar pressure (pressure in the alveoli).
- Intrapleural pressure:
 - Pressure in the intrapleural space.
 - Pressure is negative, due to lack of air in the intrapleural space.

Transpulmonary Pressure

- Pressure difference across the wall of the lung.
- Intrapulmonary pressure – intrapleural pressure.
 - Keeps the lungs against the chest wall.

Intrapulmonary and Intrapleural Pressures (continued)

- During inspiration:
 - Atmospheric pressure is $>$ intrapulmonary pressure (-3 mm Hg).
- During expiration:
 - Intrapulmonary pressure ($+3$ mm Hg) is $>$ atmospheric pressure.

Physical Properties of the Lungs

- Ventilation occurs as a result of pressure differences induced by changes in lung volume.
- Physical properties that affect lung function:
 - Compliance.
 - Elasticity.
 - Surface tension.

Elasticity

- Tendency to return to initial size after distension.
- High content of elastin proteins.
 - Very elastic and resist distension.
 - Recoil ability.
- Elastic tension increases during inspiration and is reduced by recoil during expiration.

Surface Tension

- Force exerted by fluid in alveoli to resist distension.
 - Lungs secrete and absorb fluid, leaving a very thin film of fluid.
 - This film of fluid causes surface tension.
 - Fluid absorption is driven (osmosis) by Na^+ active transport.
 - Fluid secretion is driven by the active transport of Cl^- out of the alveolar epithelial cells.
- H_2O molecules at the surface are attracted to other H_2O molecules by attractive forces.
 - Force is directed inward, raising pressure in alveoli.

Quiet Inspiration

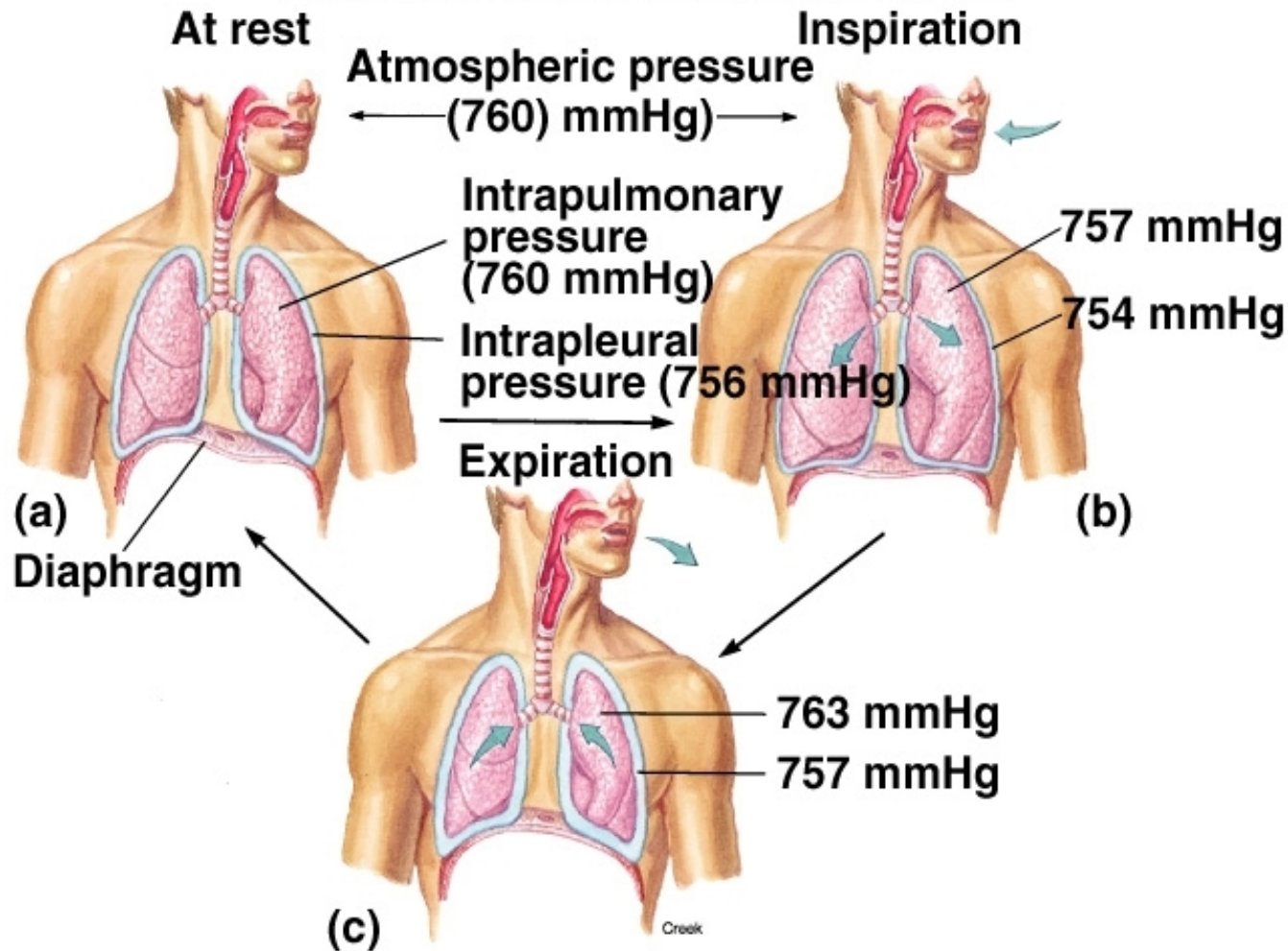
- Active process:
 - Contraction of diaphragm, increases thoracic volume vertically.
- Parasternal and external intercostals contract, raising the ribs; increasing thoracic volume laterally.
- Pressure changes:
 - Alveolar changes from 0 to -3 mm Hg.
 - Intrapleural changes from -4 to -6 mm Hg.
 - Transpulmonary pressure = $+3$ mm Hg.

Expiration

- Quiet expiration is a passive process.
 - After being stretched by contractions of the diaphragm and thoracic muscles; the diaphragm, thoracic muscles, thorax, and lungs recoil.
 - Decrease in lung volume raises the pressure within alveoli above atmosphere, and pushes air out.
- Pressure changes:
 - Intrapulmonary pressure changes from -3 to $+3$ mm Hg.
 - Intrapleural pressure changes from -6 to -3 mm Hg.
 - Transpulmonary pressure = $+6$ mm Hg.

Pulmonary Ventilation

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Lung volumes and capacities

4 lung volumes:

tidal (~500 ml)

inspiratory reserve (~3100 ml)

expiratory reserve (~1200 ml)

residual (~1200 ml)

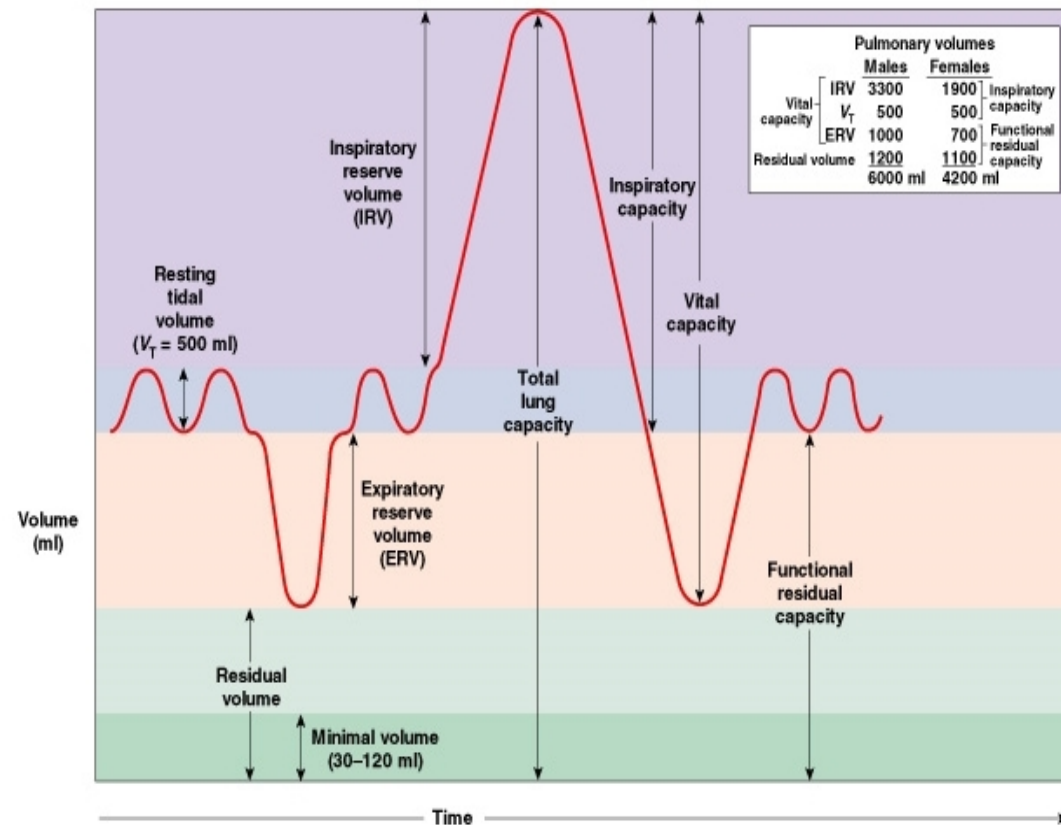
4 lung capacities

inspiratory (~3600 ml)

functional residual (~2400 ml)

vital (~4800 ml)

total lung (~6000 ml)



Terms Used to Describe Lung Volumes and Capacities

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Table 16.3 Terms Used to Describe Lung Volumes and Capacities

Term	Definition
<i>Lung Volumes</i>	The four nonoverlapping components of the total lung capacity
Tidal volume	The volume of gas inspired or expired in an unforced respiratory cycle
Inspiratory reserve volume	The maximum volume of gas that can be inspired during forced breathing in addition to tidal volume
Expiratory reserve volume	The maximum volume of gas that can be expired during forced breathing in addition to tidal volume
Residual volume	The volume of gas remaining in the lungs after a maximum expiration
<i>Lung Capacities</i>	Measurements that are the sum of two or more lung volumes
Total lung capacity	The total amount of gas in the lungs after a maximum inspiration
Vital capacity	The maximum amount of gas that can be expired after a maximum inspiration
Inspiratory capacity	The maximum amount of gas that can be inspired after a normal tidal expiration
Functional residual capacity	The amount of gas remaining in the lungs after a normal tidal expiration

Hemoglobin

- Oxyhemoglobin:
 - Normal heme contains iron in the reduced form (Fe^{2+}).
 - Fe^{2+} shares electrons and bonds with oxygen.
- Deoxyhemoglobin:
 - When oxyhemoglobin dissociates to release oxygen, the heme iron is still in the reduced form.
 - Hemoglobin does not lose an electron when it combines with O_2 .



Digestive System

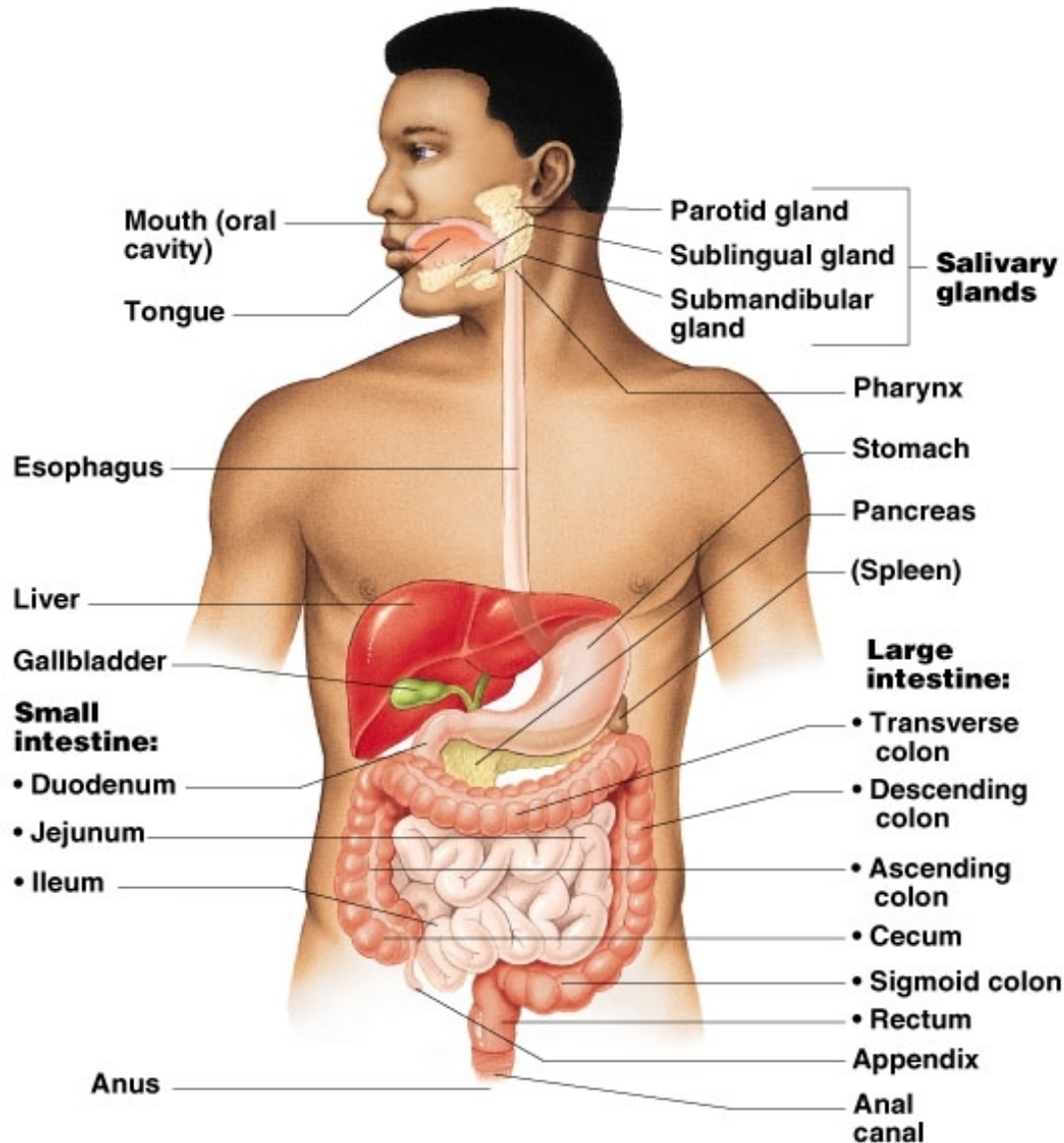
The Digestive System and Body Metabolism

- Digestion
 - Breakdown of ingested food
- Absorption
 - Passage of nutrients into the blood
- Metabolism
 - Production of cellular energy (ATP)

Organs of the Digestive System

- Two main groups
 - Alimentary canal – continuous coiled hollow tube
 - Accessory digestive organs

Organs of the Digestive System

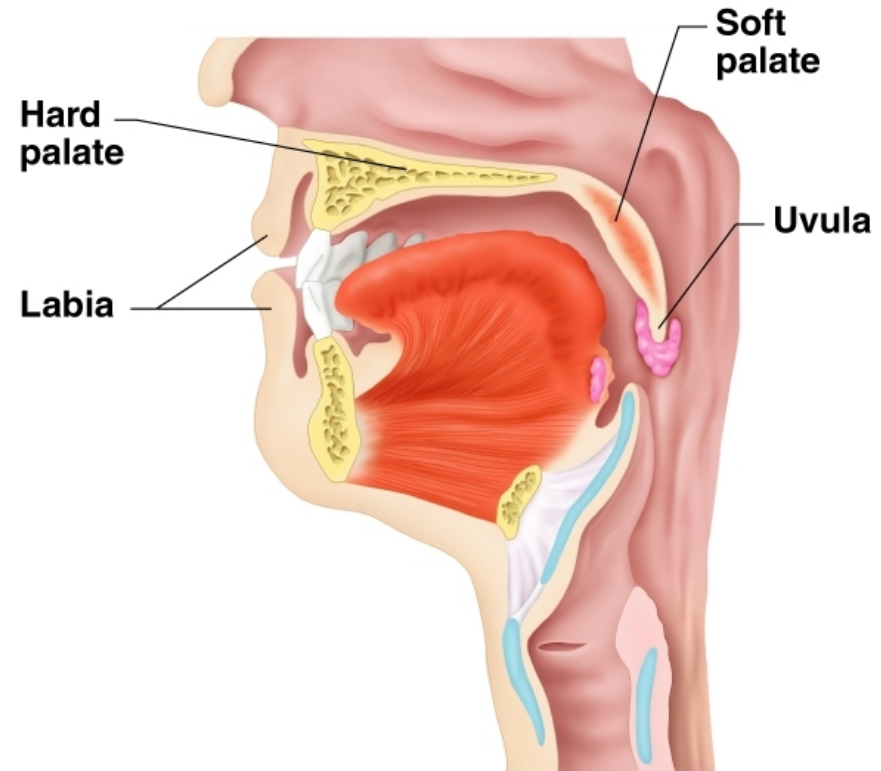


Organs of the Alimentary Canal

- Mouth
- Pharynx
- Esophagus
- Stomach
- Small intestine
- Large intestine
- Anus

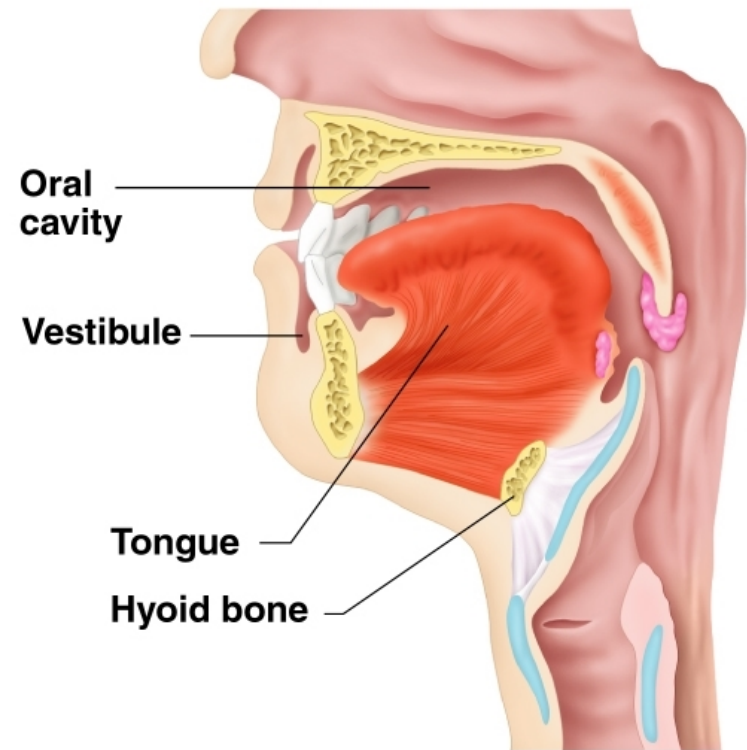
Mouth (Oral Cavity) Anatomy

- Lips (labia) – protect the anterior opening
- Cheeks – form the lateral walls
- Hard palate – forms the anterior roof
- Soft palate – forms the posterior roof
- Uvula – fleshy projection of the soft palate



Mouth (Oral Cavity) Anatomy

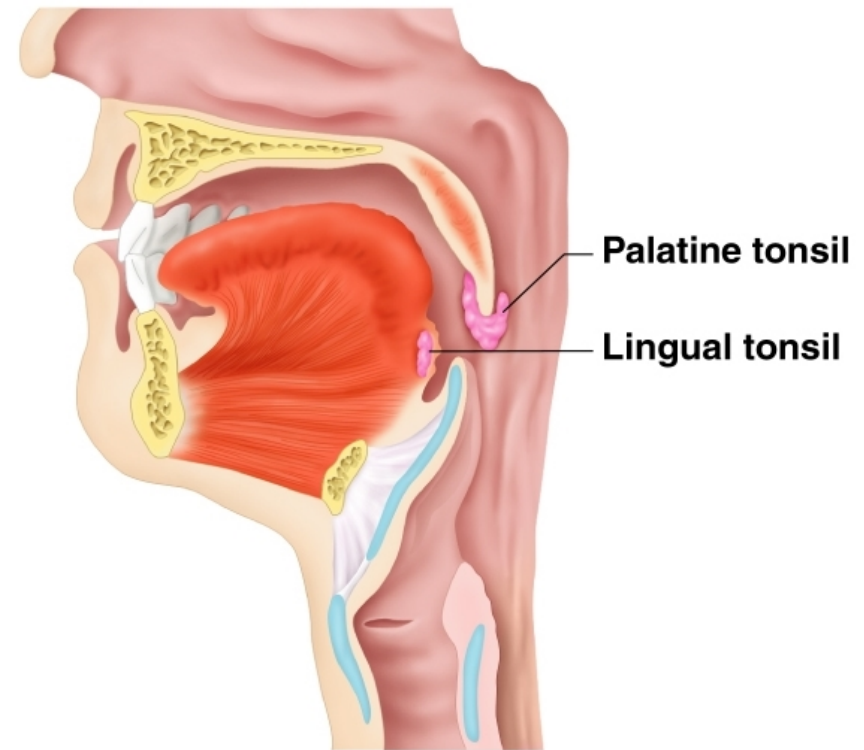
- Vestibule – space between lips externally and teeth and gums internally
- Oral cavity – area contained by the teeth
- Tongue – attached at hyoid and styloid processes of the skull, and by the lingual frenulum



(a)

Mouth (Oral Cavity) Anatomy

- Tonsils
 - Palatine tonsils
 - Lingual tonsil



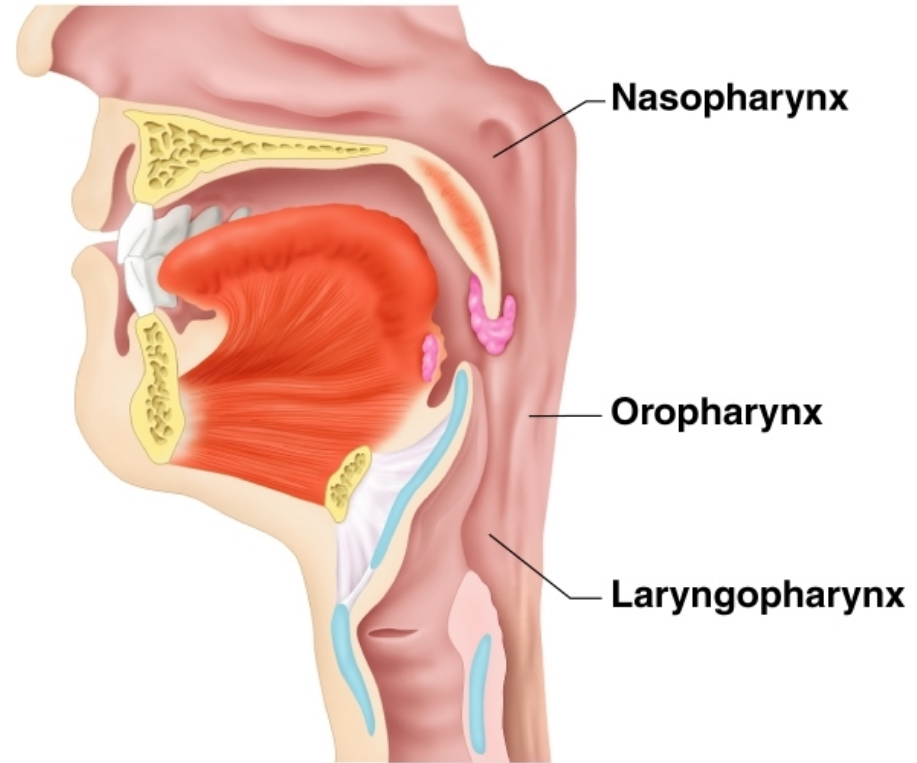
(a)

Processes of the Mouth

- Mastication (chewing) of food
- Mixing masticated food with saliva
- Initiation of swallowing by the tongue
- Allowing for the sense of taste

Pharynx Anatomy

- Nasopharynx – not part of the digestive system
- Oropharynx – posterior to oral cavity
- Laryngopharynx – below the oropharynx and connected to the esophagus



(a)

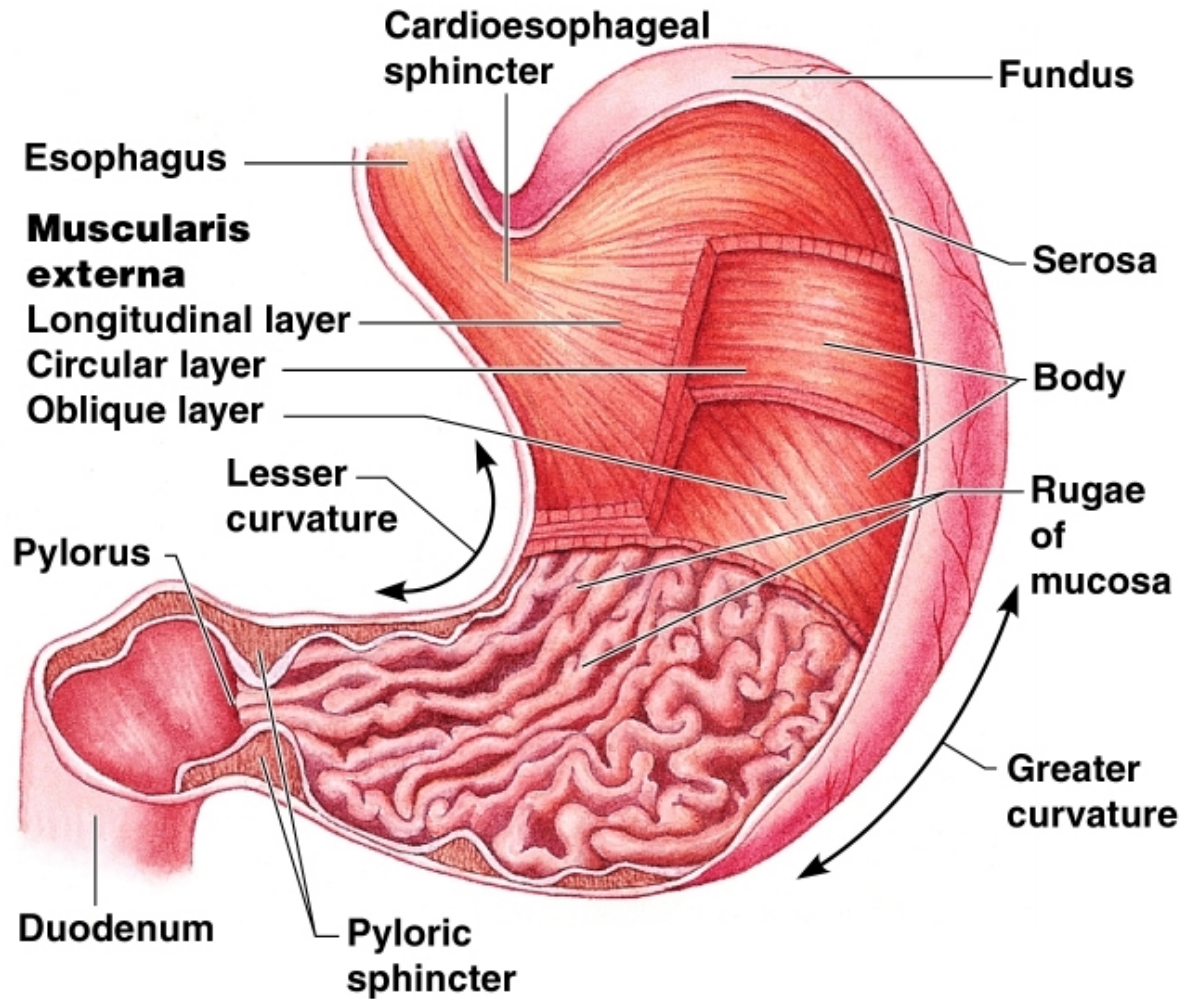
Pharynx Function

- Serves as a passageway for air and food
- Food is propelled to the esophagus by two muscle layers
 - Longitudinal inner layer
 - Circular outer layer
- Food movement is by alternating contractions of the muscle layers (peristalsis)

Esophagus

- Runs from pharynx to stomach through the diaphragm
- Conducts food by peristalsis (slow rhythmic squeezing)
- Passageway for food only (respiratory system branches off after the pharynx)

Stomach Anatomy



(a)

Figure 14.4a

Stomach Functions

- Acts as a storage tank for food
- Site of food breakdown
- Chemical breakdown of protein begins
- Delivers chyme (processed food) to the small intestine

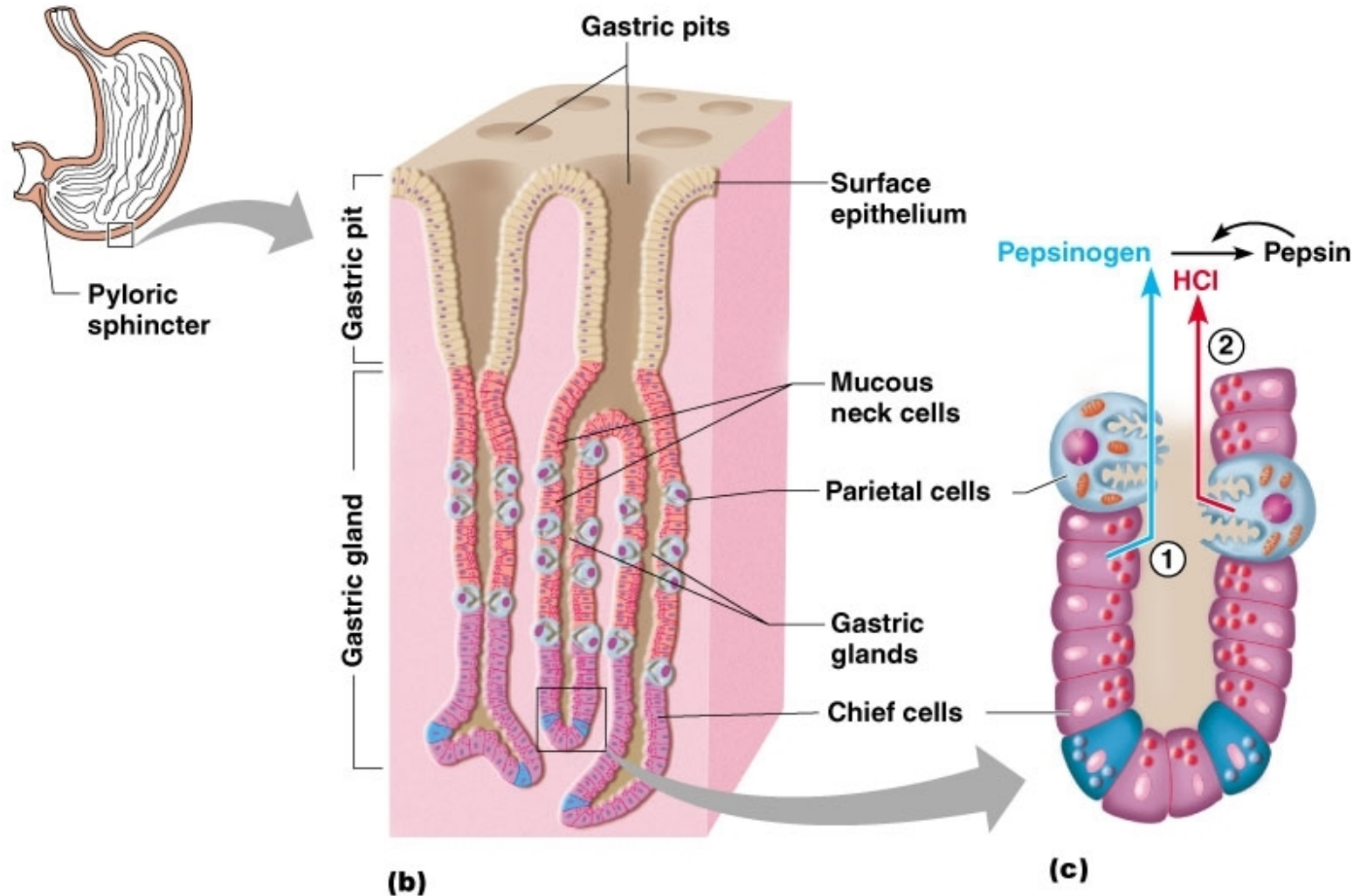
Specialized Mucosa of the Stomach

- Simple columnar epithelium
 - Mucous neck cells – produce a sticky alkaline mucus
 - Gastric glands – secrete gastric juice
 - Chief cells – produce protein-digesting enzymes (pepsinogens)
 - Parietal cells – produce hydrochloric acid
 - Endocrine cells – produce gastrin

Structure of the Stomach Mucosa

- Gastric pits formed by folded mucosa
- Glands and specialized cells are in the gastric gland region

Structure of the Stomach Mucosa



Small Intestine

- The body's major digestive organ
- Site of nutrient absorption into the blood
- Muscular tube extending from the pyloric sphincter to the ileocecal valve
- Suspended from the posterior abdominal wall by the mesentery

Subdivisions of the Small Intestine

- Duodenum
 - Attached to the stomach
 - Curves around the head of the pancreas
- Jejunum
 - Attaches anteriorly to the duodenum
- Ileum
 - Extends from jejunum to large intestine

Chemical Digestion in the Small Intestine

- Source of enzymes that are mixed with chyme
 - Intestinal cells
 - Pancreas
- Bile enters from the gall bladder

Structures Involved in Absorption of Nutrients

- Absorptive cells
- Blood capillaries
- Lacteals (specialized lymphatic capillaries)

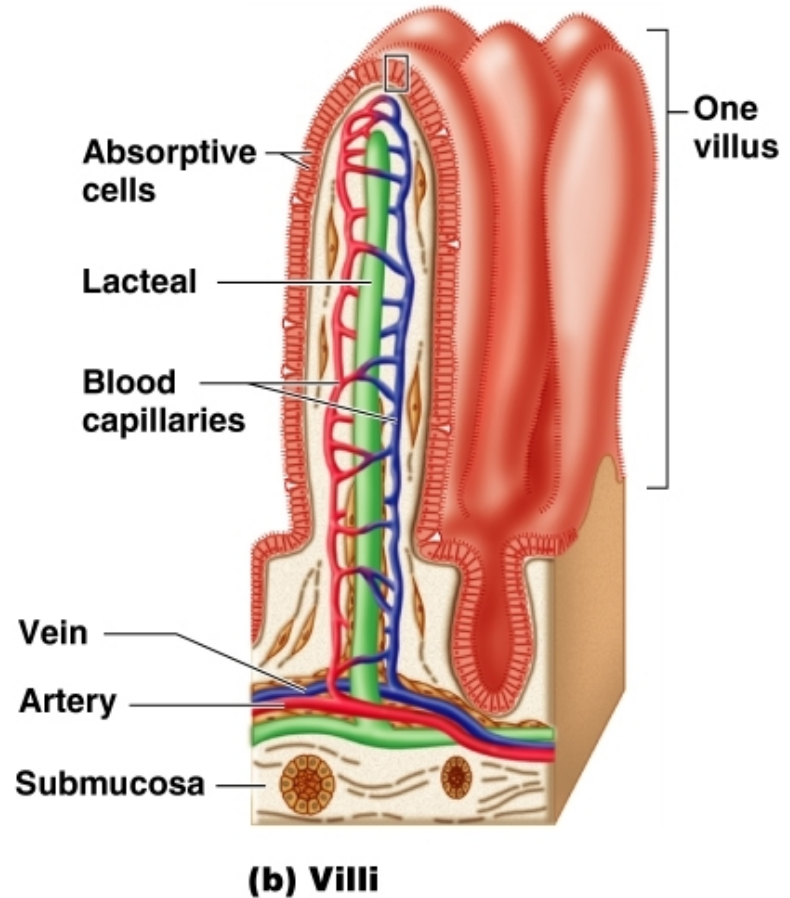


Figure 14.7b

Folds of the Small Intestine

- Called circular folds or plicae circulares
- Deep folds of the mucosa and submucosa
- Do not disappear when filled with food
- The submucosa has Peyer's patches (collections of lymphatic tissue)

Digestion in the Small Intestine

- Enzymes from the brush border
 - Break double sugars into simple sugars
 - Complete some protein digestion
- Pancreatic enzymes play the major digestive function
 - Help complete digestion of starch (pancreatic amylase)
 - Carry out about half of all protein digestion (trypsin, etc.)

Digestion in the Small Intestine

- Pancreatic enzymes play the major digestive function (continued)
 - Responsible for fat digestion (lipase)
 - Digest nucleic acids (nucleases)
 - Alkaline content neutralizes acidic chyme

Absorption in the Small Intestine

- Water is absorbed along the length of the small intestine
- End products of digestion
 - Most substances are absorbed by active transport through cell membranes
 - Lipids are absorbed by diffusion
- Substances are transported to the liver by the hepatic portal vein or lymph

Propulsion in the Small Intestine

- Peristalsis is the major means of moving food
- Segmental movements
 - Mix chyme with digestive juices
 - Aid in propelling food

Large Intestine

- Larger in diameter, but shorter than the small intestine
- Frames the internal abdomen

Large Intestine

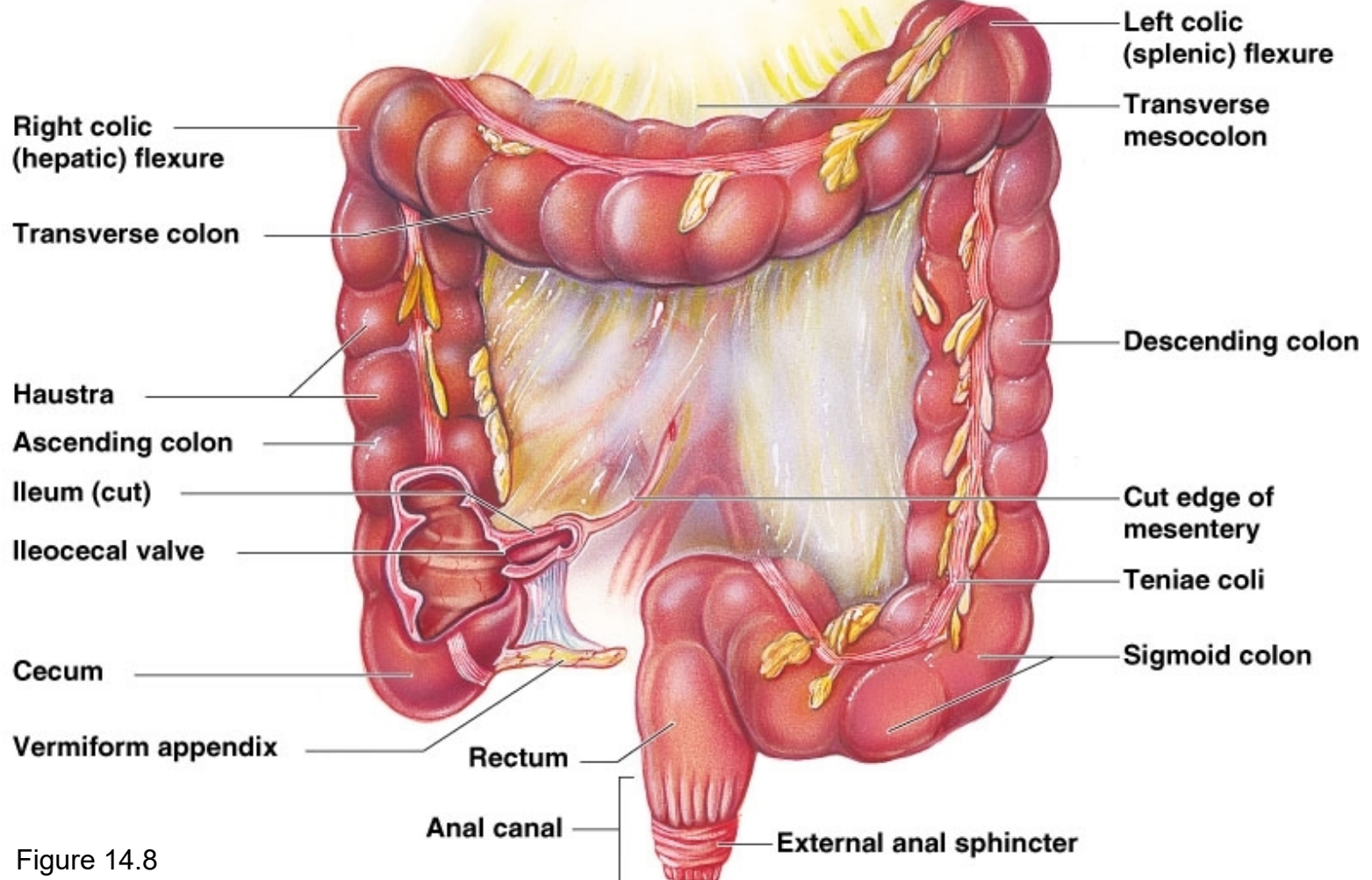


Figure 14.8

Functions of the Large Intestine

- Absorption of water
- Eliminates indigestible food from the body as feces
- Does not participate in digestion of food
- Goblet cells produce mucus to act as a lubricant

Structures of the Large Intestine

- Cecum – saclike first part of the large intestine
- Appendix
 - Accumulation of lymphatic tissue that sometimes becomes inflamed (appendicitis)
 - Hangs from the cecum

Structures of the Large Intestine

- Colon
 - Ascending
 - Transverse
 - Descending
 - S-shaped sigmoidal
- Rectum
- Anus – external body opening

Structures of the Large Intestine

- Colon
 - Ascending
 - Transverse
 - Descending
 - S-shaped sigmoidal
- Rectum
- Anus – external body opening

Food Breakdown and Absorption in the Large Intestine

- No digestive enzymes are produced
- Resident bacteria digest remaining nutrients
 - Produce some vitamin K and B
 - Release gases
- Water and vitamins K and B are absorbed
- Remaining materials are eliminated via feces

Propulsion in the Large Intestine

- Sluggish peristalsis
- Mass movements
 - Slow, powerful movements
 - Occur three to four times per day
- Presence of feces in the rectum causes a defecation reflex
 - Internal anal sphincter is relaxed
 - Defecation occurs with relaxation of the voluntary (external) anal sphincter

Accessory Digestive Organs

- Salivary glands
- Teeth
- Pancreas
- Liver
- Gall bladder

Salivary Glands

- Saliva-producing glands
 - Parotid glands – located anterior to ears
 - Submandibular glands
 - Sublingual glands

Saliva

- Mixture of mucus and serous fluids
 - Helps to form a food bolus
- Contains salivary amylase to begin starch digestion
- Dissolves chemicals so they can be tasted

Teeth

- The role is to masticate (chew) food
- Humans have two sets of teeth
 - Deciduous (baby or milk) teeth
 - 20 teeth are fully formed by age two

Pancreas

- Produces a wide spectrum of digestive enzymes that break down all categories of food
- Enzymes are secreted into the duodenum
- Alkaline fluid introduced with enzymes neutralizes acidic chyme
- Endocrine products of pancreas
 - Insulin
 - Glucagons

Bile

- Produced by cells in the liver
- Composition
 - Bile salts
 - Bile pigment (mostly bilirubin from the breakdown of hemoglobin)
 - Cholesterol
 - Phospholipids
 - Electrolytes

Role of the Liver in Metabolism

- Several roles in digestion
- Detoxifies drugs and alcohol
- Degrades hormones
- Produce cholesterol, blood proteins (albumin and clotting proteins)
- Plays a central role in metabolism

Gall Bladder

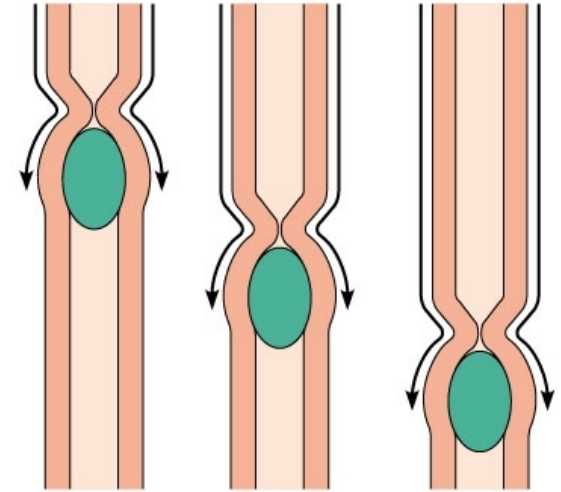
- Sac found in hollow fossa of liver
- Stores bile from the liver by way of the cystic duct
- Bile is introduced into the duodenum in the presence of fatty food
- Gallstones can cause blockages

Processes of the Digestive System

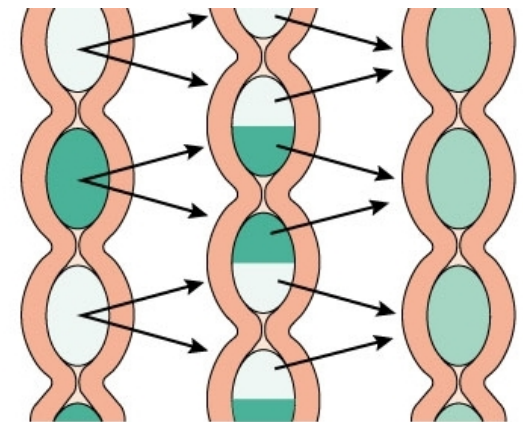
- Ingestion – getting food into the mouth
- Propulsion – moving foods from one region of the digestive system to another

Processes of the Digestive System

- Peristalsis – alternating waves of contraction
- Segmentation – moving materials back and forth to aid in mixing



(a)



(b)

Processes of the Digestive System

- Mechanical digestion
 - Mixing of food in the mouth by the tongue
 - Churning of food in the stomach
 - Segmentation in the small intestine

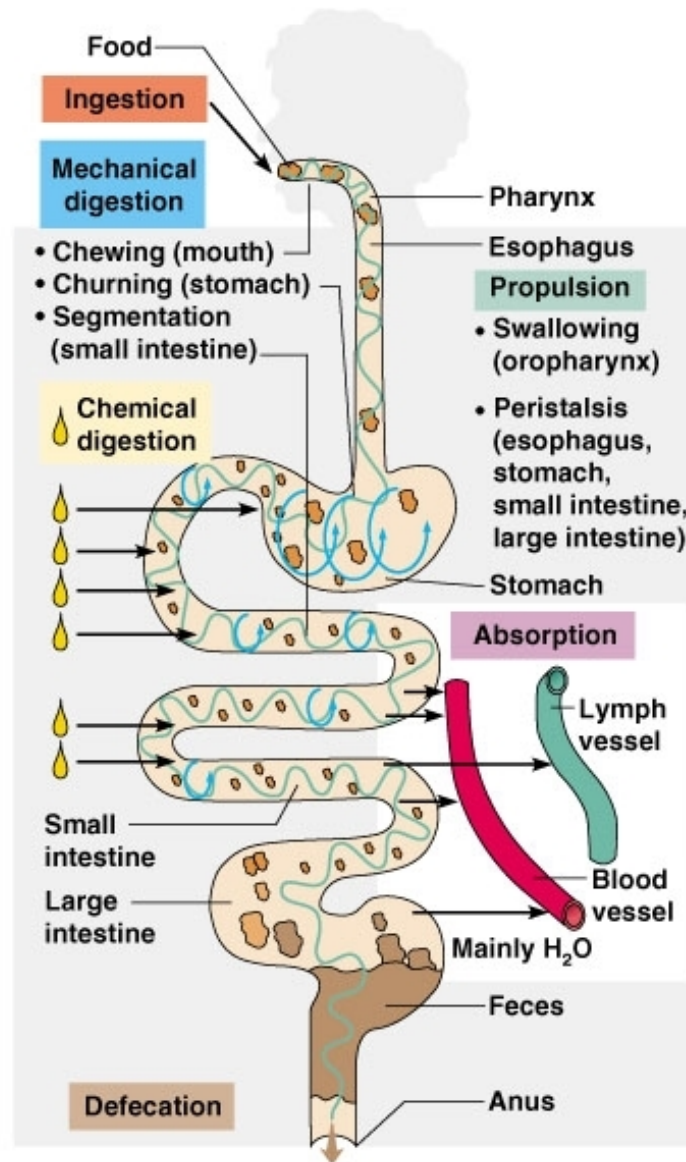
Processes of the Digestive System

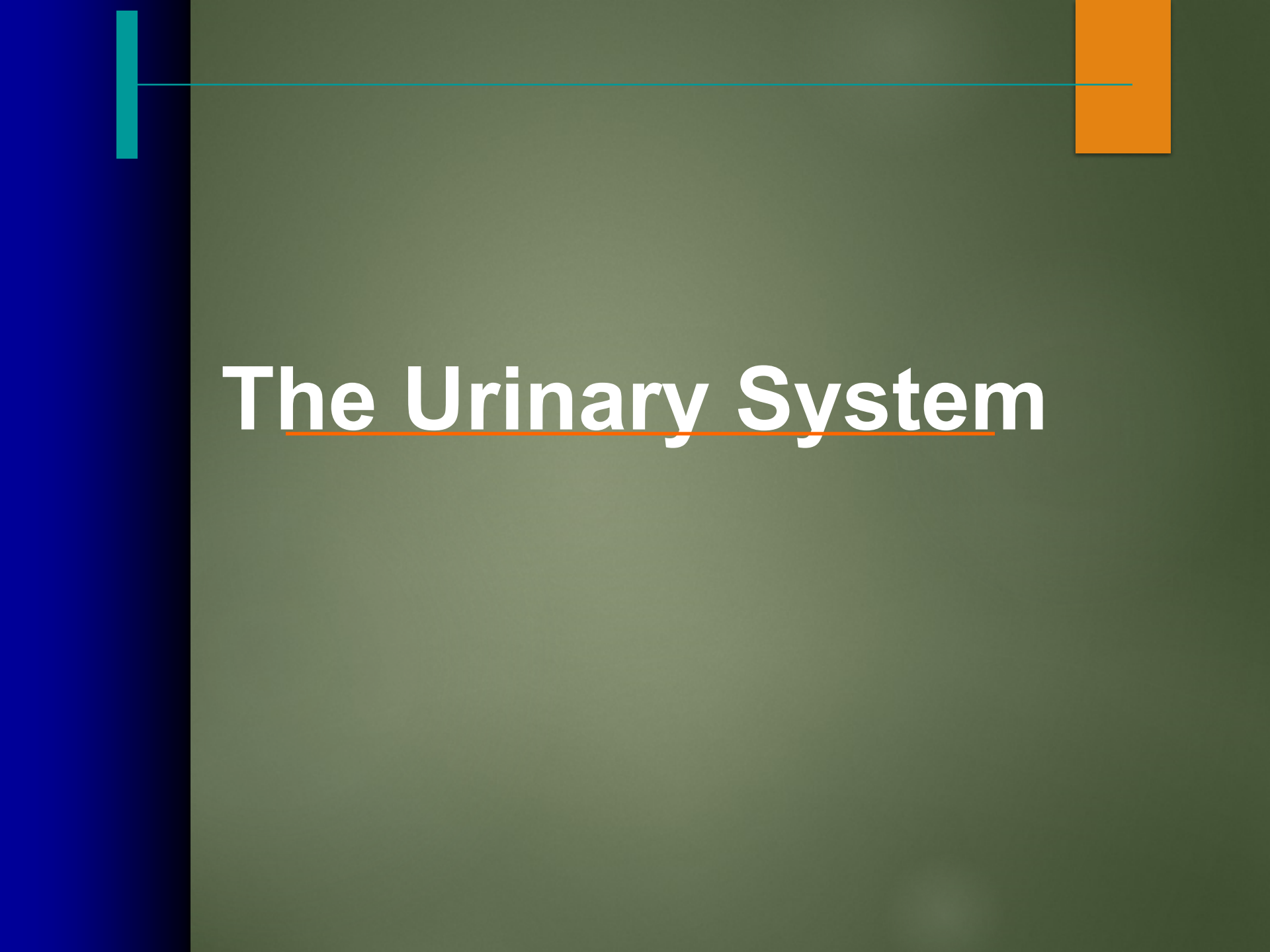
- Chemical Digestion
 - Enzymes break down food molecules into their building blocks
 - Each major food group uses different enzymes
 - Carbohydrates are broken to simple sugars
 - Proteins are broken to amino acids
 - Fats are broken to fatty acids and alcohols

Processes of the Digestive System

- Absorption
 - End products of digestion are absorbed in the blood or lymph
 - Food must enter mucosal cells and then into blood or lymph capillaries
- Defecation
 - Elimination of indigestible substances as feces

Processes of the Digestive System





The Urinary System

Functions of the Urinary System

- ▼ Excrete wastes & foreign substances such as:
 - ▼ Nitrogenous wastes
 - ▼ Toxins
 - ▼ Drugs
- ▼ Maintain blood osmolarity
 - ▼ Maintain water balance
 - ▼ Regulates blood-ion composition
 - ▼ Regulate blood glucose

Functions of the Urinary System

- ▼ Regulate blood volume (affects blood pressure)
 - ▼ Conserve/eliminate water
- ▼ Regulate blood pressure
 - ▼ Secretes & stores enzyme RENIN
 - ▼ which promotes the production of the protein angiotensin
 - ▼ Which increases blood pressure

Functions of the Urinary System

- ▼ Regulates blood pH by maintaining proper balance between acids & bases –
 - ▼ Excretes excess H^+ ions
 - ▼ Conserves bicarbonate ions (HCO_3^-)

Functions of the Urinary System

- ▼ Produces hormone ERYTHROPOIETIN
 - ▼ Stimulates erythrocyte production in bone marrow
- ▼ Activation of vitamin D
 - ▼ Allows calcium uptake from stomach to blood

Organs of the Urinary system

▼ Sketch & label:

- ▼ Kidneys
- ▼ Ureters
- ▼ Urinary bladder
- ▼ Urethra

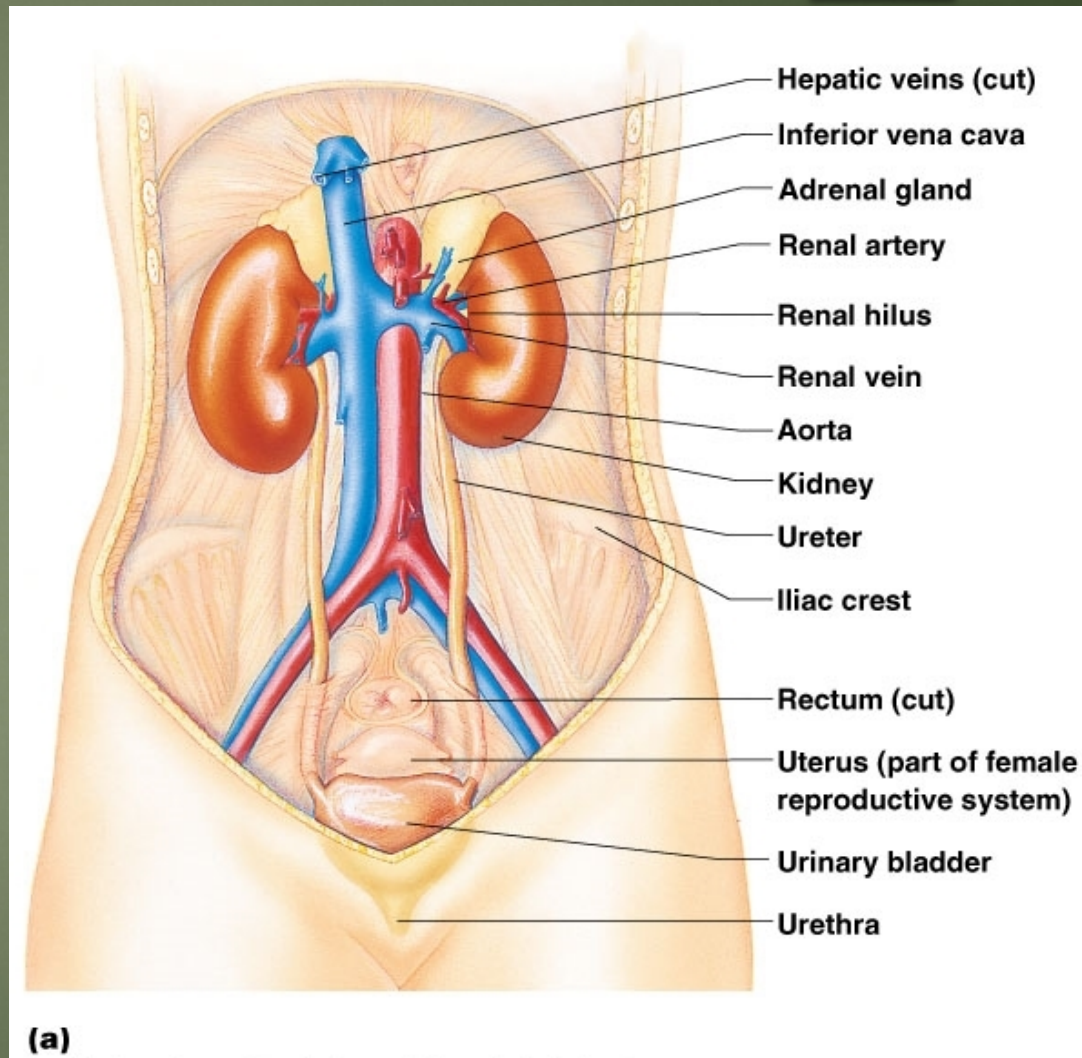
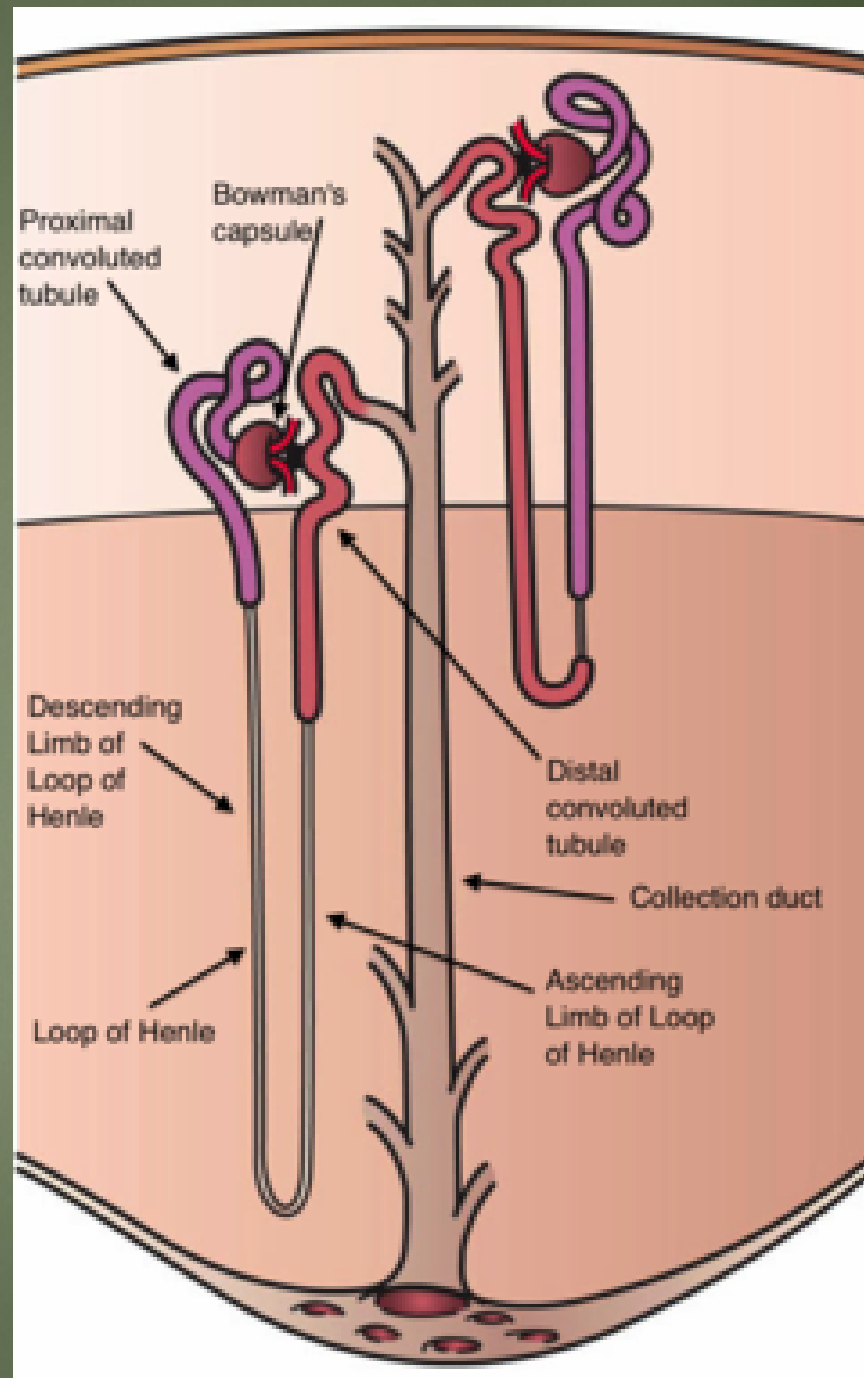


Figure 15.1a

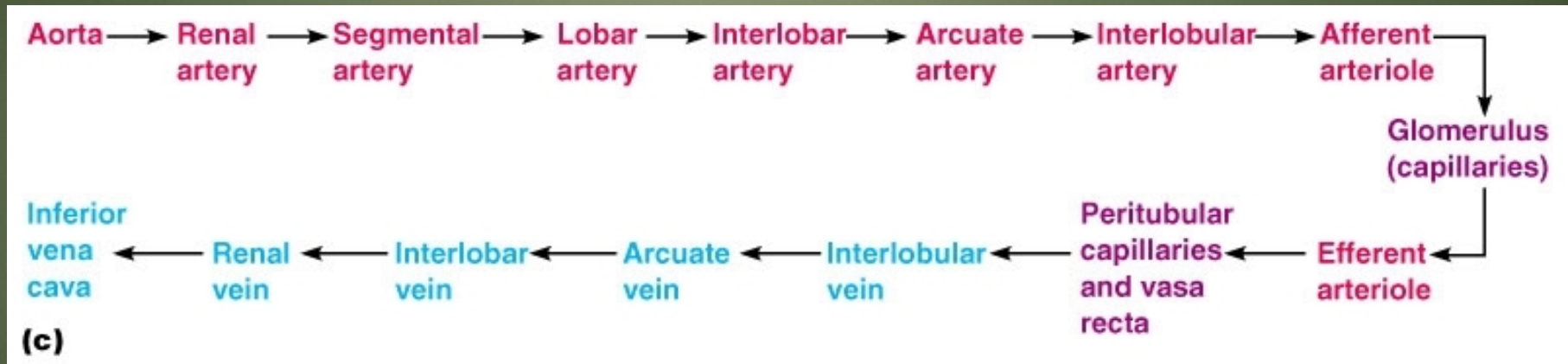
Nephron

- ▶ Nephrons – functional unit of kidney
 - ▶ Filter blood plasma
 - ▶ Form urine



Blood Flow in the Kidneys

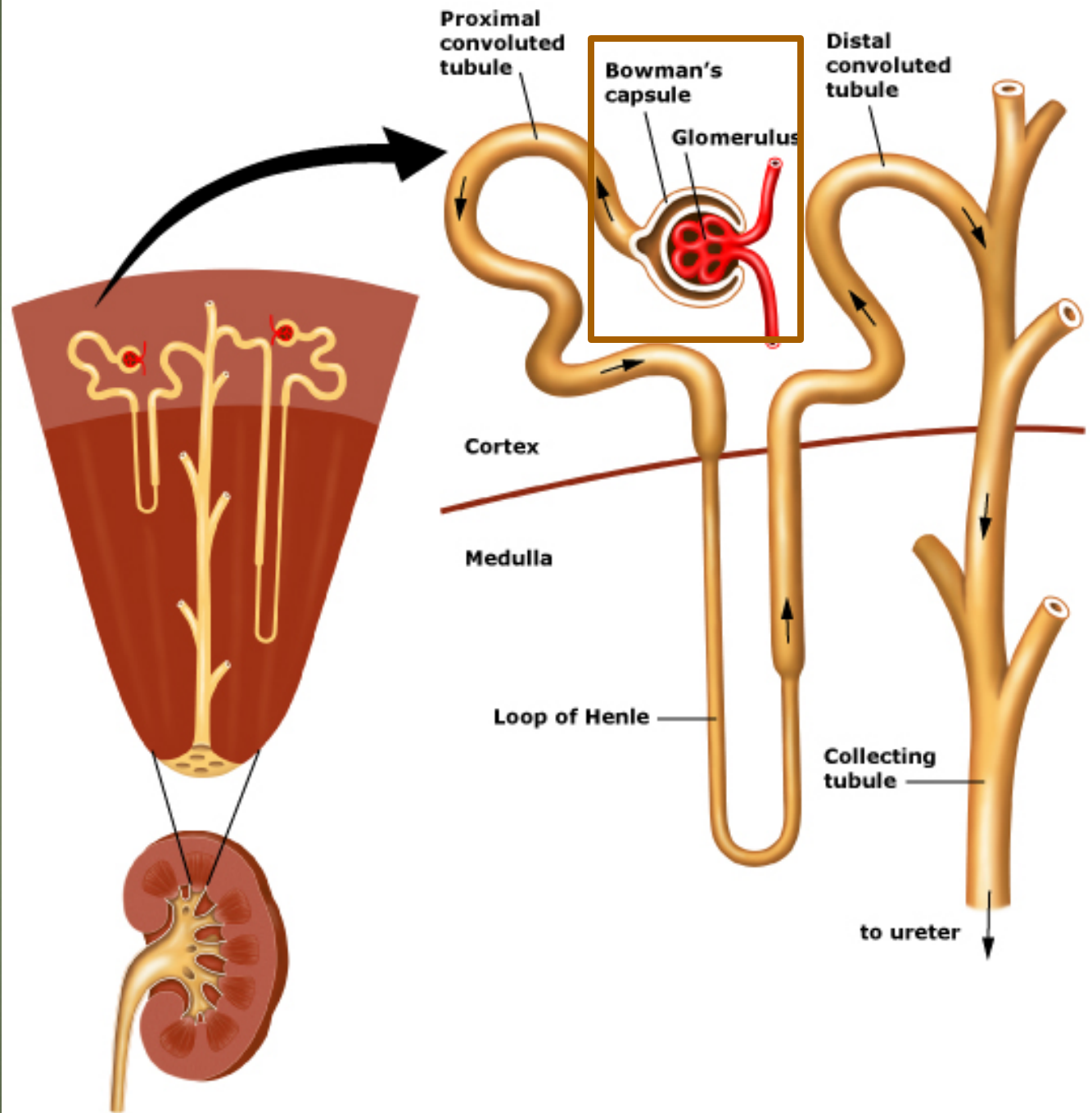
KNOW THE ORDER!!!



Nephrons

- ▼ Definition: The structural and functional units of the kidneys
- ▼ Responsible for formation of urine
- ▼ Main structures of the nephrons
 - ▼ Renal corpuscle
 - ▼ Renal tubule

Renal corpuscle is boxed.
The other structures are known as the renal tubule

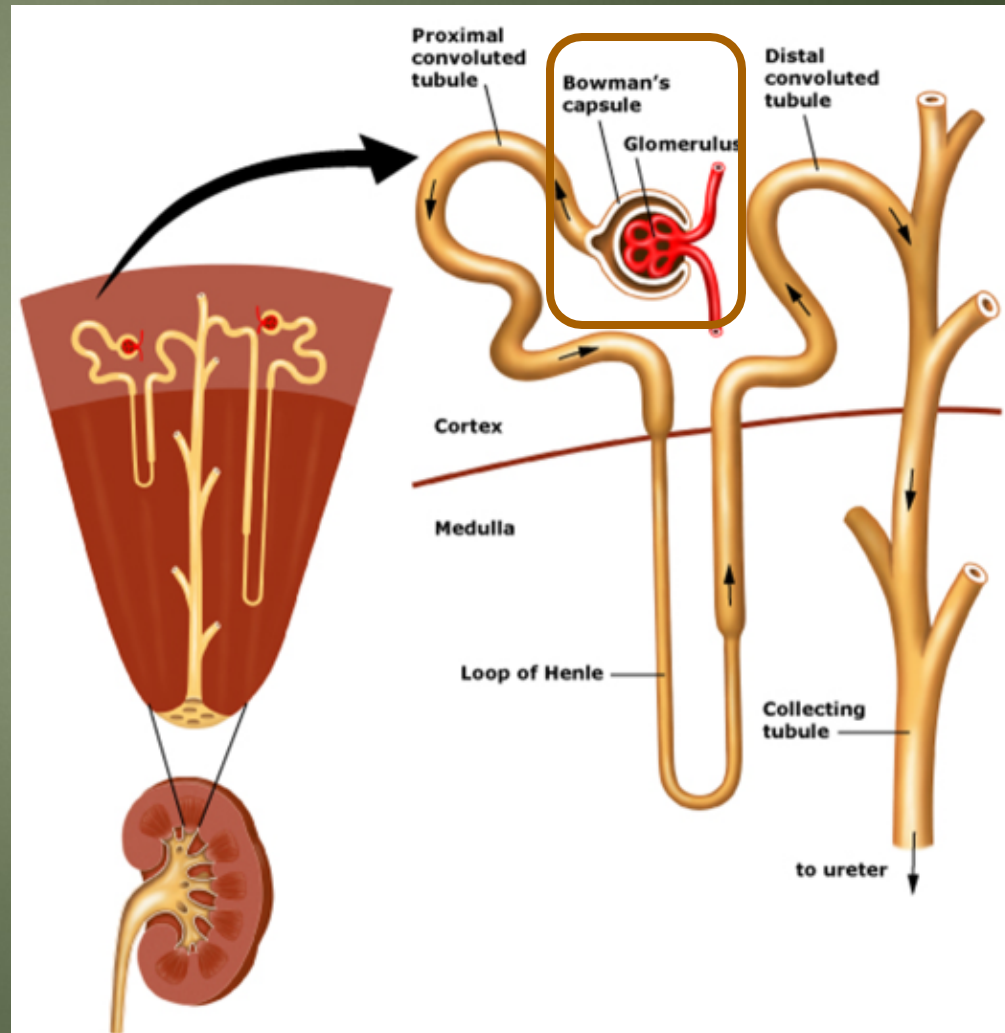


Renal corpuscle

- ▼ Functions to:
 - ▼ Filter the blood plasma
- ▼ Glomerular filtrate
 - ▼ the fluid that passes from the blood through the capillary walls of the glomeruli into Bowman's capsule.
 - ▼ It is similar to plasma but without proteins;
 - ▼ urine is formed from it.

Renal corpuscle

- Structure: consists of two parts:
 - Glomerulus
 - Bowman's capsule



Glomerulus

- ▶ Network of fenestrated capillaries
- ▶ Attached to arterioles on both sides (function to maintain BP homeostasis in capillaries)
 - ▶ Large afferent arteriole – incoming blood
 - ▶ Narrow efferent arteriole – outgoing blood

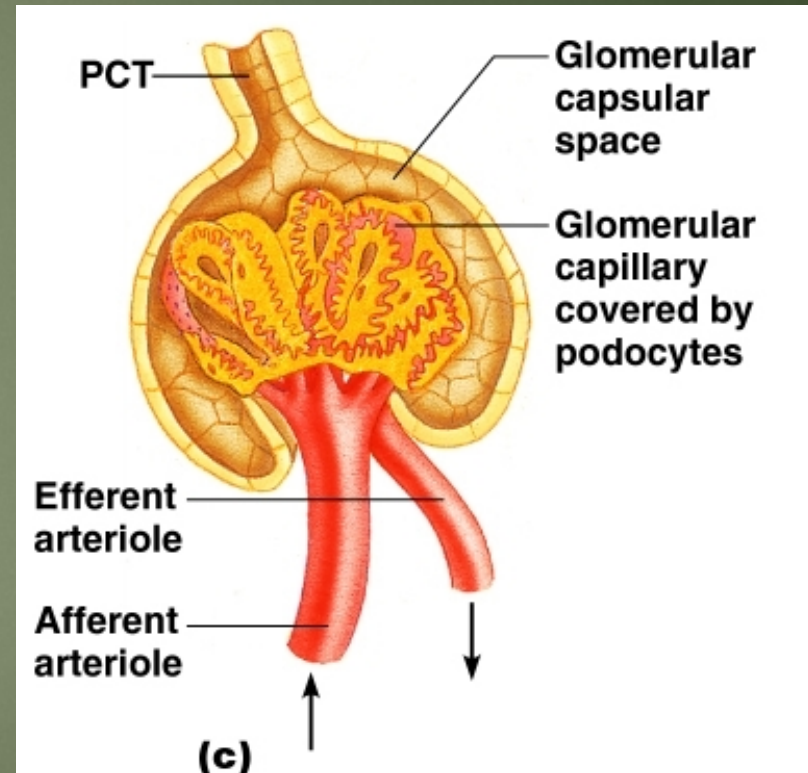


Figure 15.3c

2. Proximal convoluted tubule:

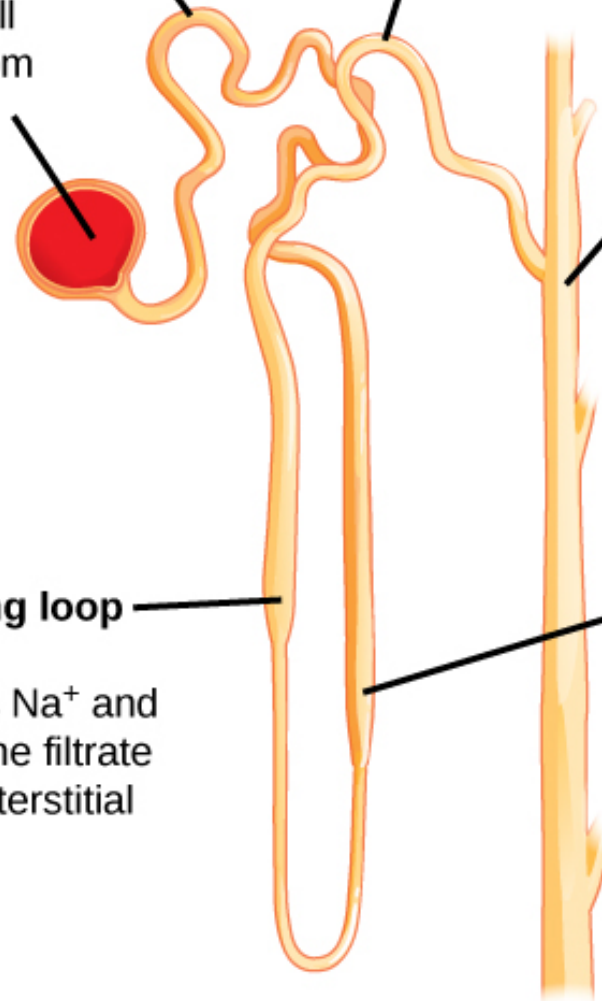
reabsorbs ions, water, and nutrients; removes toxins and adjusts filtrate pH

5. Distal tubule:

selectively secretes and absorbs different ions to maintain blood pH and electrolyte balance

1. Glomerulus:

filters small solutes from the blood



6. Collecting duct:

reabsorbs solutes and water from the filtrate

4. Ascending loop of Henle:

reabsorbs Na^+ and Cl^- from the filtrate into the interstitial fluid

3. Descending loop of Henle:

aquaporins allow water to pass from the filtrate into the interstitial fluid

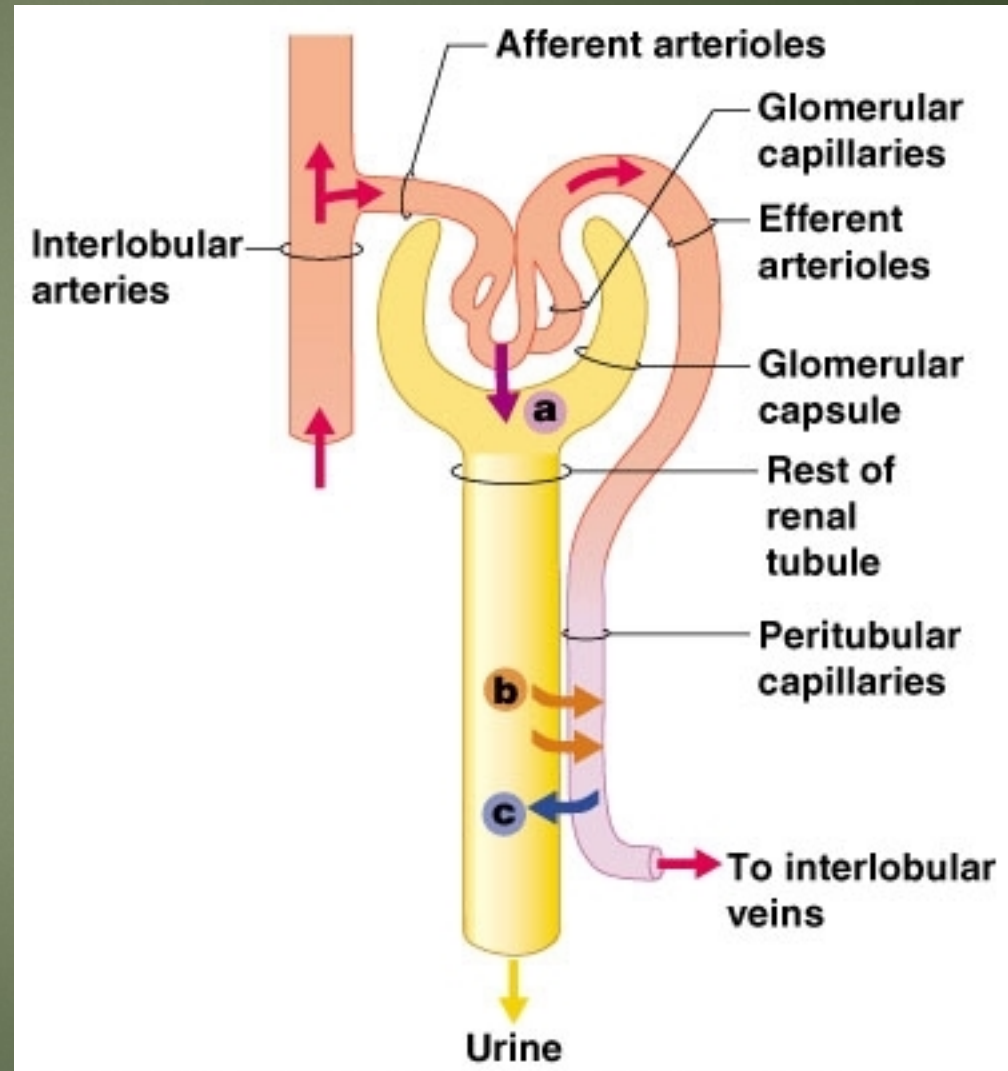
Fluid Flow

- **Flow of glomerular filtrate: from filtrate to exit from the body**
 - glomerular capsule → PCT → nephron loop →
 - DCT → collecting duct → papillary duct →
 - minor calyx → major calyx → renal pelvis →
 - ureter → urinary bladder → urethra

Renal Physiology

Urine Formation Processes

- ▼ Glomerular Filtration
 - ▼ plus
- ▼ Tubular Reabsorption
 - ▼ minus
- ▼ Tubular Secretion
 - ▼ Equals
- ▼ Urinary excretion



Definitions to KNOW

- ▼ Glomerular filtration
 - ▼ Movement of water, small solutes & ions across a filtration membrane - out of glomerular capillary blood plasma into capsular space.
 - ▼ Dependent on net filtration pressure (NFP)
 - ▼ Balance between opposing hydrostatic pressure and opposing colloidal pressures

Filtration Facts

- ▼ Nonselective passive process
 - ▼ Blood cells cannot pass out to the capillaries
 - ▼ Plasma proteins are not filtered out of blood due to size & charge

Definitions to KNOW



- ▼ (Glomerular) **Capillary hydrostatic pressure - HP_{gc}**
 - ▼ Pressure exerted by the blood in the glomerular capillaries – acts to push water & solutes out of the blood into the capsular space
- ▼ (Capsular) **Interstitial hydrostatic pressure - HP_{cs}**
 - ▼ Pressure exerted by the filtrate in the capsular space – acts to push water & solutes out of the capsule into the blood
 - ▼ Always lower than HP_{gc} since filtrate is exiting through PCT

Definitions to KNOW

- ▼ (Glomerular) **Capillary colloidal osmotic pressure - OP_{gc}**
 - ▼ Pressure exerted by the suspended proteins in the blood – acts to pull water back into blood
- ▼ (Capsular) **Interstitial colloidal osmotic pressure - OP_{cs}**
 - ▼ Pressure exerted by the suspended proteins in the capsule – acts to pull water back into the capsule
 - ▼ Not a factor in a HEALTHY kidney

Factors affecting glomerular filtration rate (GFR)

- ▼ Autoregulation
 - ▼ Local blood flow regulation through vasoconstriction/vasodilation of arterioles by the juxtaglomerular cells
- ▼ Hormonal regulation – discussed later
 - ▼ RAAS
 - ▼ ANP and BNP
- ▼ Autonomic regulation
 - ▼ Direct effect by SNS – arteriole vasoconstriction – decreased GFR

Definitions to KNOW

- ▼ **Tubular reabsorption (into blood)**
 - ▼ Absorption of substances from tubules back into blood
 - ▼ Most occurs in the PCT.
 - ▼ These are the good substances that we do NOT want excreted.
 - ▼ Amount of water excreted depends on hydration level of body.

Definitions to KNOW

▼ **Tubular secretion (out of blood)**

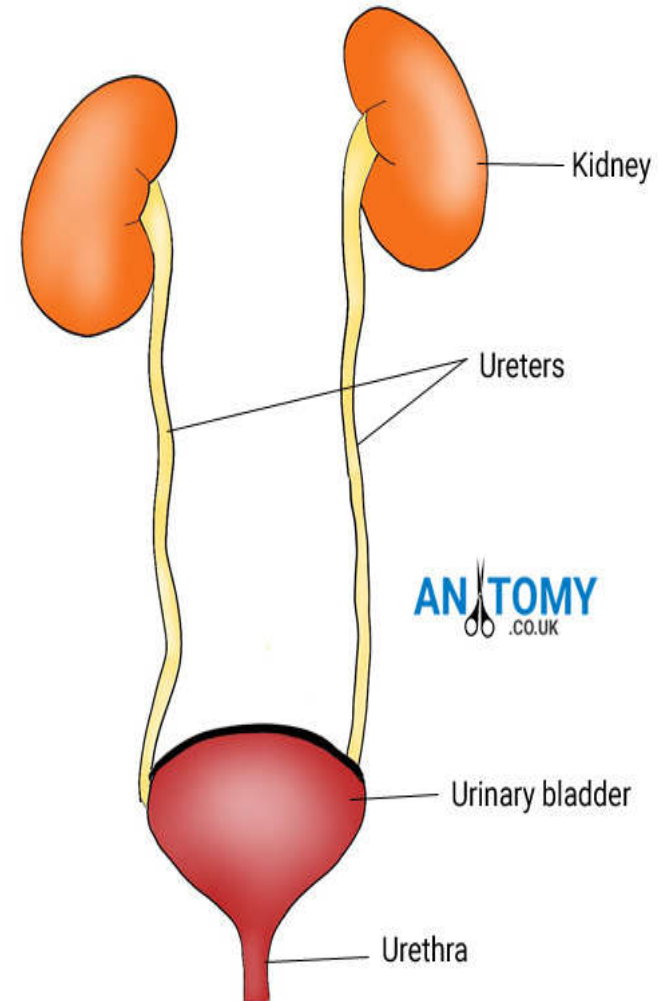
- ▼ Movement of substances from blood into tubules for excretion
- ▼ Substances still in excess in body such as, hydrogen and potassium ion, & creatine
- ▼ Materials left in renal tubule move toward ureter



Urine Transportation, Storage, & Elimination

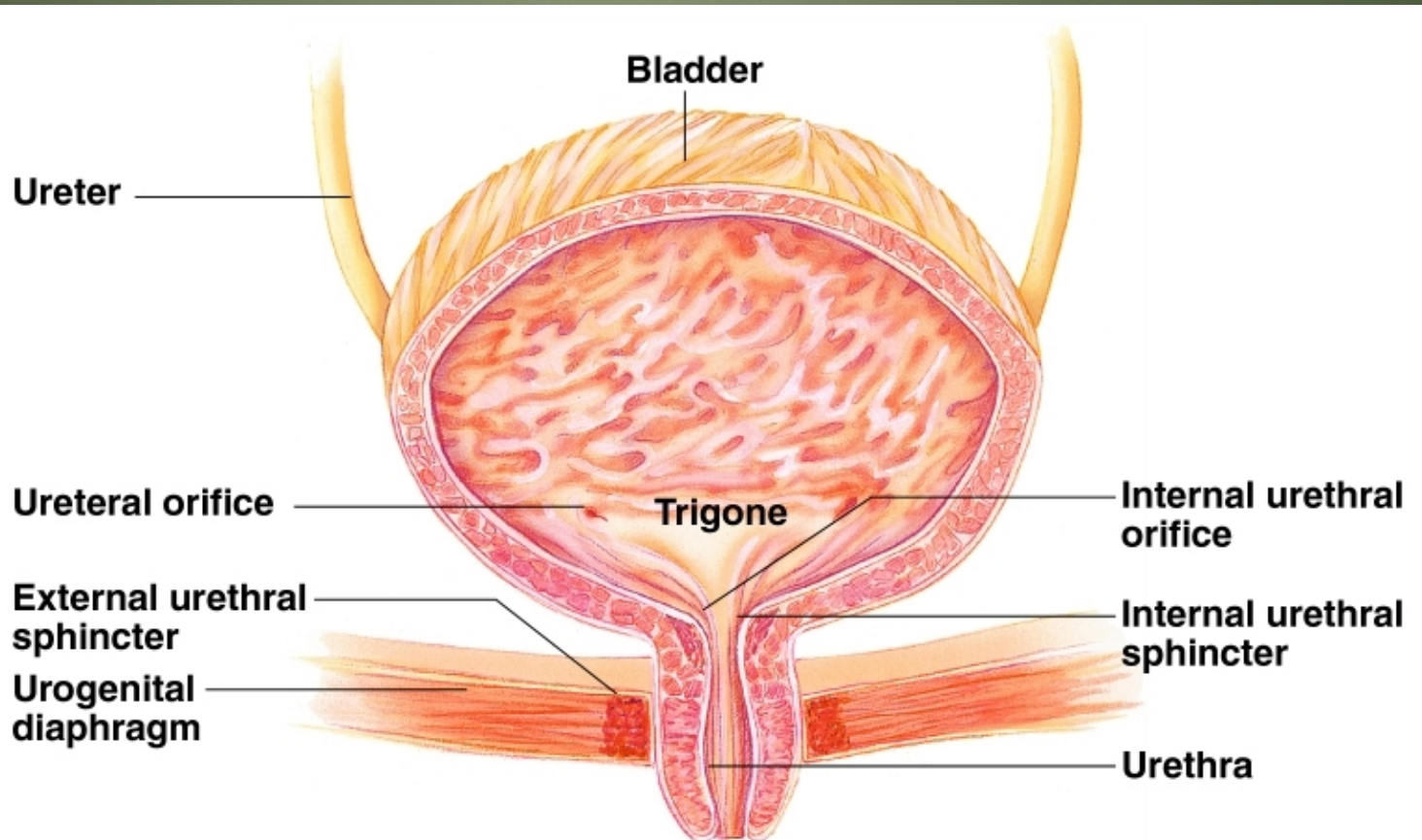
Ureters

- ▶ Slender tubes that transport urine from the kidney to the urinary bladder
- ▶ Peristalsis aids gravity in urine transport



Urinary Bladder

- ▼ Hollow, muscular sac
- ▼ Temporarily stores urine



Urethra

- ▼ Thin-walled tube that carries urine from the urinary bladder to the outside of the body by peristalsis
- ▼ Release of urine is controlled by two sphincters
 - ▼ Internal urethral sphincter (involuntary)
 - ▼ External urethral sphincter (voluntary)

Micturition (Voiding)

- ▼ The act of urination
- ▼ Both sphincter muscles must open to allow voiding
 - ▼ Volume exceeds 200-400 ml - The internal urethral sphincter is relaxed as stretch receptors send signal to SC (spinal reflex)
 - ▼ The external urethral sphincter (composed of skeletal muscle) must be voluntarily relaxed

Age Related Changes

- ▼ Degenerative changes in kidneys
- ▼ Kidney atrophy due to loss of glomeruli due to fewer functional nephrons
- ▼ Renal tubules thicken
- ▼ Loss of elasticity of ureters, urinary bladder, and urethra

Effects on Urinary System

- ▼ GFR declines
- ▼ Reduced reabsorption of sodium, glucose, and other molecules
- ▼ Slowed medication clearance
- ▼ Incontinence

Endocrine System

Messenger Molecules



- ❑ Local hormones: chemicals that allow cells adjacent to one another to communicate.
- ❑ Endocrine hormones: chemicals released to the blood to stimulate a response in distant cells.
- ❑ Neurotransmitters: chemicals that cross the synaptic gap between two neurons.

Local hormones



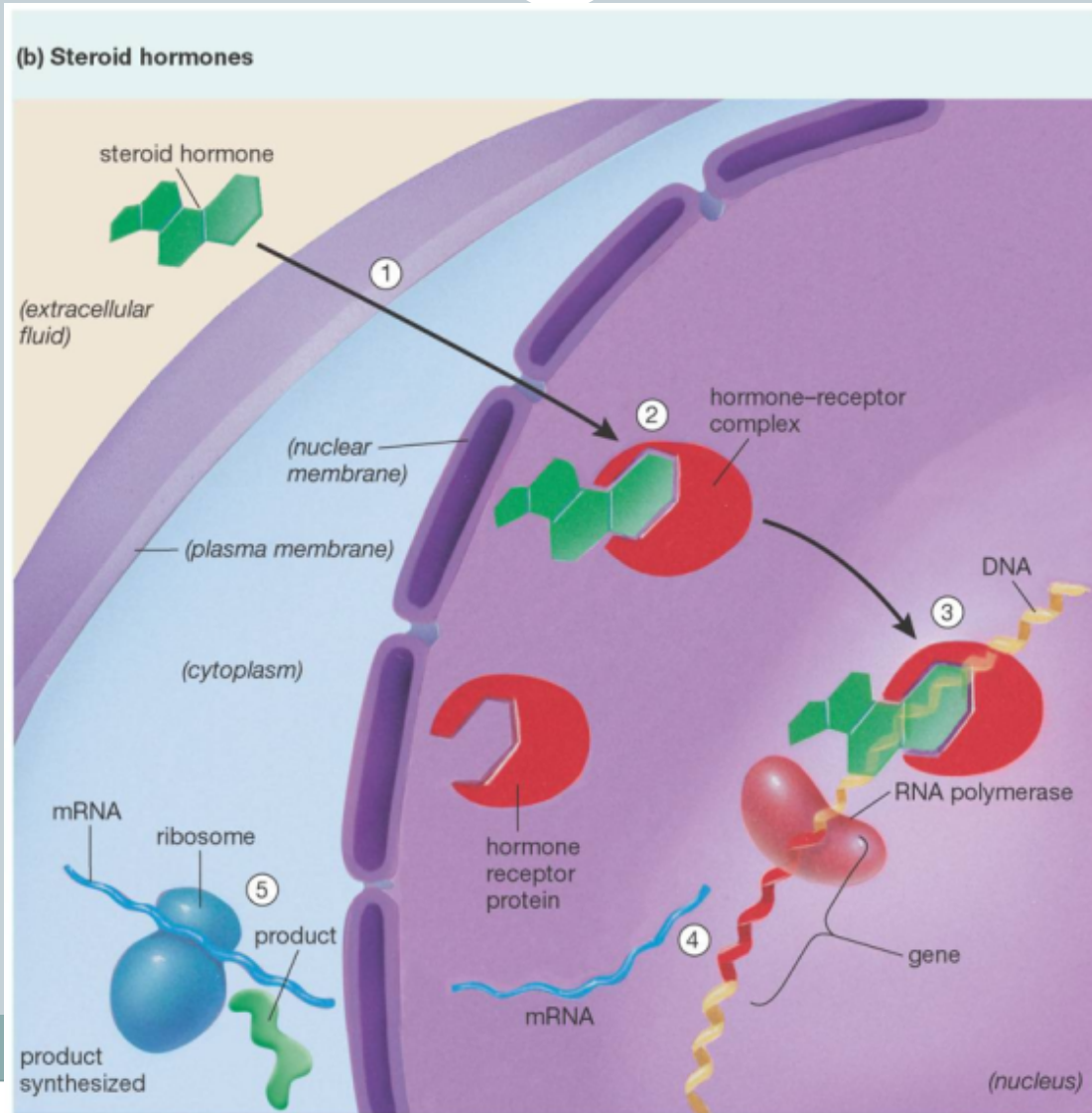
- Secreted by most, if not all, cells.
- Used to communicate between nearby cells.
- Example: Prostaglandins, which cause multiple effects, including stimulus of pain receptors near injured cells, uterine contractions, and more.

Endocrine hormones



- Produced by endocrine (“ductless”) glands and secreted into the bloodstream.
- Endocrine hormones may affect a wide array of target cells to produce multiple effects.
- Many endocrine hormones are involved in maintaining homeostasis.
- Two types: peptides (small proteins) and steroids (lipids).

Steroid Hormones



Role of the Hypothalamus



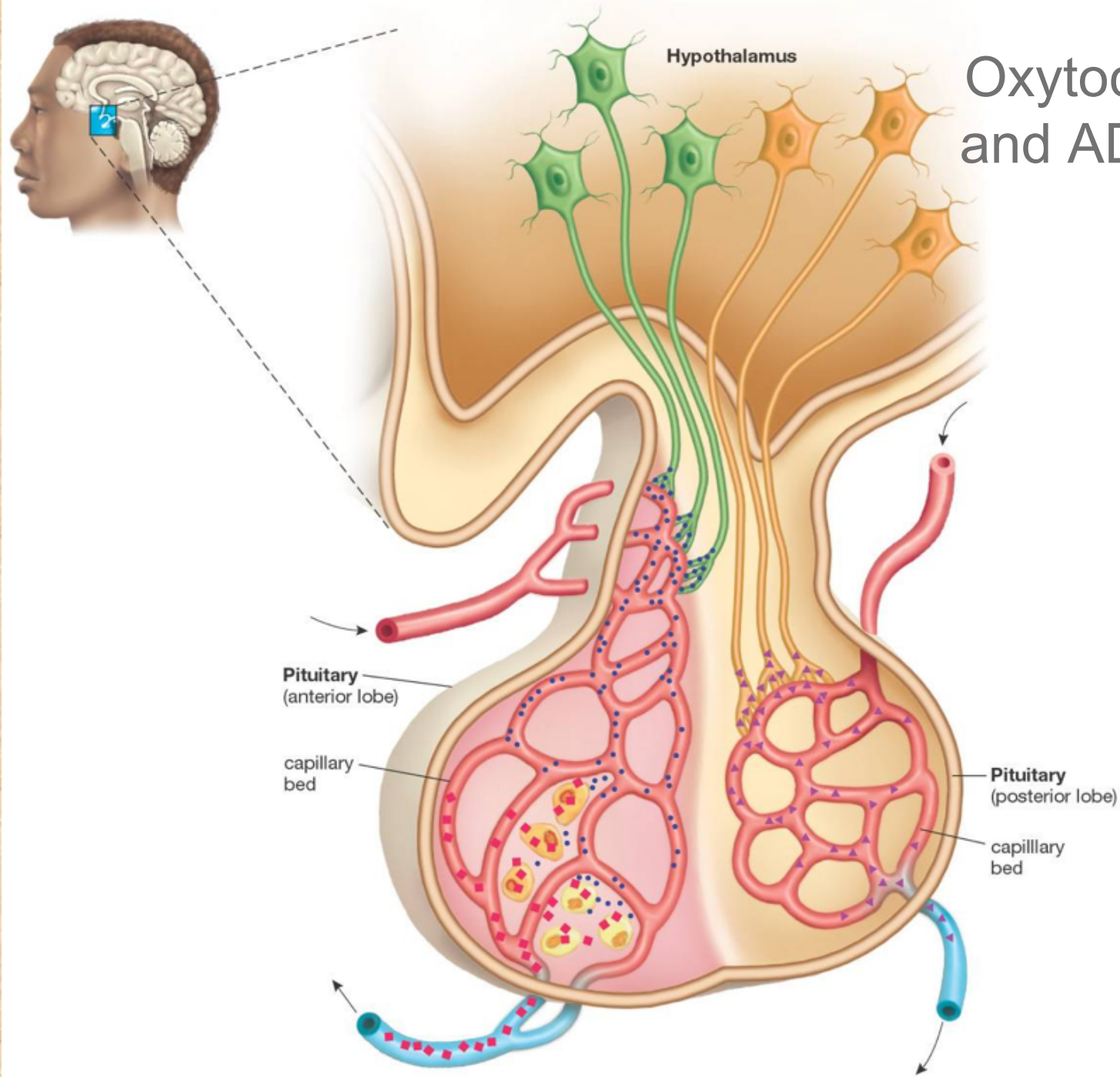
- The thalamus receives sensory information, relays some to the hypothalamus.
- Hypothalamus monitors the body for temperature, pH, other conditions.
- Hypothalamus signals pituitary gland if conditions need to be corrected, if there is an emergency to respond to, or other conditions.

Role of the Pituitary



- The pituitary is the “master gland” that signals other glands to produce their hormones when needed.
- The anterior lobe of the pituitary receives signals from the hypothalamus, and responds by sending out the appropriate hormone to other endocrine glands.
- The posterior pituitary receives oxytocin or antidiuretic hormone (ADH) from the hypothalamus, relays them to the body as necessary.

Oxytocin and ADH



Pituitary Hormones



Pituitary Hormone	Functions
Follicle-stimulating hormone (FSH)	Stimulates egg maturation in the ovary and release of sex hormones.
Lutenizing hormone	Stimulates maturation of egg and of the corpus luteum surrounding the egg, which affects female sex hormones and the menstrual cycle.
Thyroid-stimulating hormone	Stimulates the thyroid to release thyroxine.
Adrenocorticotropic hormone	Causes the adrenal gland to release cortisol.
Melanocyte-stimulating hormone	Stimulates synthesis of skin pigments.
Growth hormone	Stimulates growth during infancy and puberty.
Antidiuretic hormone	Signals the kidney to conserve more water.
Oxytocin	Affects childbirth, lactation, and some behaviors.

Endocrine Hormones



Gland	Hormones	Functions
Thyroid	Thyroxine	Regulates metabolism
	Calcitonin	Inhibits release of calcium from the bones
Parathyroids	Parathyroid hormone	Stimulates the release of calcium from the bones.
Islet cells (in the pancreas)	Insulin	Decreases blood sugar by promoting uptake of glucose by cells.
	Glucagon	Increases blood sugar by stimulating breakdown of glycogen in the liver.
Testes	Testosterone	Regulates sperm cell production and secondary sex characteristics.
Ovaries	Estrogen	Stimulates egg maturation, controls secondary sex characteristics.
	Progesterone	Prepares the uterus to receive a fertilized egg.
Adrenal cortex	Epinephrine	Stimulates “fight or flight” response.
Adrenal medulla	Glucocorticoids	Part of stress response, increase blood glucose levels and decrease immune response.
	Aldosterone	Regulates sodium content in the blood.
	Testosterone (in both sexes)	Adult body form (greater muscle mass), libido.
Pineal gland	Melatonin	Sleep cycles, reproductive cycles in many mammals.

Homeostasis and Hormones



□ Examples:

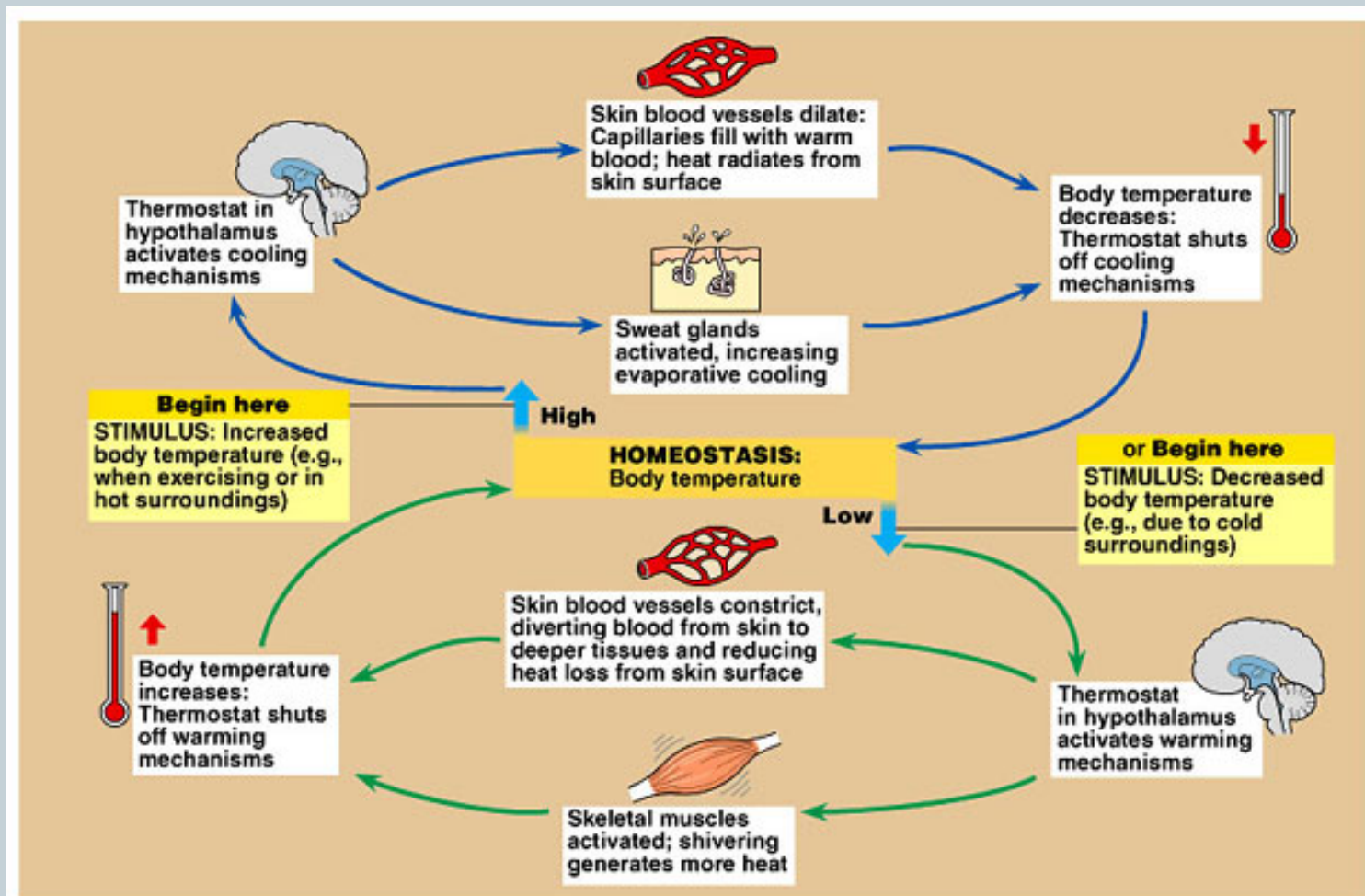
- | Thyroid and temperature control
- | Thyroid, Parathyroid, and calcium
- | Pancreas and glucose control

On your own paper...



- Knowing the following:
 - | The thyroid produces thyroxine, which increases metabolism.
 - | Thyroid-stimulating hormone is released by the pituitary.
- Sketch a negative feedback loop that controls body temperature regulation.

Temperature Control



On your own paper...

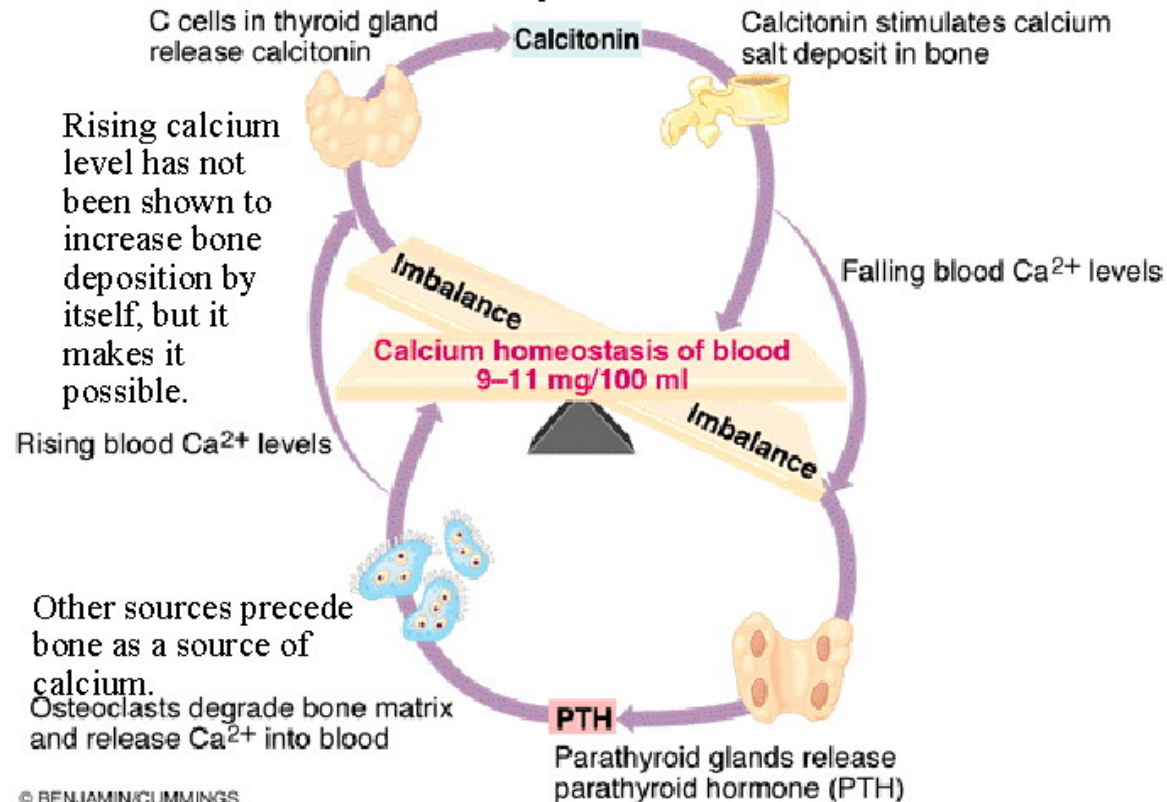


- Knowing the following:
 - | Calcitonin from the thyroid inhibits calcium release from the bones.
 - | Parathyroid hormone stimulates release of calcium from the bones.
- Sketch a negative feedback loop that controls blood calcium level.

Blood Calcium



Important for bone growth in children.
It is now being given to older adults to help reverse osteoporosis



On your own paper...

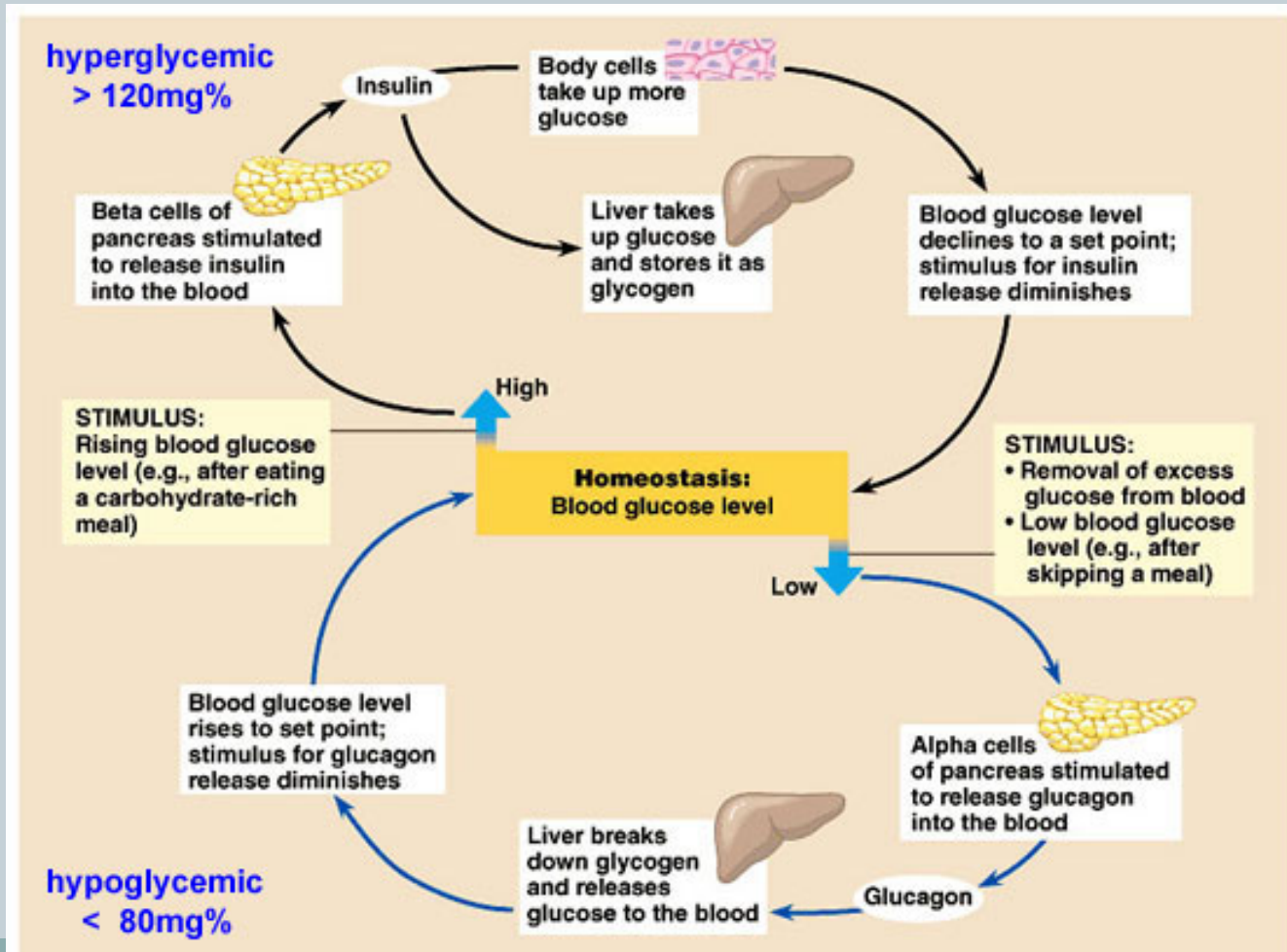


☐ Knowing that:

- | Insulin from the pancreas lowers blood glucose by stimulating body cells to take up glucose.
- | Glucagon from the pancreas increases blood sugar by stimulating the liver to break down glycogen into glucose.

☐ Sketch a negative feedback loop that controls blood sugar.

Blood Sugar Control



Assignt



- Summarize the roles of the pituitary and other endocrine glands in maintaining homeostasis in the human body.

Other hormone roles



- Controlling sleep cycles (melatonin)
- Controlling reproductive cycles (melatonin, sex hormones)
- Growth (growth hormone)
- Responding to stress or emergencies (epinephrine and other hormones)

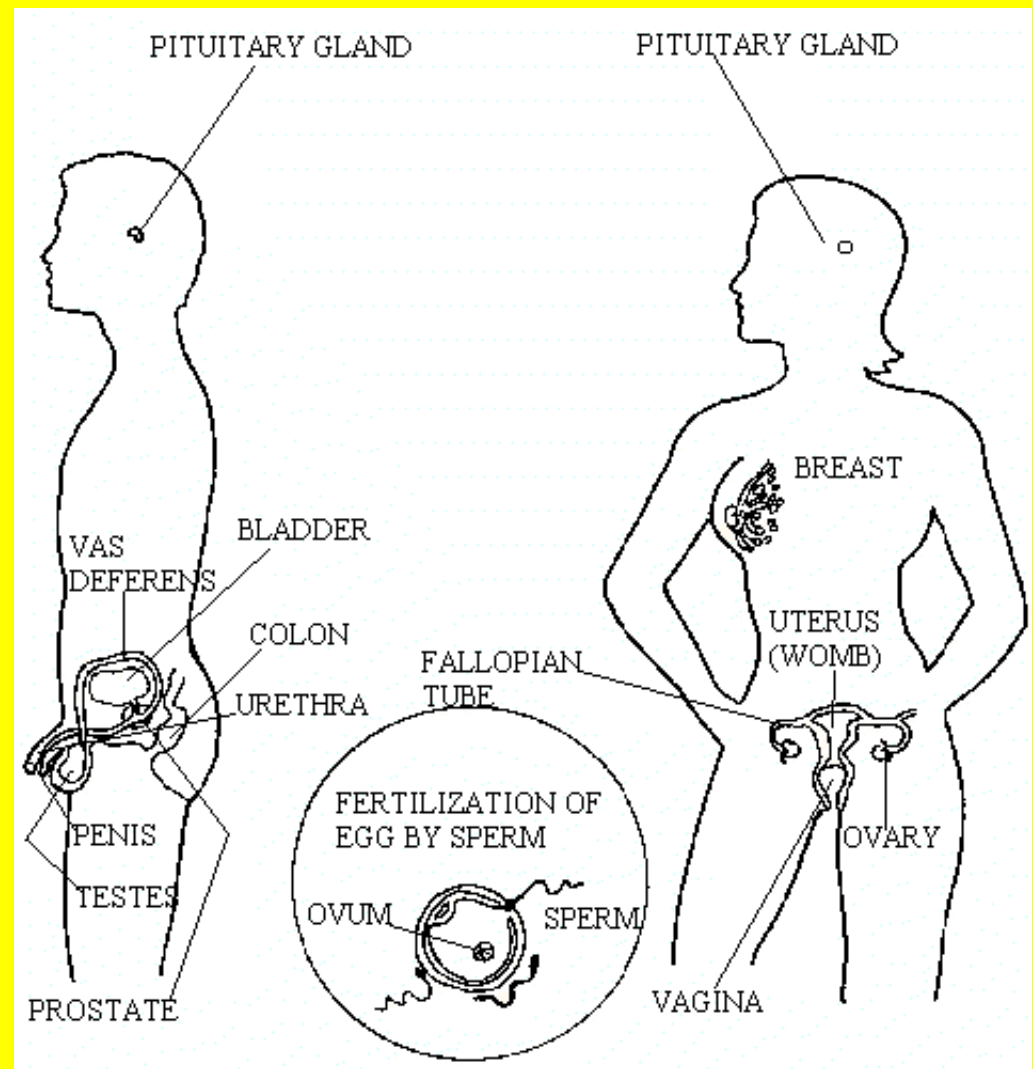
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Reproductive System

REPRODUCTIVE SYSTEM

What is the reproductive system?

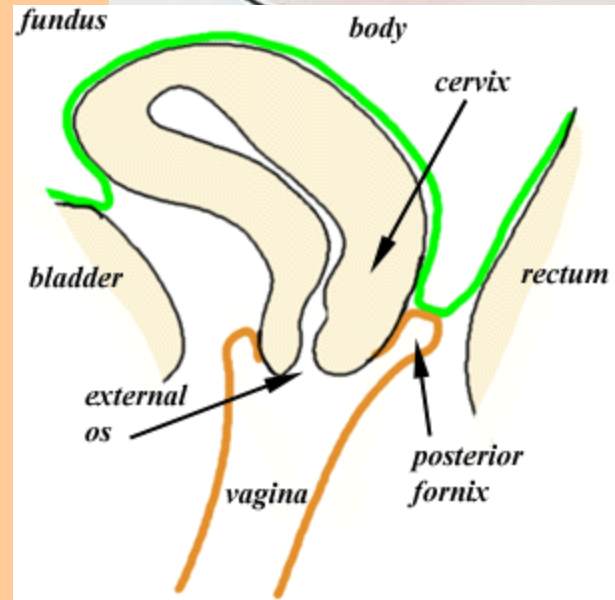
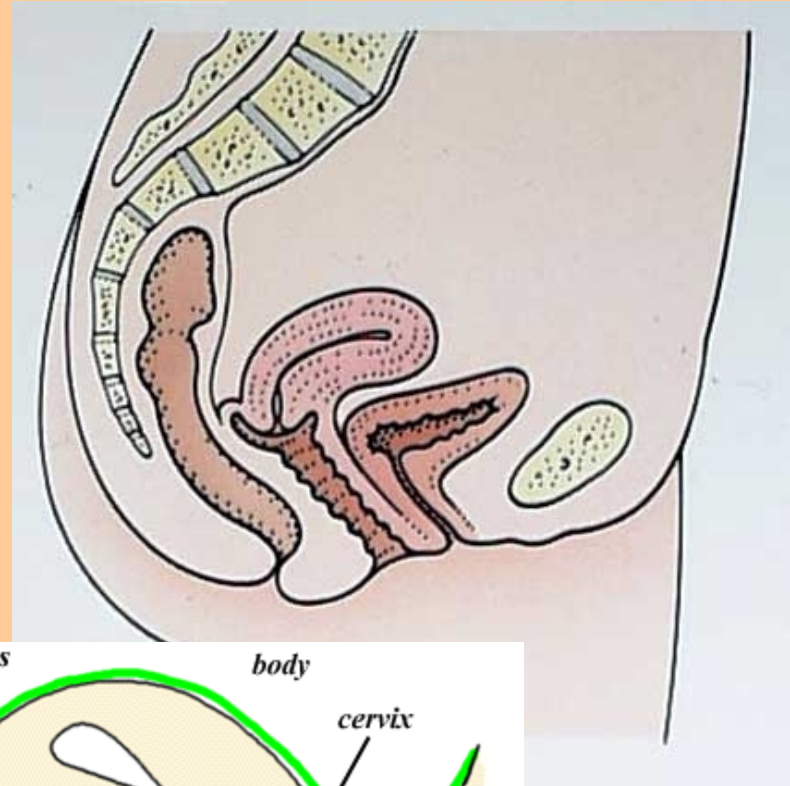
The reproductive systems in both the male and female consist of primary and secondary sex organs and sex glands.



The primary function of the reproductive systems is to perpetuate the species through sexual or germ cell fertilization and reproduction.

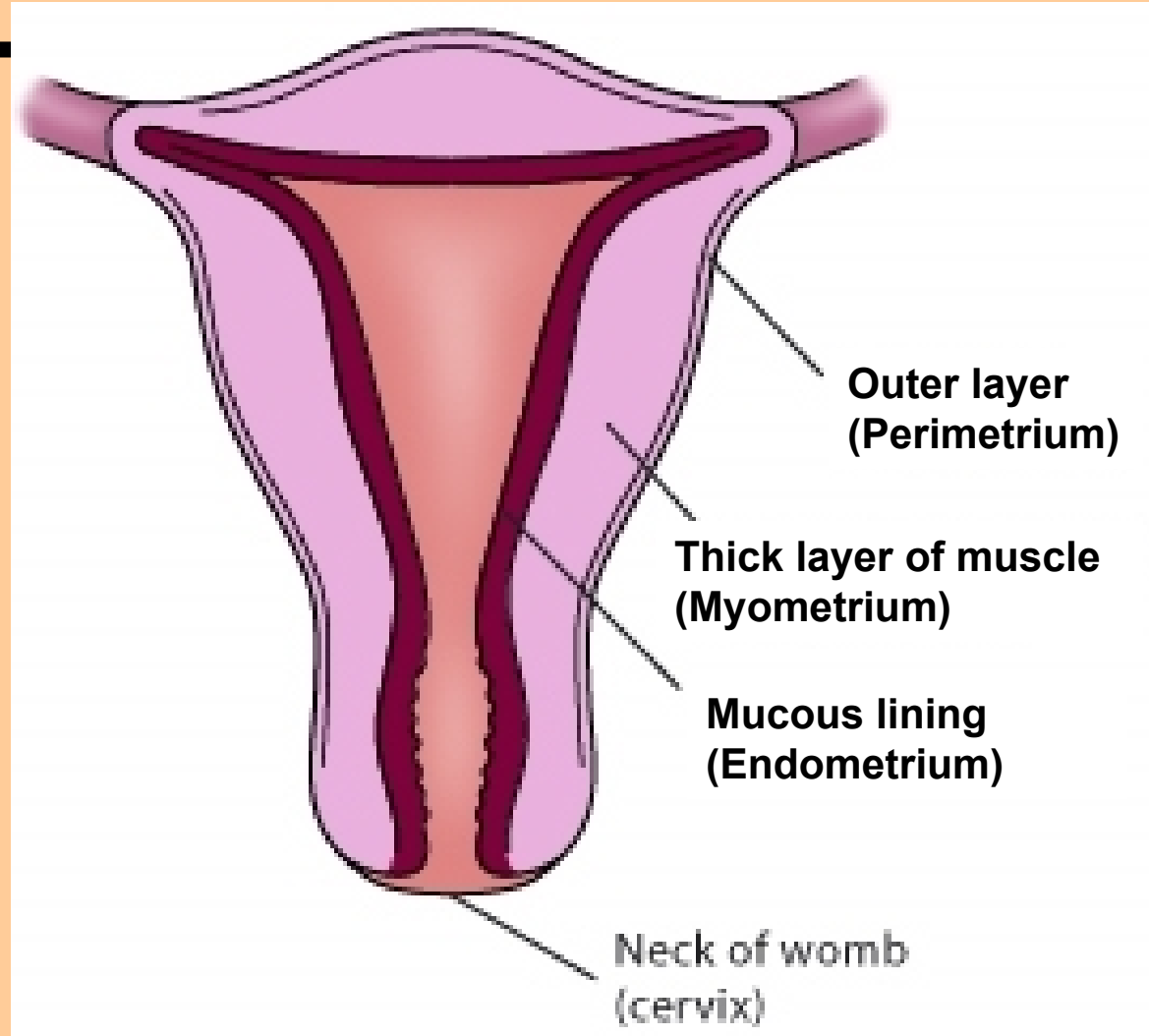
Female reproductive system: the uterus...

The uterus is a hollow, muscular, pear-shaped organ about the size of a woman's clenched fist. The top is tipped forward in a normal 'anteflexion' position. It can be divided into the body or corpus, and the bottom cervix. The rounded top portion, above the fallopian tubes, is called the fundus.



Female reproductive system: the uterus...

The uterus wall has 3 layers: the outside layer called the perimetrium, the muscular inside layer called the myometrium, and the mucous membrane lining the uterus called the endometrium.



The cervical opening is about the diameter of a pencil.

Female reproductive system: the uterus...

**The uterus has 3
functions:**

- 1. The endometrium sheds the lining of the uterus every 21 to 40 days by menstruation**
- 2. It provides a place for the protection and nourishment of the fetus during pregnancy**
- 3. It contracts during labor to expel the fetus**



Female reproductive system: the fallopian tubes...

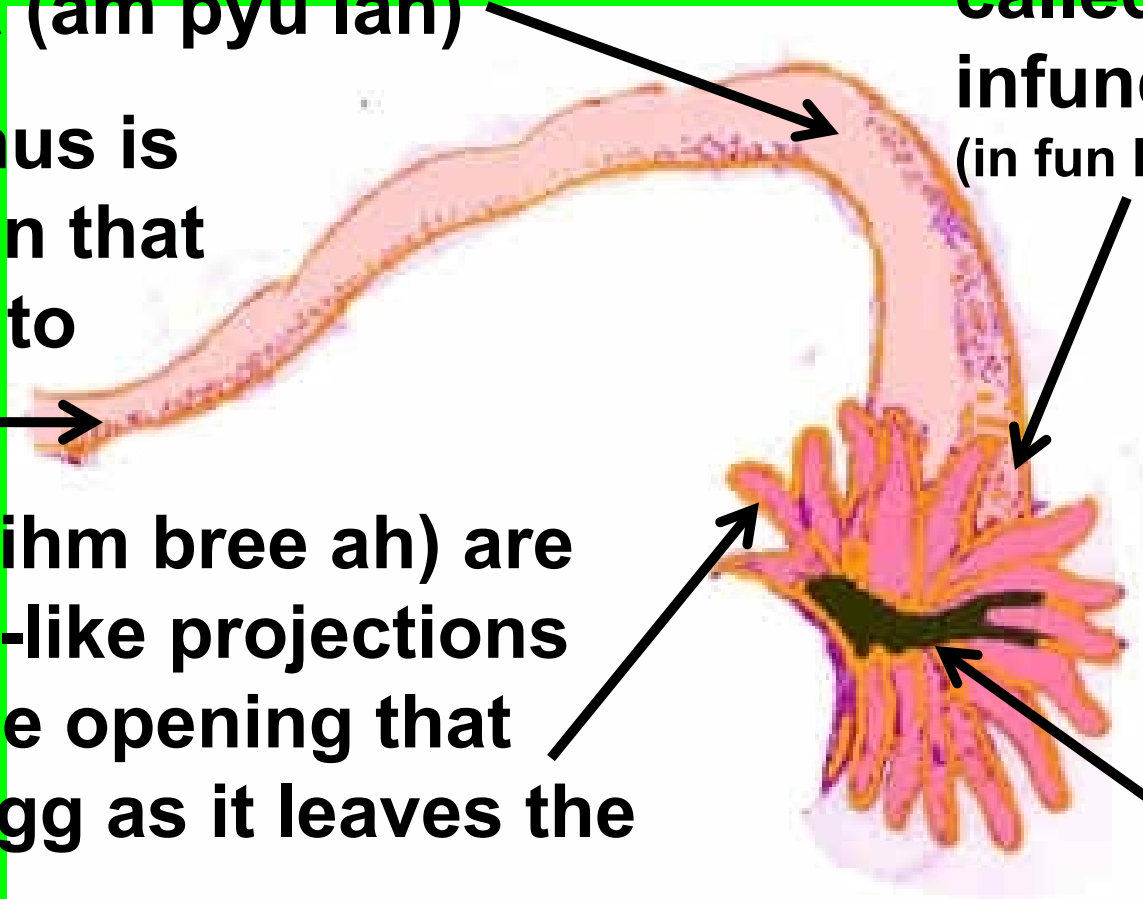
Tube widens to form the
ampulla (am pyu lah)

End of tube is
called the
infundibulum
(in fun DIB yū lum)

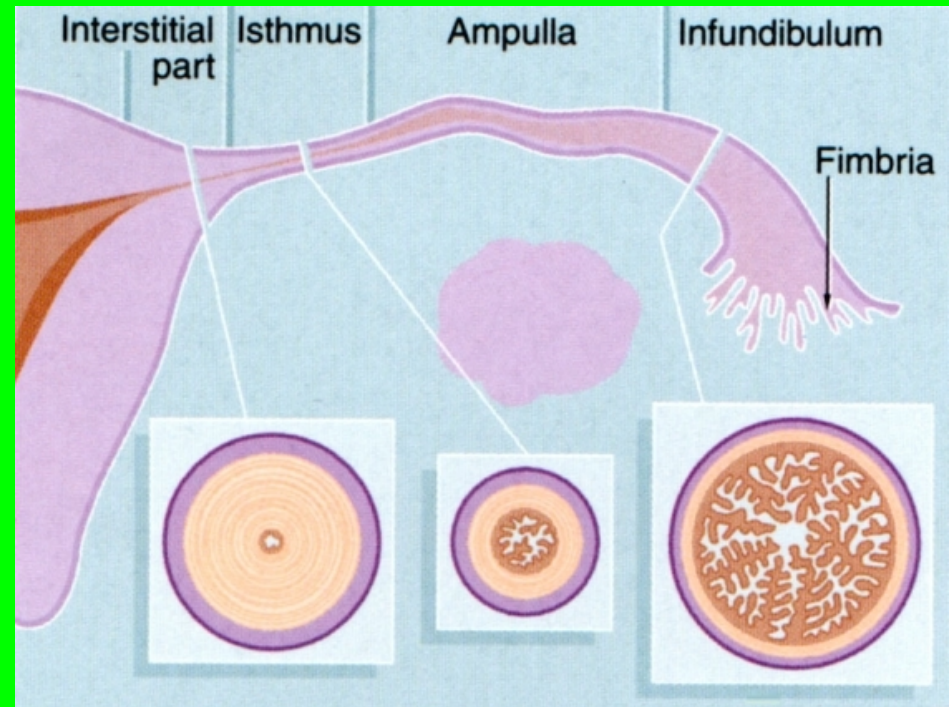
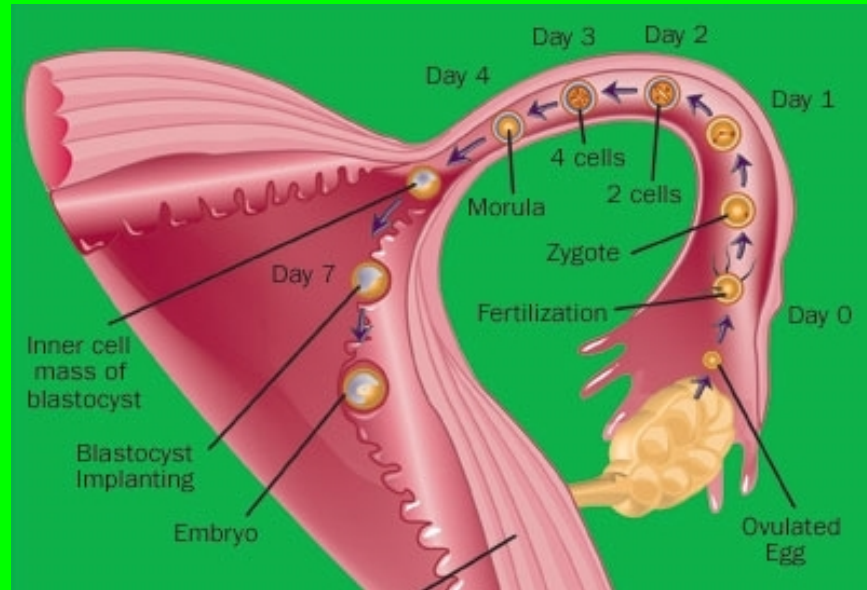
The isthmus is
the portion that
connects to
uterus

Fimbria (fihm bree ah) are
the finger-like projections
around the opening that
trap the egg as it leaves the
ovary

Opening
is called
the
ostium
(ah stē um)



Female reproductive system: the fallopian tubes...



The fallopian tube is 4-6 inches long. The egg, released from the ovary, is captured by the fimbria and brought into the fallopian tube. The egg is moved along inside the tube by muscular contractions and the waving action of cilia. It takes an egg about 3-4 days to travel the length of the tube. If an egg is fertilized, it occurs here.

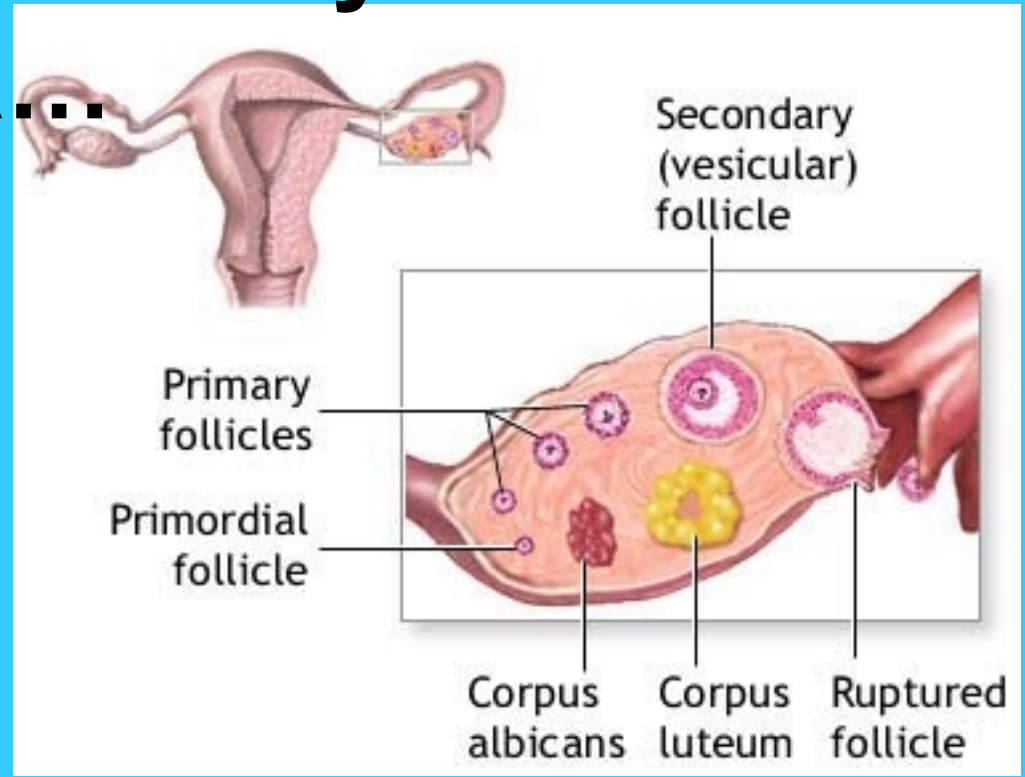
Female reproductive system: the ovaries....

The two ovaries are attached to each side of the uterus by a ligament. They are oval-shaped, about the size of a large olive, and lie close to the fimbria at the end of the fallopian tubes.

Each ovary is filled, already at birth, with egg-containing sacs called follicles. Each egg is called an ovum.

Female reproductive system: the ovaries and ova...

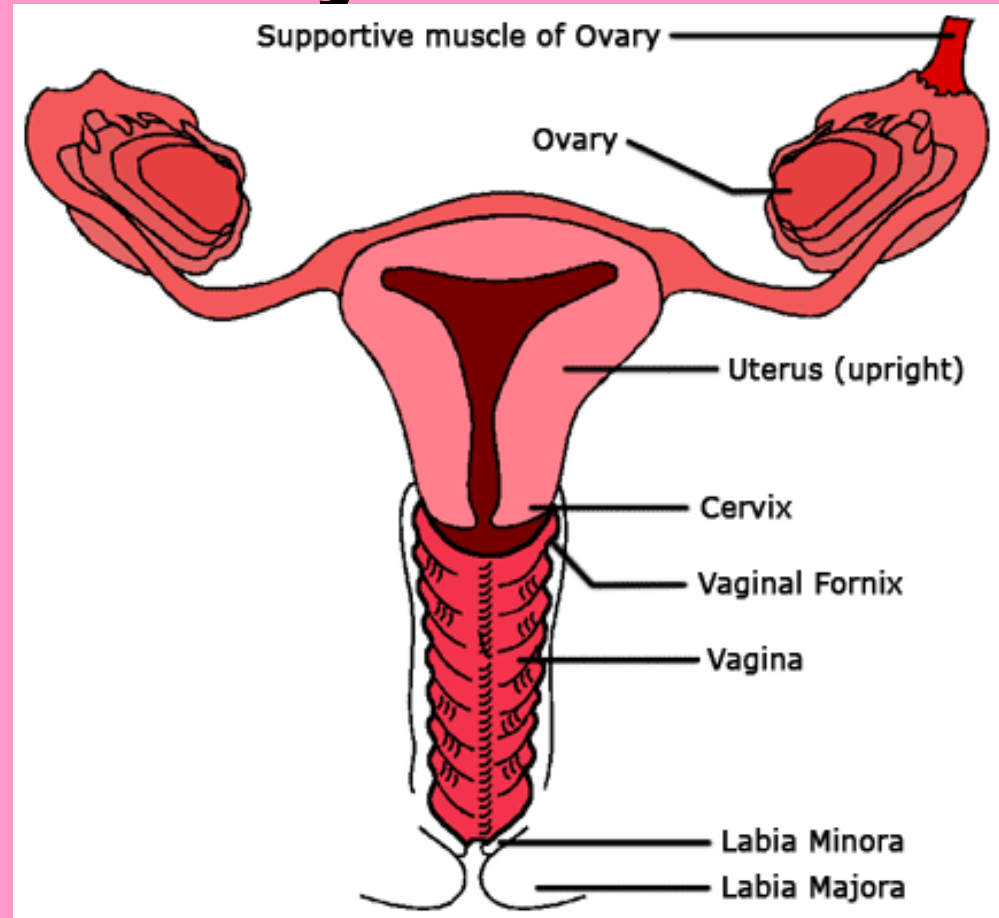
Once every 21 days, one follicle in one ovary ripens. This mature follicle is a graafian (GRAW fee un) follicle. The follicle ruptures in response to hormones from the pituitary gland, releasing the ovum/egg... a process called ovulation.



After the follicle ruptures, it becomes a mass of yellow cells called the corpus luteum. This is a temporary, progesterone-producing structure.

Female reproductive system: the vagina...

The vagina extends from the cervix to the outside of the body. It is a 3 ½ inch long muscular tube that expands in length and width during sexual arousal.



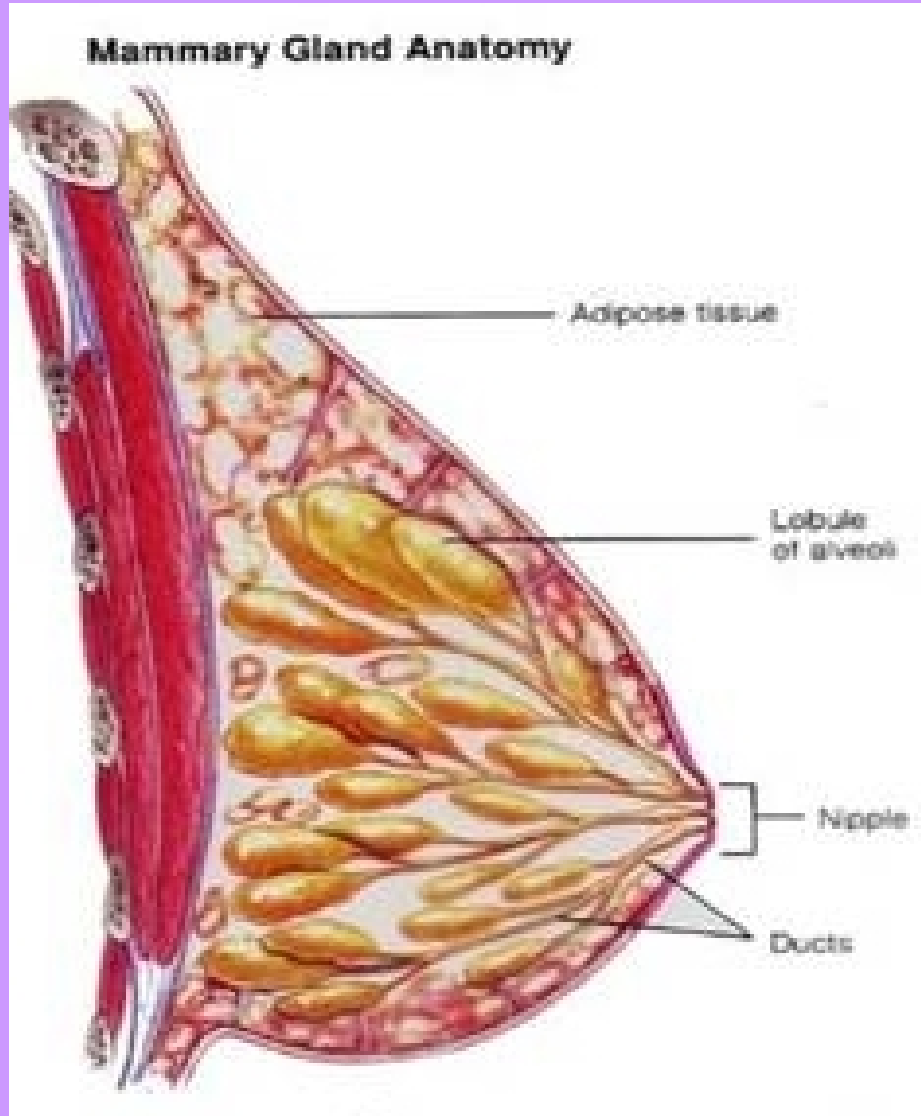
The vagina is the female organ for copulation (sexual intercourse), receiving the seminal fluid from the male penis. It is also a passageway for menstruation or the birth of a fetus.

Female reproductive system: the vulva...

The vulva is 5 organs making up the external genitalia of the female:

- 1. Mons pubis: triangular-shaped pad of fatty tissue over the pubis bone, covered with pubic hair**
- 2. Labia majora: 2 large folds of adipose tissue on the sides of the vaginal opening**
- 3. Labia minora: 2 smaller folds of adipose tissue on the inside of the labia majora**
- 4. Vestibule: area between labia with openings for the vagina, urethra, and two excretory ducts for Bartholin's glands (provide lubricant)**
- 5. Clitoris: sensitive fold of tissue partially covered by hood**

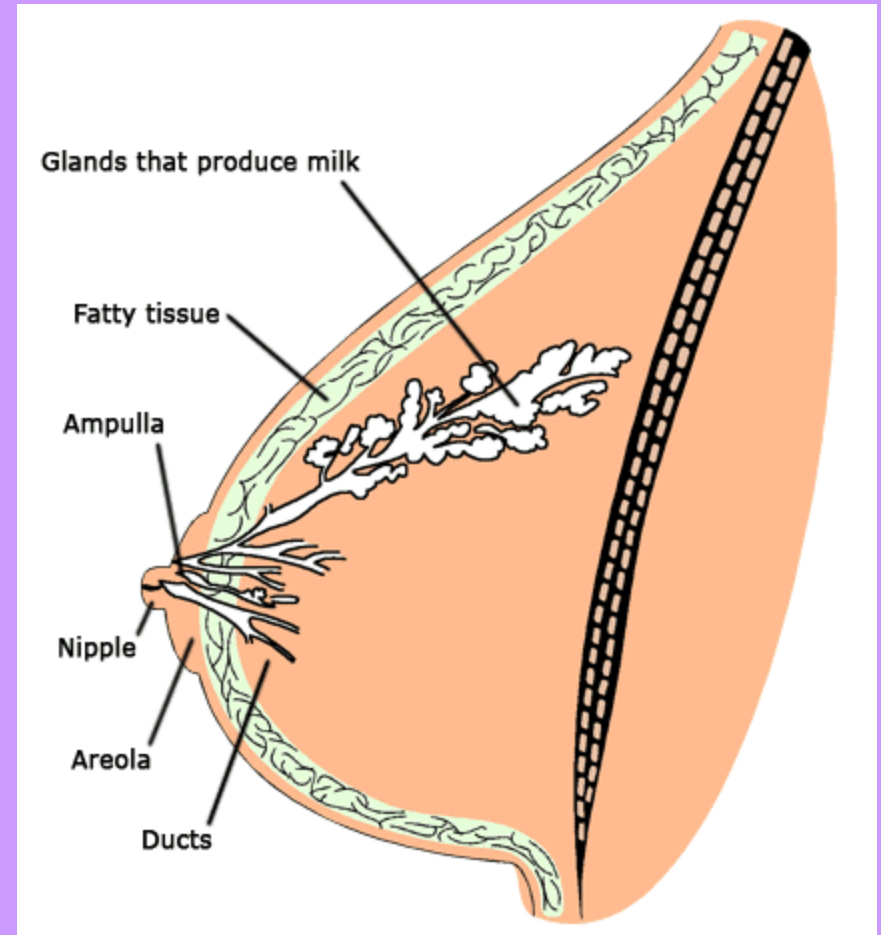
Female reproductive system: the breasts...



The breasts are the mammary glands, varying in size according to age, heredity, and the amount of fatty tissue present. Each breast has 15-20 glandular lobes separated by connective tissue. After childbirth, the pituitary gland stimulates these lobules with the hormone prolactin... and they produce milk.

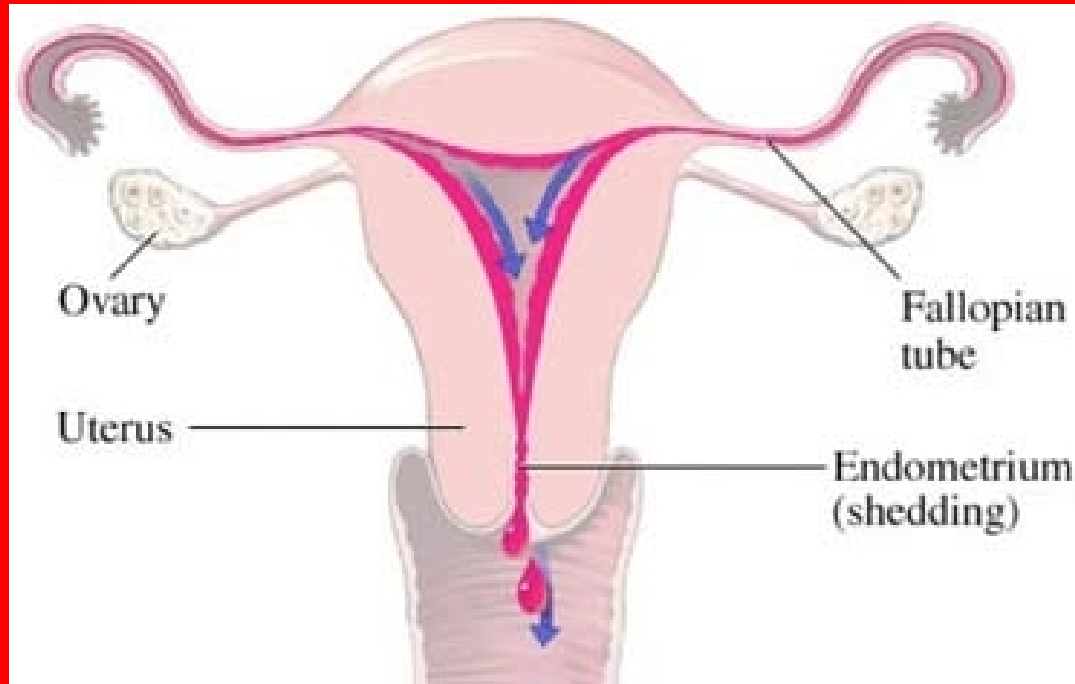
Female reproductive system: the breasts...

The dark-colored circle at the tip of the breast is called the areola (ah REE ah lah). It contains sebaceous glands to keep the skin conditioned. In the center is the nipple, where ducts from the lobules open. The first secretion from the breast is not a true milk, but a thin yellowish substance called colostrum.



Colostrum contains nutrients and the mother's immunities that can protect baby.

Female reproductive system: the menstrual cycle...

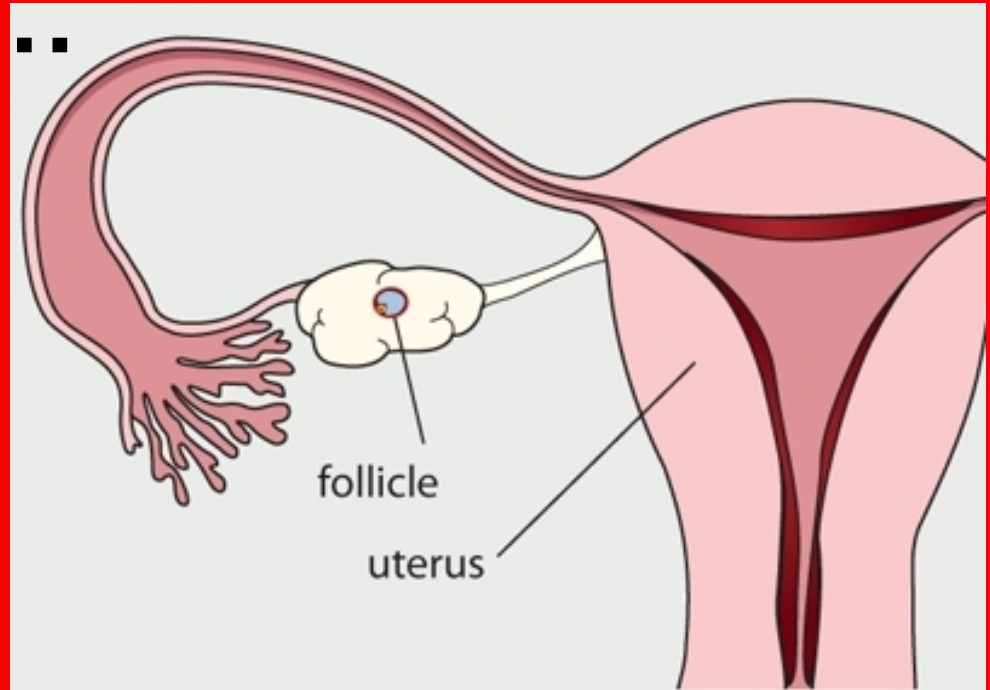


The onset of the menstrual cycle, or menarche, begins at puberty and ceases at menopause. The cycle has 3 phases:

The follicular phase is characterized by menstruation, when the thickened lining of the endometrium is shed because no egg was fertilized or implanted. Just prior to this phase is a premenstrual period characterized by hormonal and physical changes.

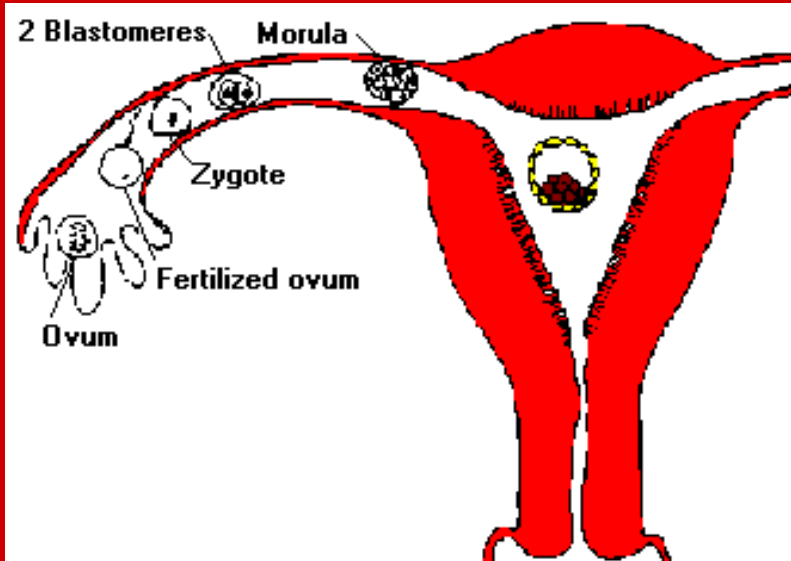
Female reproductive system: the menstrual cycle...

The ovulatory phase comes next. Estrogen is the hormone produced by the ovaries, which stimulates the maturation of a follicle and thickens the endometrium.



The luteal phase follows ovulation, and is characterized by the development of the corpus luteum, the secretion of progesterone, the preparation of the endometrium for implantation of a fertilized egg, and the formation of a thick mucus to block the cervix once the egg passes out of the fallopian tube.

Female reproductive system: pregnancy...



At the moment of conception, a single sperm with 23 chromosomes (carrying genetic information from the father) penetrates/fertilizes a single egg with 23

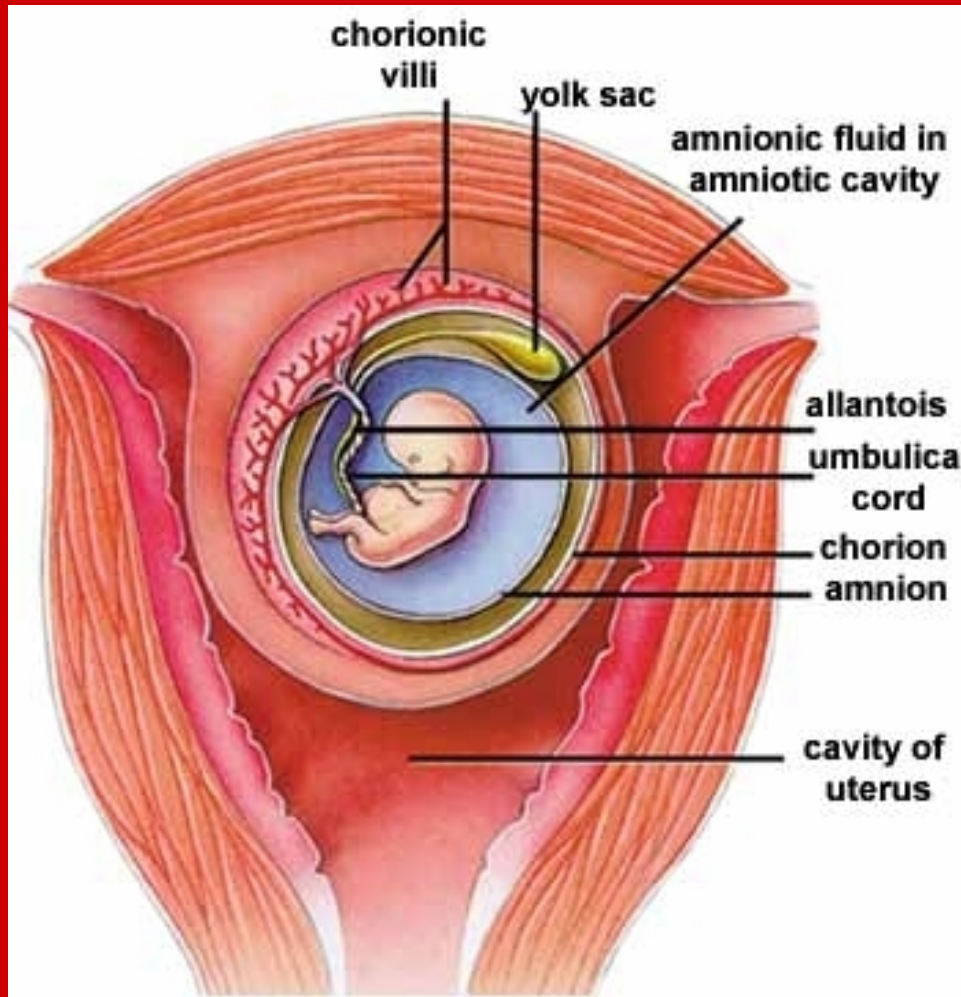
The fertilized egg implants in the uterus.

chromosomes (carrying genetic information from the mother).

Progesterone production increases to signal a pregnancy; it can be detected in urine and blood.

The resulting cell, a zygote, now has 46 chromosomes. The cell begins dividing and is also called a blastocyst. From week 2-8, it is called an embryo.

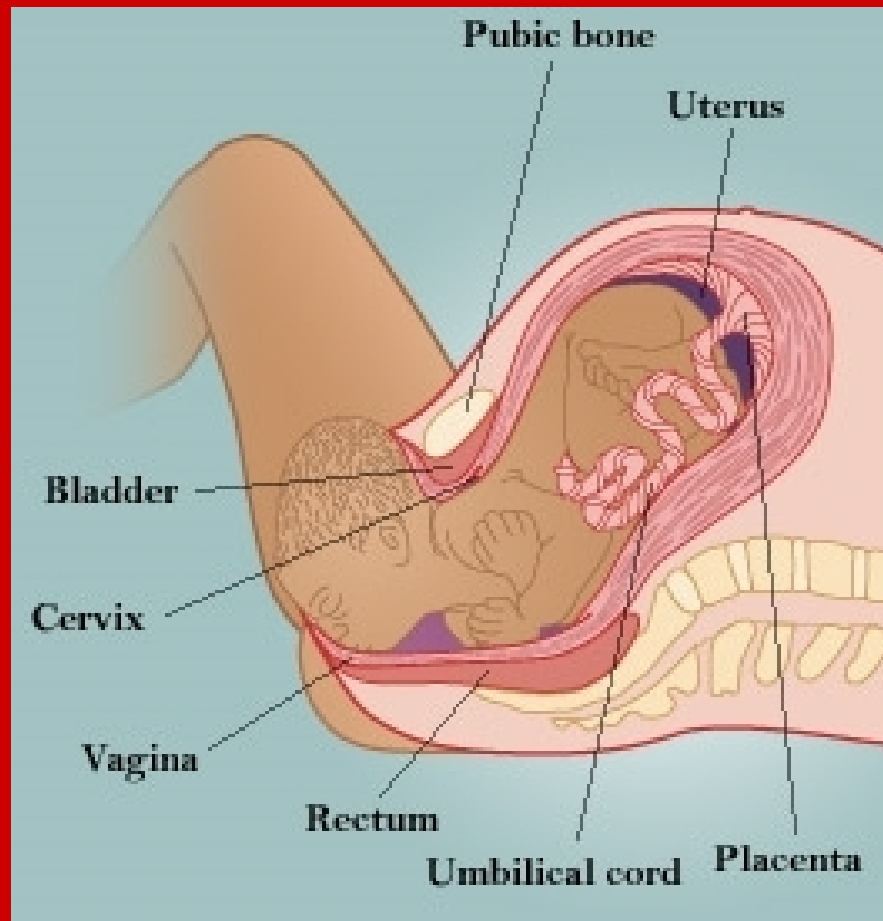
Female reproductive system: pregnancy...



The embryo is suspended in an amniotic sac surrounded by fluid during the 280 day gestation period. The umbilical cord attached at the navel connects it to the placenta, where it gets nutrients and oxygen. During the 1st trimester, all parts of the embryo are formed.

During the second trimester all parts start to function; during the last trimester the embryo is now called a fetus and the main task is growth.

Female reproductive system: pregnancy...



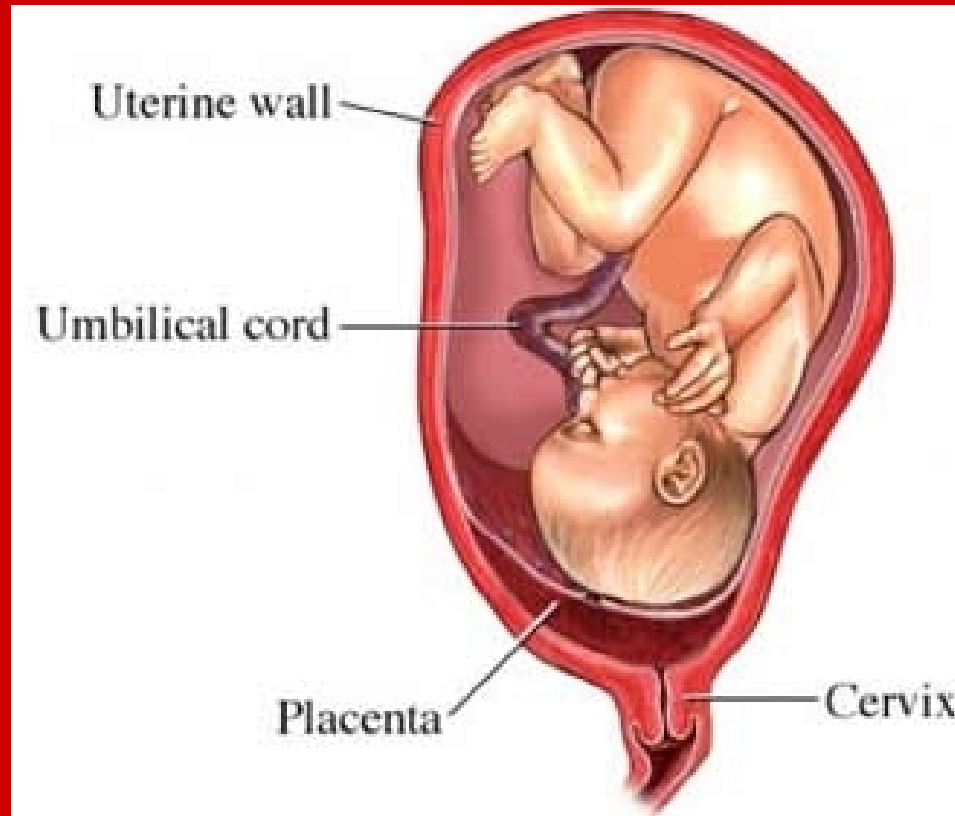
Labor, characterized by muscle contractions, dilation (to 10 cm) and effacement (thinning) of the cervix, and expulsion of the mucous plug that formed in the cervix, signals the onset of parturition... the childbirth process. The cephalic, or head-first delivery, is the most common. Breech is a

The umbilical cord is cut and clamped, and placenta (afterbirth) is delivered following birth of the baby.

backward presentation; Caesarian is delivery through an incision in the abdomen.

Female reproductive system: pregnancy...

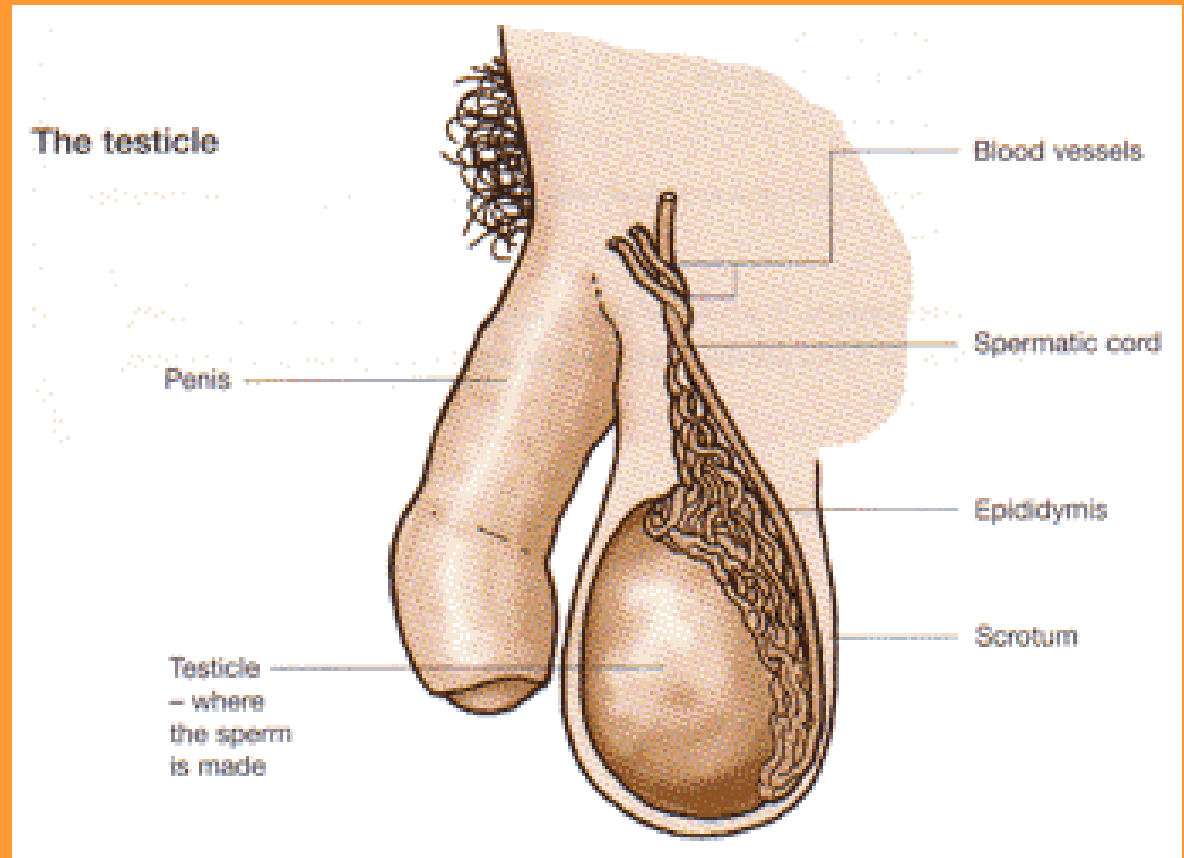
Pregnancy can be complicated by certain conditions:



1. Placenta previa is the development of the placenta over the opening of the cervix
2. Preeclampsia is a pregnancy- induced hypertension (high blood pressure)
3. Spontaneous abortion or miscarriage is the loss of a fetus during the first 20 weeks, often due to abnormalities, trauma, or lifestyle choices.

Male reproductive system: scrotum

The scrotum is a pouch of skin suspended from the perineal area and divided into two sacs... each containing one testicle and epididymis.



The scrotal tissue contracts in the absence of sufficient heat, pulling the testes up closer to the body where the temperature maintains the viability of sperm.

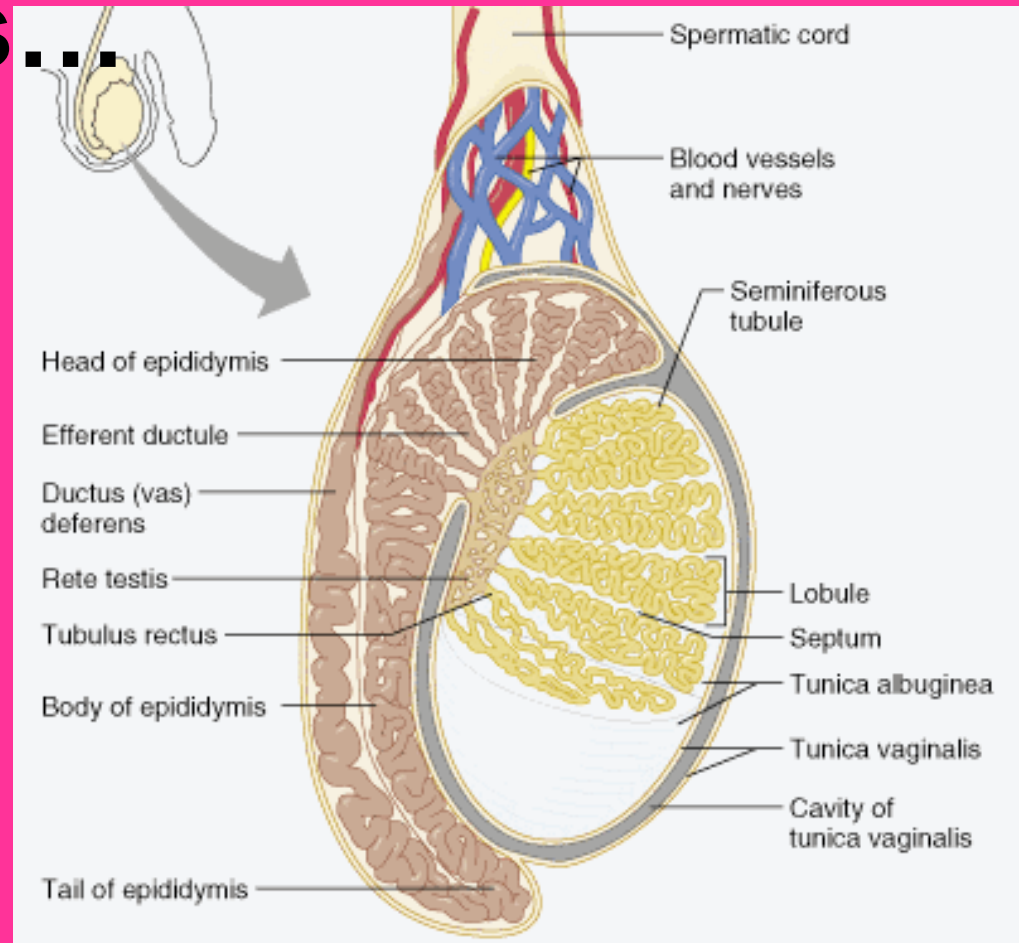
Male reproductive system: penis

The urethra runs the length of the penis from the bladder to the meatus (me-AY-tus). The flaccid penis fills with blood during arousal, causing an erection and allowing for the function of copulation/intercourse.

A lubricant, smegma, is produced under the prepuce on the penis. This foreskin may be circumcised/removed for hygiene or religious reasons. It covers the glans/head that is filled with nerve endings.

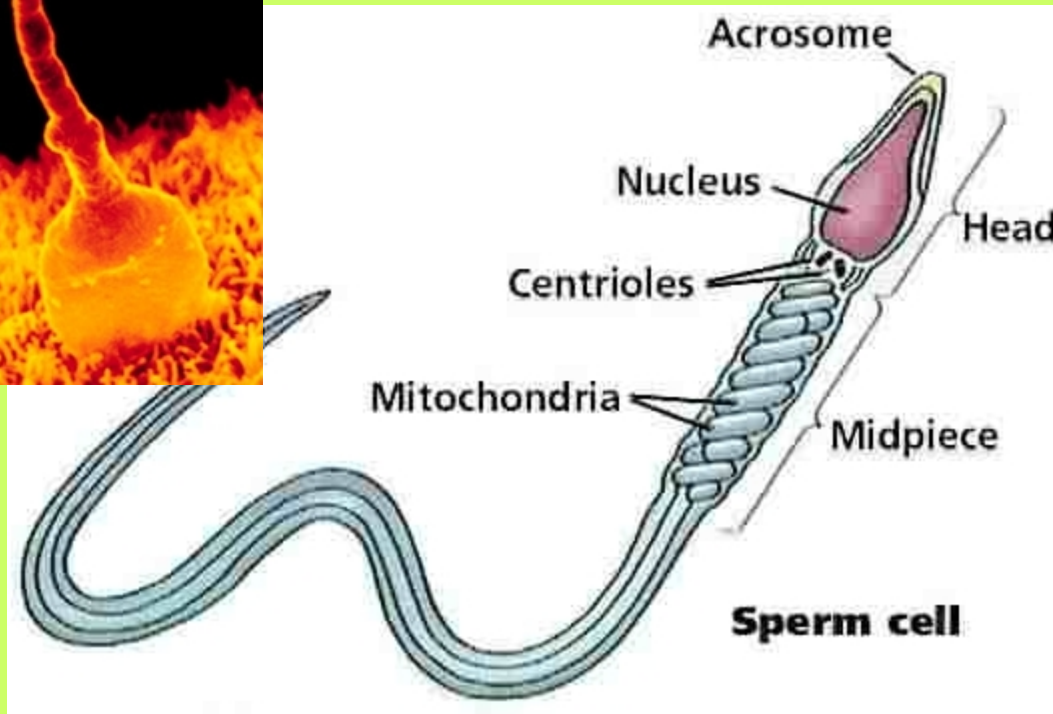
Male reproductive system: testes and epididymis

The oval-shaped testes are located in the scrotum, and are each divided into 250 lobules. Coiled within the lobule are seminiferous tubes where the sperm are formed.



The mature sperm are stored in the epididymis, a tube 13-20 feet in length, coiled and lying on the side of each testicle.

Male reproductive system: sperm...

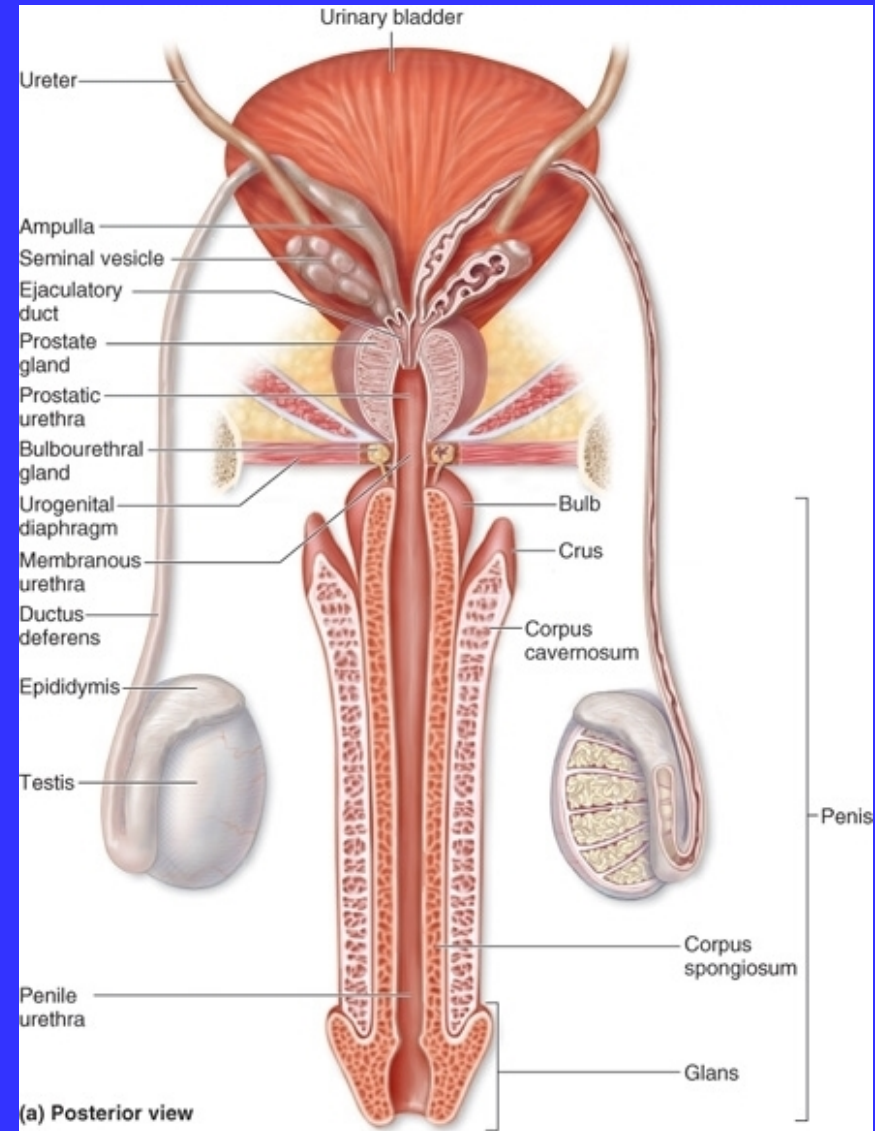


The acrosome (AK roh zome) covering the head of the sperm contains enzymes that help it penetrate the ova. The head carries the genetic material. The midpiece supplies energy. The tail or flagellum (flah JELL um) provides motility.

Sperm carry either an X/female OR Y/male chromosome. Since all ova carry the X/female chromosome, the male sperm does influence the baby's sex.

Male reproductive system: vas deferens and seminal vesicles...

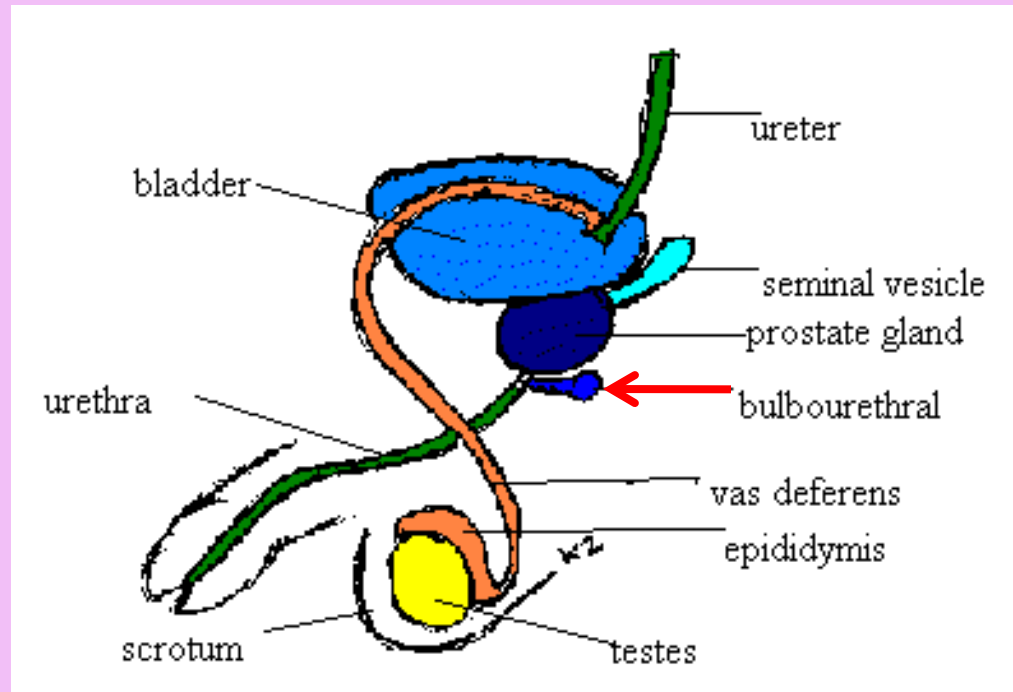
The vas deferens are muscular tubes transporting sperm from the epididymis to the seminal vesicles. The seminal vesicles produce a slightly alkaline fluid that mixes with the sperm and becomes part of the semen ejaculated during the male orgasm. It protects the sperm from the acidity of the urethra.



Male reproductive system: Prostate and Cowper's glands

The prostate gland also secretes an alkaline fluid that mixes with the sperm as part of the semen.

Enlargement of the prostate, benign prostatic hyperplasia, obstructs the flow of urine through the urethra.



The Cowper's or bulbourethral glands (bull boh yur REE thral) produce another mucous- alkaline secretion in the urethra, excreted from the glands just before ejaculation.